114 "HADLEY" REFLECTING TELESCOPE

Assembly and use guide

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SECTION LIST

PART 1: PARTS TO BUY

1.1: MIRRORS AND EYEPIECES

Optics – you need a mirror pair, and eyepieces:

Hadley only accepts 1.25" eyepieces (for reasoning, see wiki for details)

Mirrors — the **primary/secondary** set is commonly found on Amazon, eBay, and AliExpress.

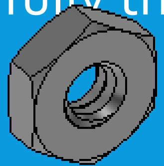
- Look for a "d114/f900mm" set that includes a 25mm elliptical flat + spherical primary
- Typical price approximately $30 USD.

Eyepieces — come standard in 1.25" and 2", listed by focal length (mm). Shorter focal lengths provide higher magnification, but a smaller view of the sky. It pays to have at least one "high" and "low" power. **See Appendix A.**

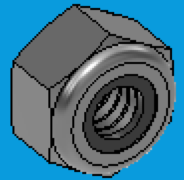
- **Low Power:** 25mm plossl (better & sharper view) OR 23mm aspheric (~$5 to 10 USD cheaper, slight blur)
- **High Power:** "66°/68° UWA" or "TMB Planetary" in 6mm or 9mm. (AVOID short focal lengths of plossl and aspheric, as at short lengths these distort light badly)

1.2: PARTS LIST: BOLTS AND SCREWS

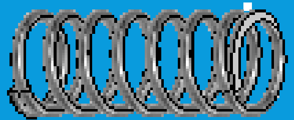
- **Screws:**
  All hardware is in #10 (imperial). There are metric remixes; these instructions still apply otherwise.
  Quantities are matched to typical box quantities for some commonly-used sizes to ensure margin for loss and excess for modifications or mounts. Screws should all be fully threaded.



- #10-24 nuts - **50x**

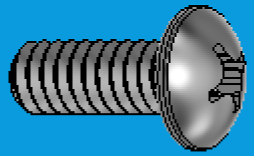


- #10-24 nuts (nylon locknuts) - **3x**

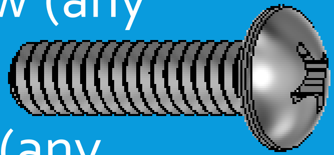


- Spring, diameter able to fit over 10-24 screw, 0.75" to 1" or 20-25mm -**4x**

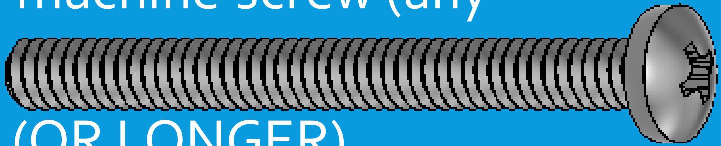
- #10-24, ½" long machine screw (any head) - **50x**



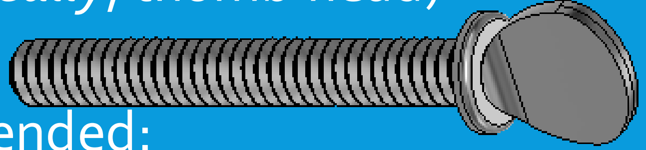
- #10-24, 3/4" long machine screw (any head) - 8**x**



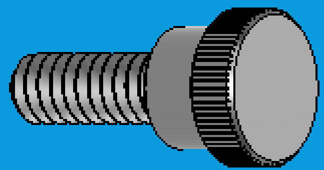
- #10-24, 2" long machine screw (any head) - 4**x**



- #10-24, 1" long (OR LONGER) machine screw (*ideally*, thumb head) - 3**x**



- Optional/recommended:
  #10-24, ½" long thumb screws (*ideally*, NYLON)
  –2x minimum, up to 4x



1.3: RODS OR TUBES

- "Hadley 114" is built using three main rods, which should be ½" in diameter and ideally 36" long
- Excess rod length is acceptable, with a practical minimum of ~33" (850mm)
- The intended material is aluminum solid rod, but aluminum tube (2mm wall, ½" outer diameter), steel rod or tube, carbon fiber tube, and even wooden dowels have been used in builds by community members
- Outer diameter needs relatively tight tolerance to slide printed parts on
- If buying rod or tube from a hardware store or other in-person location, it is recommended to bring a test fit part (see 2.3.1) to check fit of rods with printed parts
- NOTE: ½" EMT conduit pipe is actually 0.706" outer diameter, and thus cannot be used with this project's standard models. Adjusted and remixed CAD (not currently available) would be required.

1.4: ADHESIVE FOR MIRROR ATTACHMENT

- Attaching the mirrors to the telescope requires an adhesive which cures "soft" - superglue or epoxy, for instance, will pull the mirror out of shape (on a microscopic scale). This worsens the image.
- The mirror is also susceptible to thermal expansion, further necessitating the glue cure "soft."

Any silicone-based adhesive should work well for this, as silicone adhesive cures "soft" and bonds excellently to both glass and plastic.

Examples of silicone adhesives:

- Loctite clear silicone
- RTV silicone glue
- Silicone caulk
- Silicone aquarium sealant

PART 2: PARTS TO PRINT

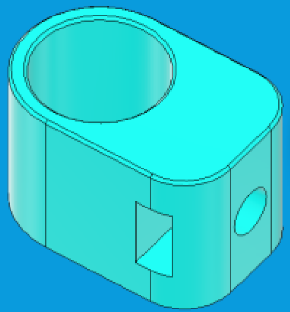
2.1: PART PRINTING GENERAL

- This section lists the parts which need to be printed to assemble a Hadley 114 telescope
- Parts marked "BLACKOUT" should be printed black if available, other parts may be printed in any color desired (see Part 3 for further discussion of blackening)
- The design allows printing without supports for most materials on common machines
- Telescope prints excellently in PLA, and has been printed successfully in PETG and ABS, though those materials have their own challenges (temperature control and dimensional accuracy)
- Do not scale the prints excessively. With proper print settings, scaling shouldn't be required, and scaling more than 1-2% can cause issues with fit of purchased parts like eyepieces and nuts
- Recommended print settings:
  - 30% cubic infill
  - 4 perimeters (0.4mm nozzle) OR 3 perimeters (0.6mm nozzle)
  - Whatever layer height makes you happy & works with your nozzle

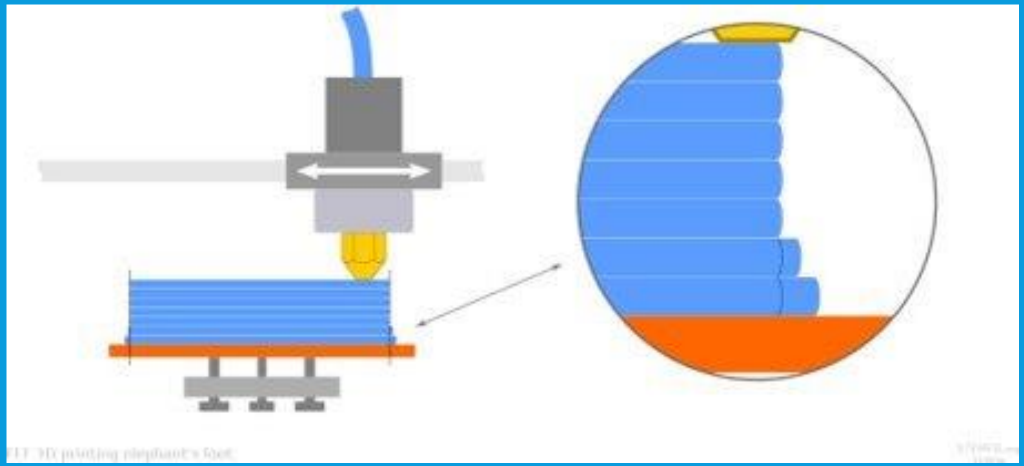
Note: The way to stronger prints is "more perimeters" rather than higher infill.

2.2: TEST PRINT & ANALYSIS

- Test print shown can be used as a quick check of rods acquired from a store or vendor before printing all parts
- Rod should fit smoothly into large hole with some slack, gap of 0.5 to 1mm
- Also check a #10-24 nut and screw fit pocket and through-hole respectively
- With screw tightened into nut, test print should lock in place on rod



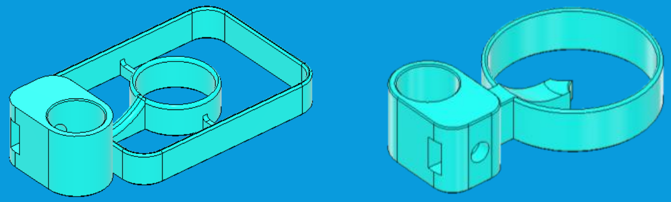
- If fit is poor, check for over-extrusion in print settings, or "elephant's foot" in initial layers of print which can be adjusted in settings or fixed by sanding/filing the lip of the hole



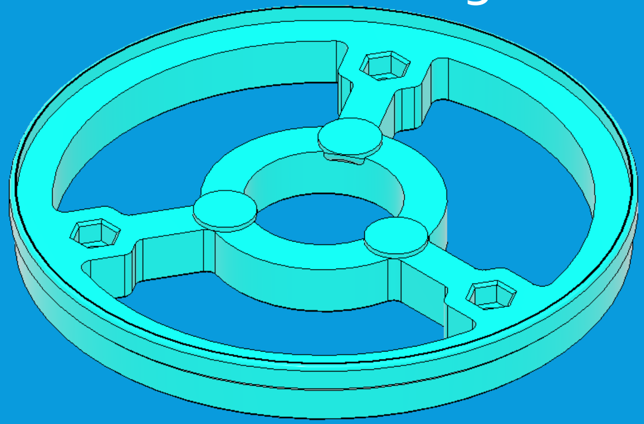
https://commons.wikimedia.org/wiki/File:3D_printing_calibration_elephant_foot.svg

2.3.1: PRINTED PART LIST

- **Sights** (lower and upper)**:** Orientation matters, pay mind so sight needles are not floating on your printbed.

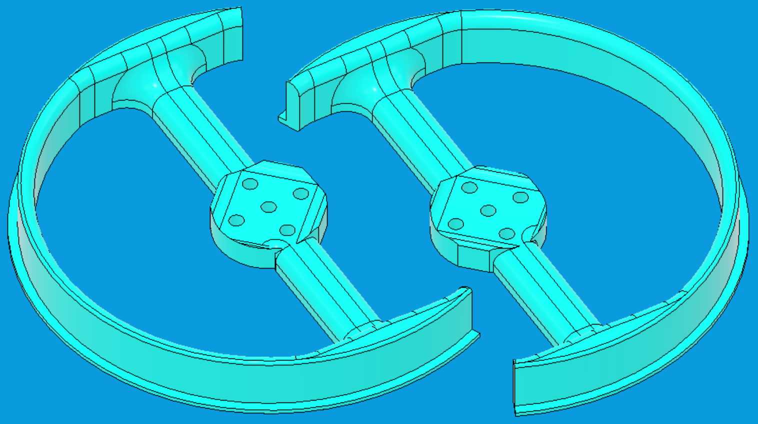


- **Mirror Cell:** Note that a pre-made STL includes both sights and the cell.



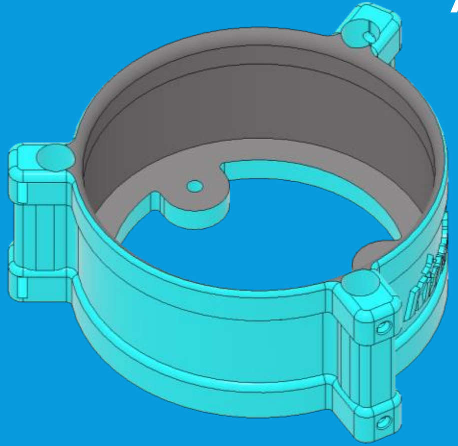
- **Bearings:** These are mirrors of each other; print the combined file, or individually. Just don't print the same one twice!

  These carry the weight of the whole telescope, so print with extra perimeters for strength! In my prints, I use 5 walls with a 0.6mm nozzle.

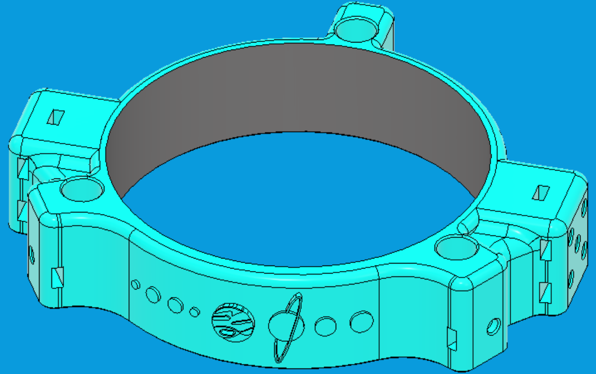


2.3.2: PRINTED PART LIST

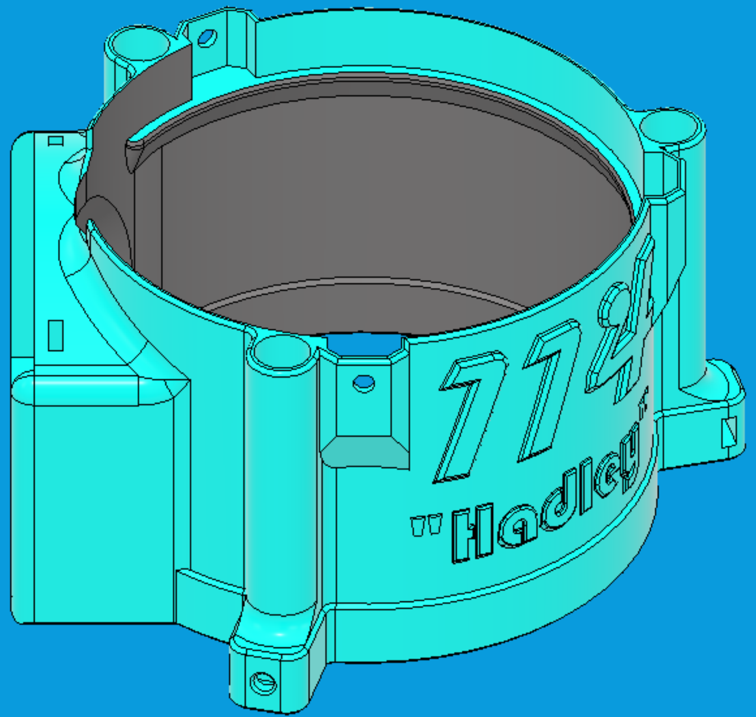
- **Lower Tube Assembly (LTA) Housing**



- **Middle Tube Assembly (MTA) Ring**

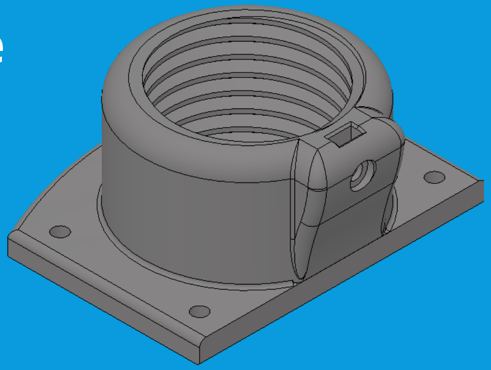


- **Upper Tube Assembly (UTA) Housing**
- The three on this page are the parts you want interior-blackened

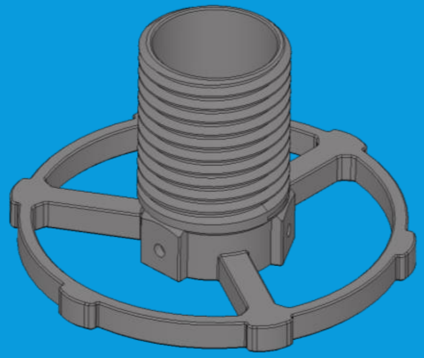


2.3.3: PRINTED PART LIST (BLACK)

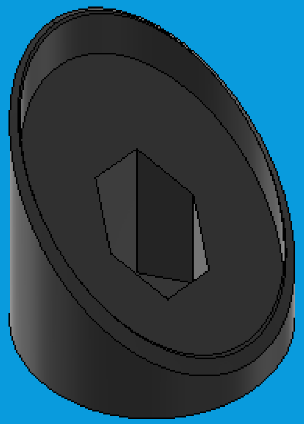
- **Focuser Base**



- **Focuser Screw/drawtube:** The focuser base and tube are challenging prints, make sure your printer is up for accuracy on the threads to ensure they screw together. Consider a finer layer height



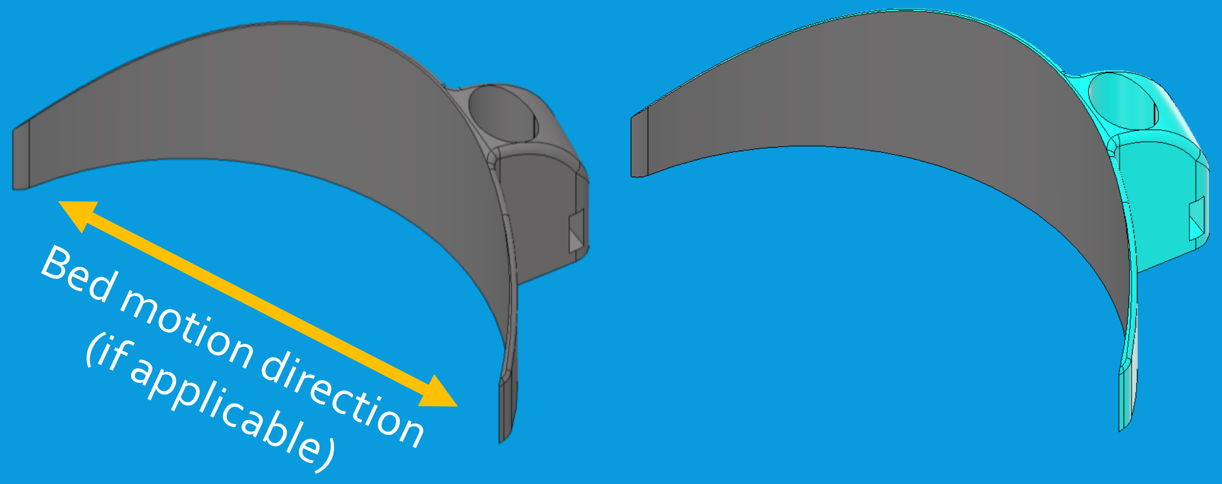
- **Secondary mirror holder:** print this part with thick walls as well. It is subject to a lot of stresses



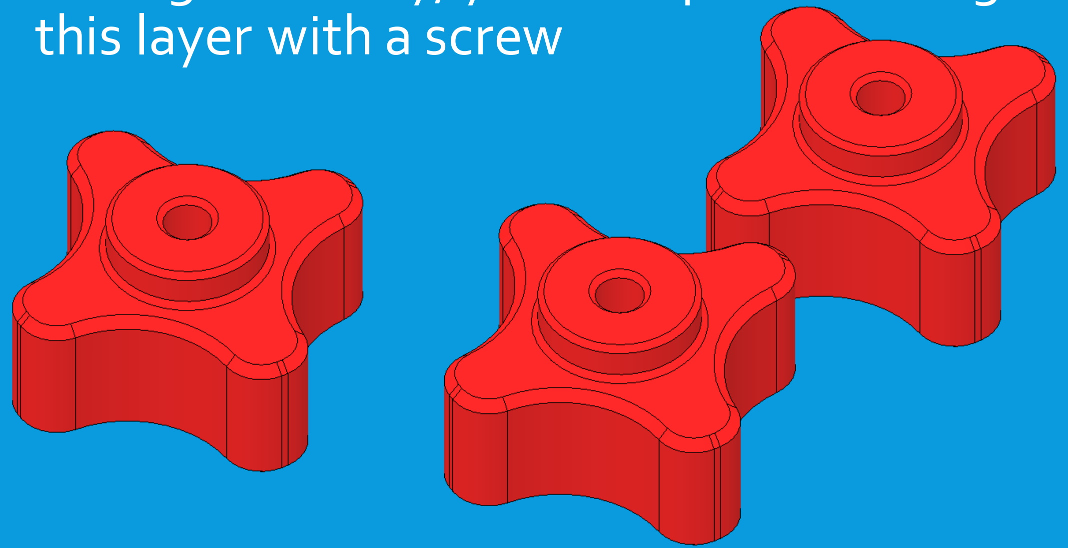
BLACKOUT: These parts should all be printed in black filament! Glossy is acceptable, matte is slightly preferable

2.3.4: PRINTED PART LIST

- **Baffle:** if you have an i3 ("bed-slinger") printer, be sure to orient this **Parallel** to the y-axis motion. Your bed adhesion needs to be good for this piece
- This part can be colored, but the inside face needs to be black

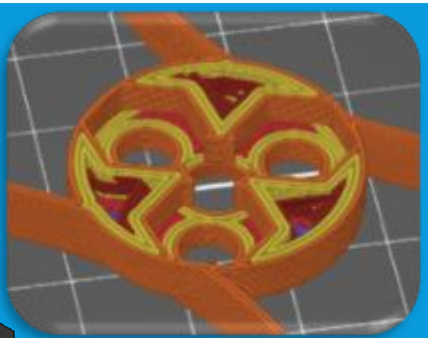


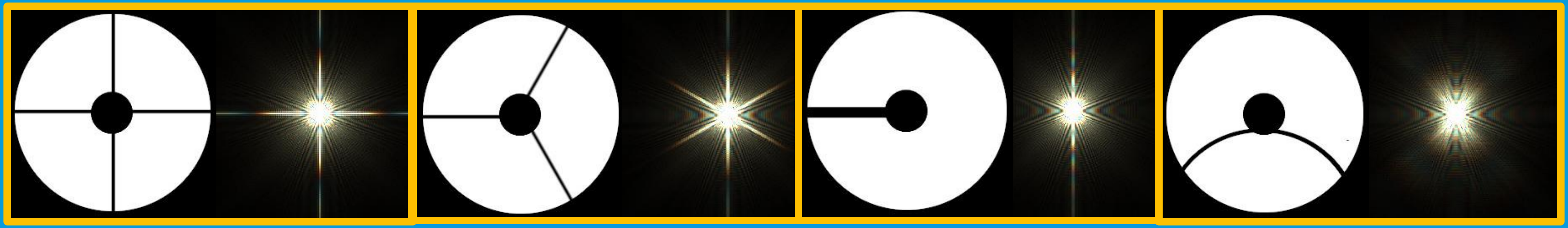
- **Knobs (print three of these), OR** you may purchase #10-24 wing nuts, thumb nuts or through-knobs instead
- These prints have a "sacrificial" layer for clean bridging above a hexagonal cavity
- During assembly, you will "punch through" this layer with a screw



2.3.5: PRINTED SPIDER

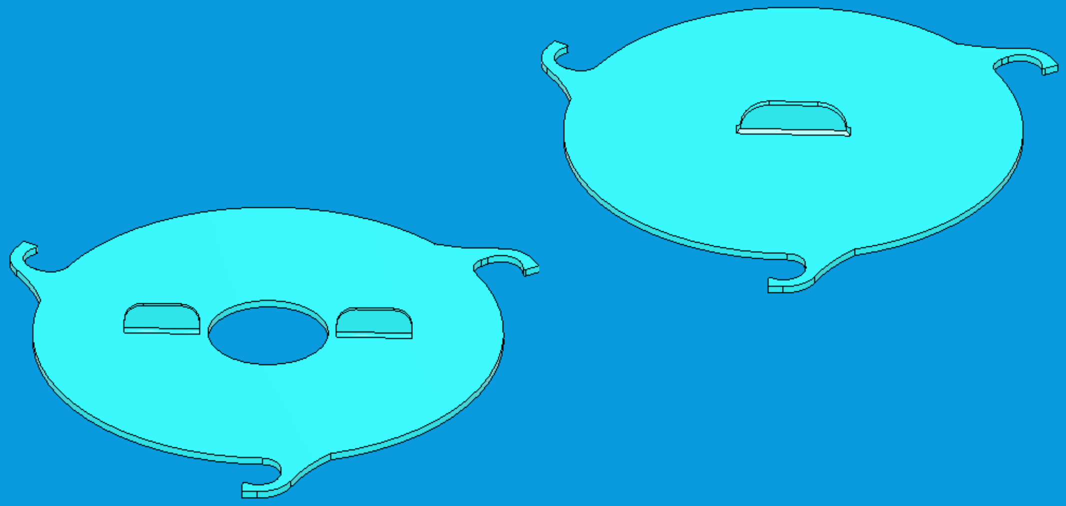
- **Spider** (PICK ONE) There are several to choose from. The spider selected changes how stars will appear (see below). This effect is minor for planets and the moon, and spiders can be interchanged on Hadley 114
- This guide will use the single-curved shown at right
- Unless you choose a slot-in remix, you need to add a "pause print" in your slicer application, at the top of the hexagonal cavities
- After pause, insert nut and ensure it is flush with the layer. Resume print and monitor for adhesion for next several layers

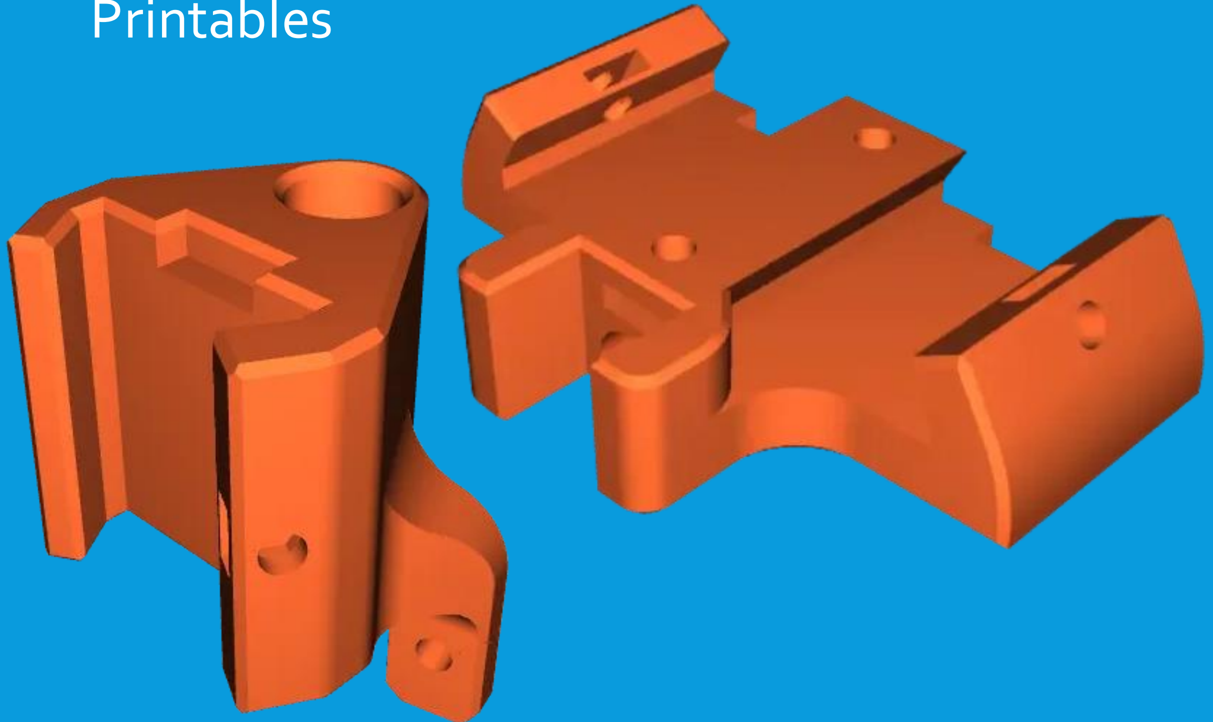




2.4: OPTIONAL ADDITIONAL PARTS

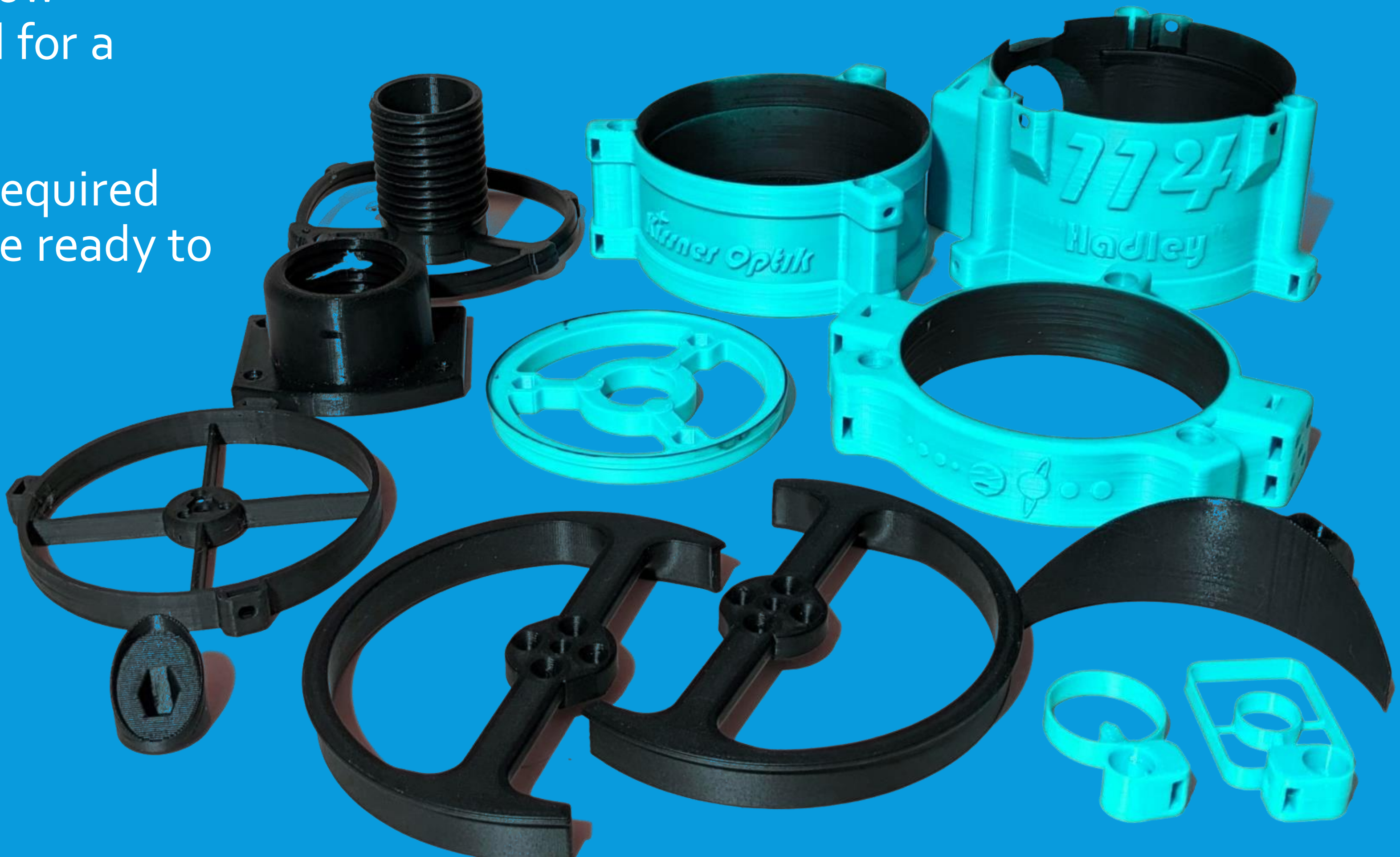
- **Mirror covers** — Covers useful to protect mirrors from dust and damage in storage, available in Hadley Remixes on Printables. Solid is for primary, holed is for over spider end of scope
- **Single or Double Dovetail** — These are standardized mounting brackets to add accessories like finder-scopes and red dot sights available in Hadley Remixes on Printables





2.5: PRINT COMPLETE

- Congratulations! You have now printed all the parts required for a Hadley 114 telescope.
- A little more preparation is required for light control, then you are ready to begin assembly.

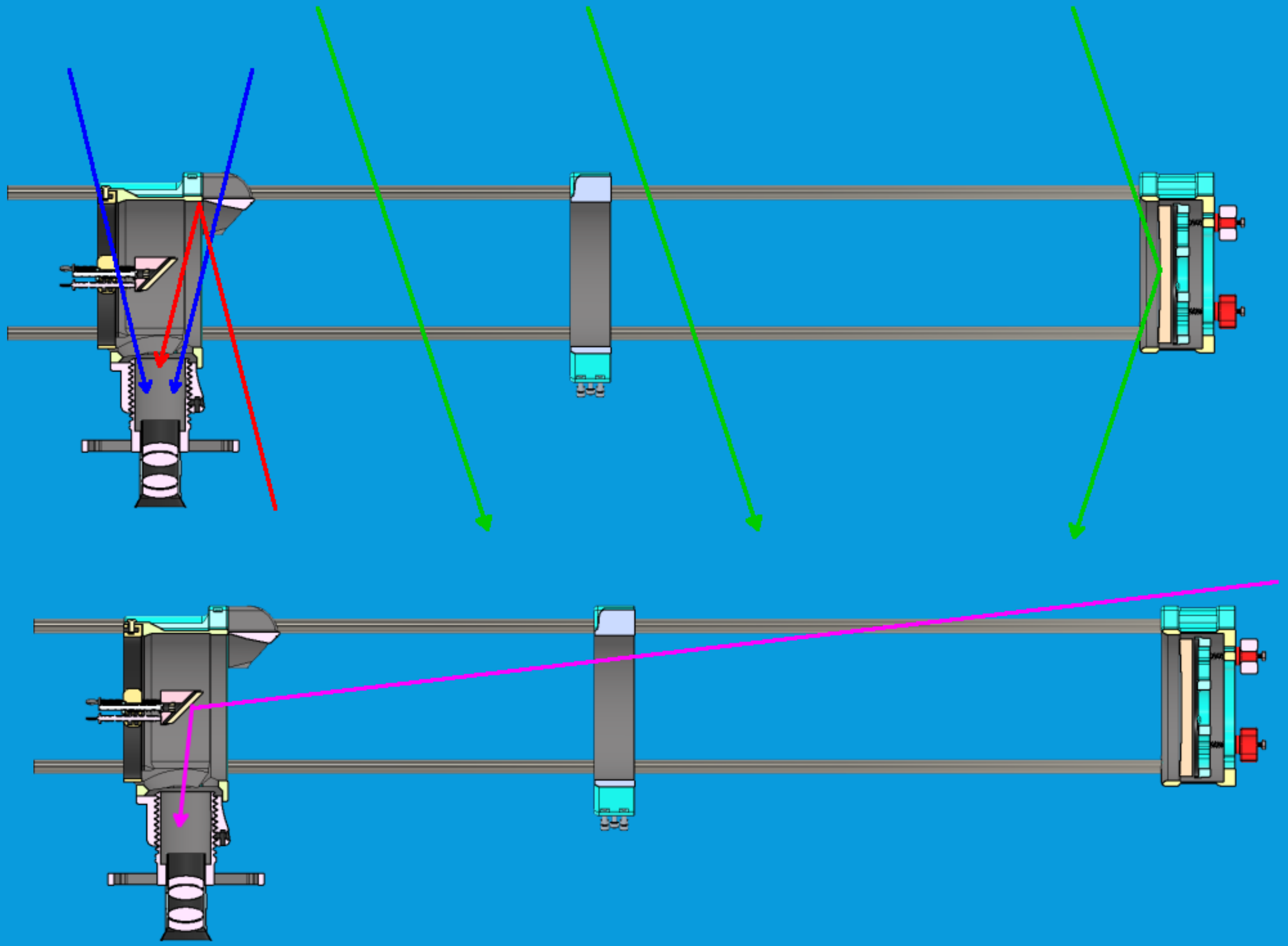


PART 3: PREPARING TO BUILD

3.1: PREPARING TO BUILD

- Before beginning assembly, a few steps are recommended to streamline the assembly process and to provide optimal results
- First, use one of the purchased rods, and validate all printed holes of size to take the rods (three each in LTA housing, UTA housing, and MTA ring) will fit the rods smoothly. Adjust any that do not
- Second is a process called "blackening," involving coloring certain parts partially black if not already printed in black.
- The reasoning is explained on the next slide, but in short, blackening improves telescope function in conditions with ambient light
- This can be left for later, but is easiest to do before beginning assembly, as it avoids excessive disassembly later

3.2: OPEN TRUSS TELESCOPES AND LIGHT



Hadley is an open-truss telescope, lacking a full tube to save on prints & simplify assembly. This works as most light (green at left) passes through the tube without reaching the eyepiece.

Any extra light entering the eyepiece (red, blue lines) worsens contrast in your view. Direct paths (blue) are blocked by design of UTA & baffle.

Stray light reflecting off the interior like the red path in the top diagram can be an issue if observing with lots of ambient light (streetlights, a neighbor's porch light, etc). Blackened interiors help – see next slide.

Pink path is not prevented, though minimal, but can be minimized by using a darker observing site and maximizing contrast between ambient light and the target. An additional baffle design may be added in modifications.

3.3: WHICH PARTS TO BLACKEN

These parts are best printed in black (matte is preferred, though glossy is acceptable) or fully blackened on all surfaces:

- Focuser drawtube (interior)
- Secondary holder
- Spider of choice
- Light baffle

These parts may be printed in any color, but should be blackened as shown on the next slide for best results:

- Upper Tube Assembly (UTA) Housing
- Lower Tube Assembly (LTA) Housing
- Middle Tube Assembly (MTA) Ring
- Light baffle (if desired to match color of upper tube assembly exterior)
- Primary mirror holder

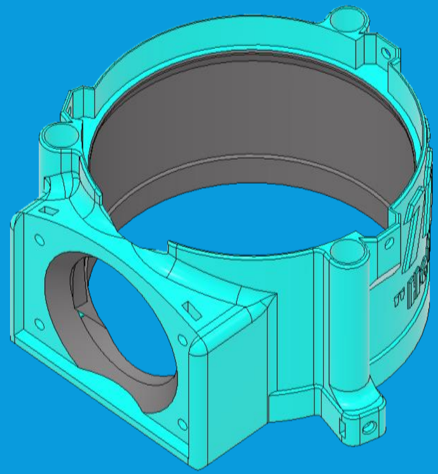
If the parts from the first list are printed black, Hadley will function if only some or even none of the second list are blackened, but the telescope will be more sensitive to stray light particularly in daytime observing. If required, blackening can be done as a later modification for all critical parts.

THIS IS A MODULAR TELESCOPE, YOU CAN ALWAYS FIX OR SWAP THINGS LATER

3.4: BLACKENING THE PARTS

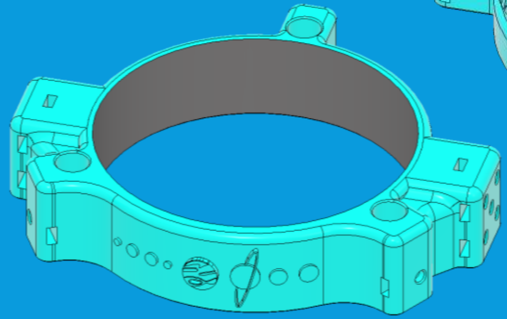
- The parts shown below should have the interior surfaces blackened with a brush and ink or paint, or masked & sprayed with black spray paint
- Printing in black (whether matte or glossy) is an acceptable substitute if you don't want a colored exterior
- **Single most important part to blacken is the UTA**

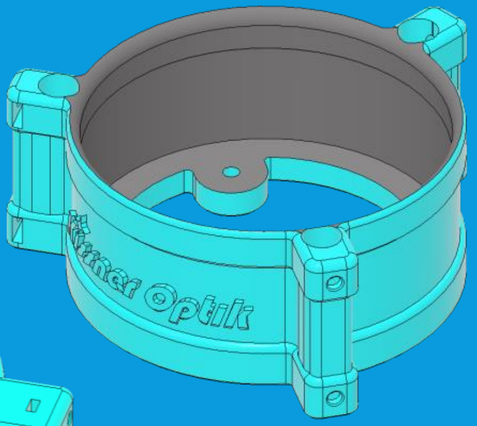
- For baffle, choice of printing in black, fully blackening all surfaces, or printing in color and blackening inside
- **Baffle is the second most critical part to blacken, even before MTA or LTA**



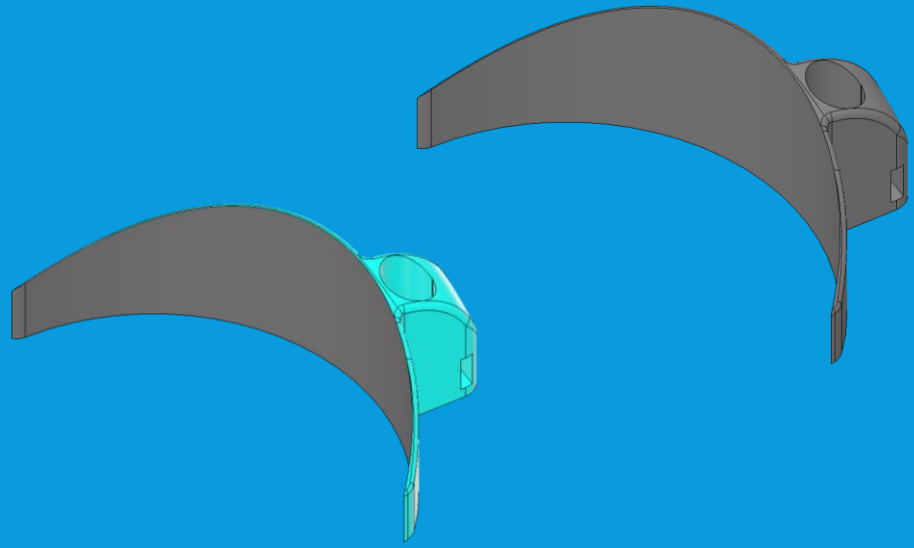
Upper Tube Assembly

Middle Tube Assembly





Lower Tube Assembly



3.5: PREPARATIONS COMPLETE

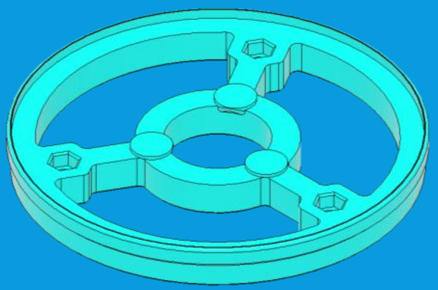
- Congratulations! With the parts listed on the previous slides, you should now have everything required to assemble the tube for a Hadley 114 telescope!
- The next section will begin walking through the assembly sequence
- Hadley 114 is a modular telescope, so can be assembled several possible sequences. This is not the only possible order, but it is designed to leave the mirrors (the most delicate parts) to last for safety.

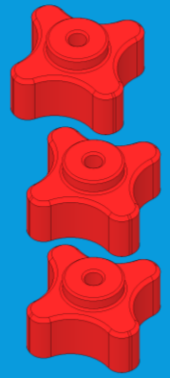
PART 4: LOWER TUBE ASSEMBLY

4.1: LOWER TUBE ASSEMBLY SEQUENCE

- Parts required for this are listed below

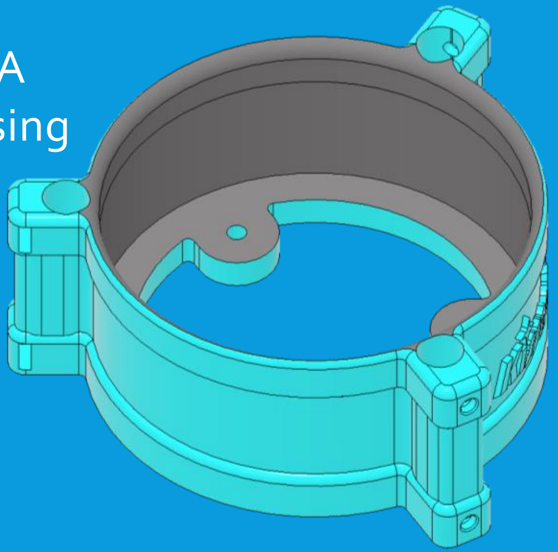
Primary Mirror Cell

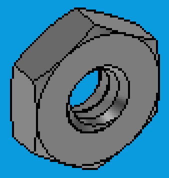




(3x) Collimation Knobs

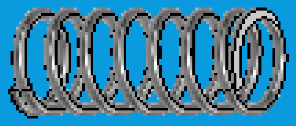
LTA Housing

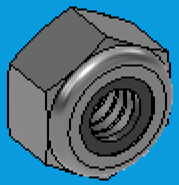




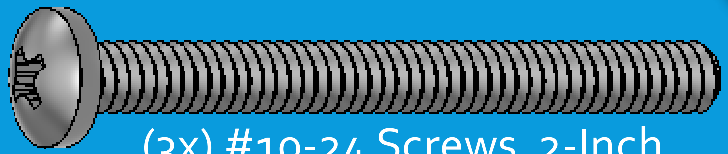
(3x) #10-24 Nuts

(3x) Springs





(3x) #10-24 Locknuts

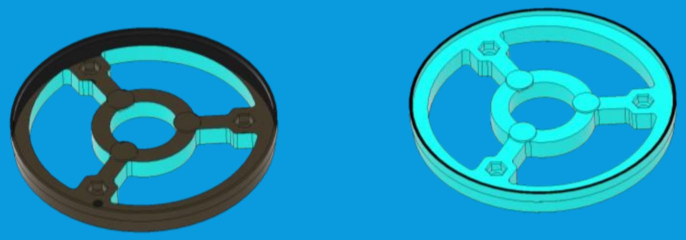


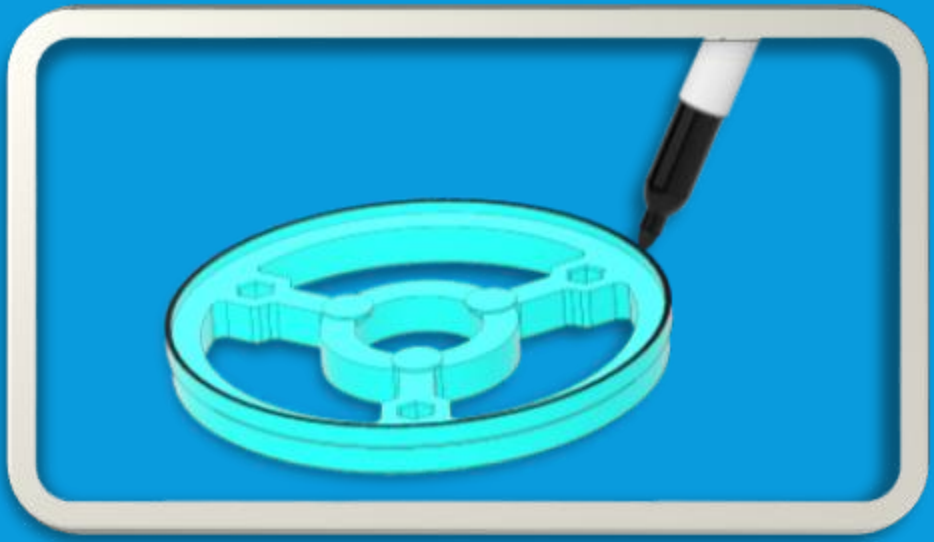
(3x) #10-24 Screws, 2-Inch

- When complete, this produces the "Lower Tube Assembly" which will support the primary mirror and attach it to the three main rods

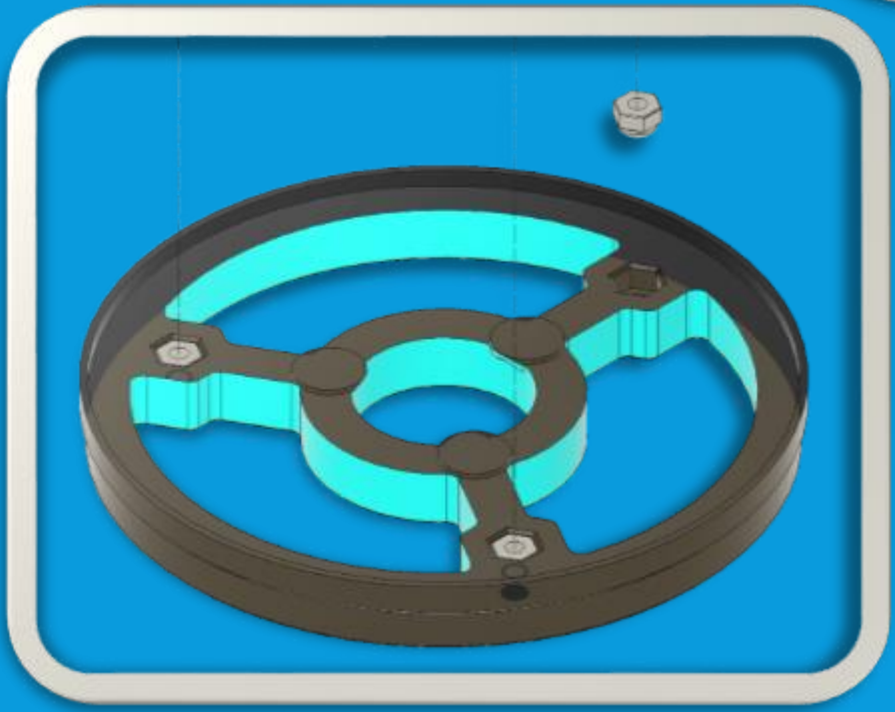
4.2: PREPARING PRIMARY MIRROR CELL

- **Blackening of the primary cell is not critical, the mirror covers most of this piece.**
- **Blackening the outer ring is sufficient**



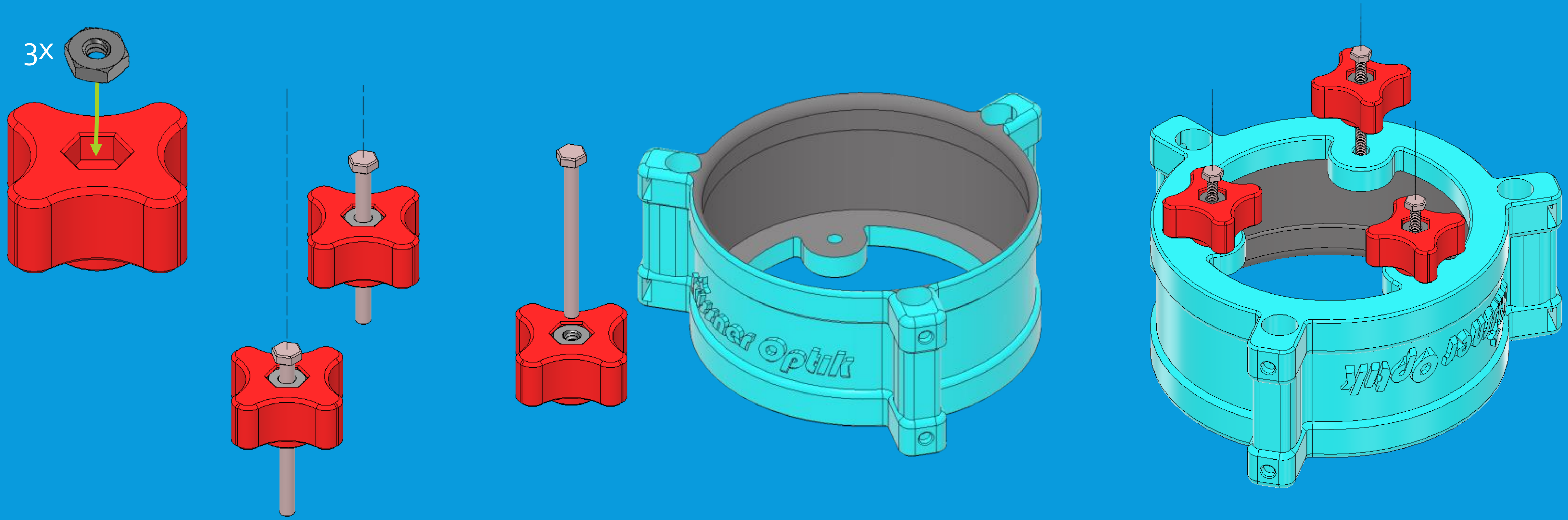


- Insert three nylon lock-nuts to the cell. (Note gasket orientation—rounded side with gasket *down*)
- If you don't have locknuts, see appendix.



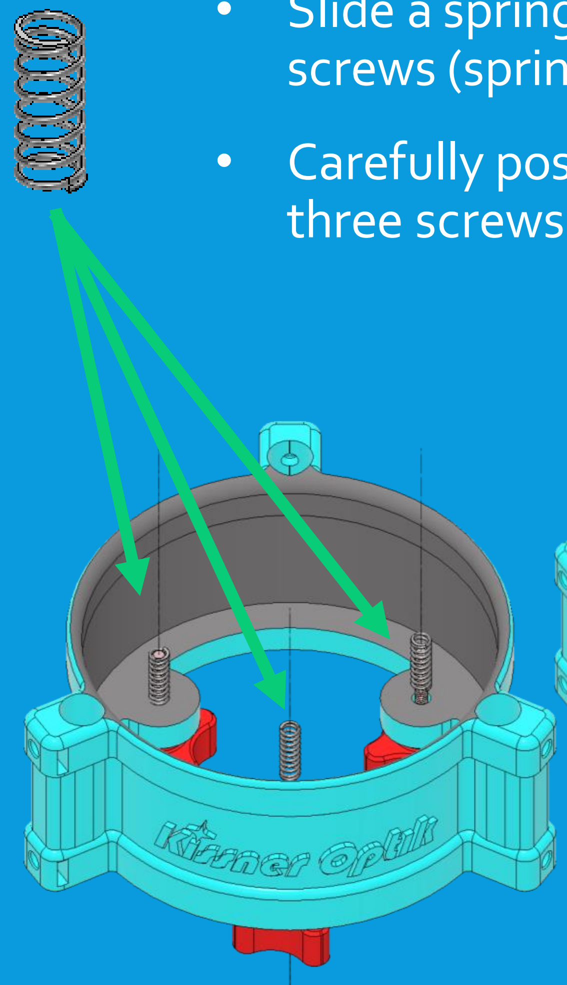
4.3: PREPARING THE PRIMARY KNOBS

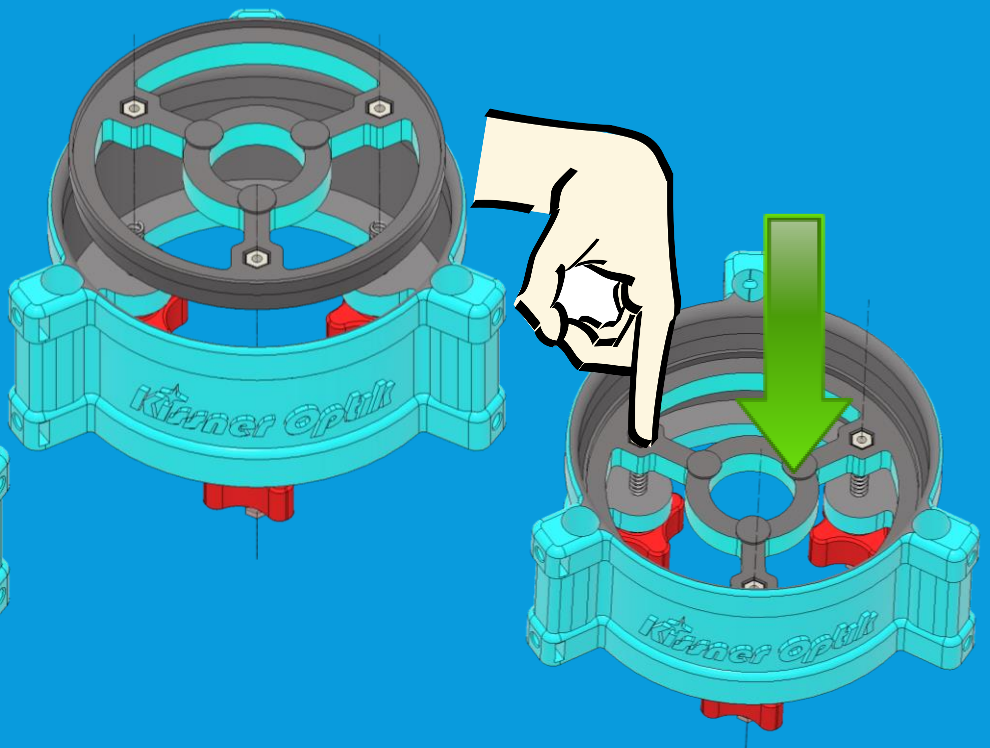
- Find the three knobs, insert one standard nut into the hexagonal pocket, then screw one of the long (2"+) machine screws through each knob
- Don't tighten the screws fully, leave space as shown.
- Find lower tube assembly
- Seat knobs
- They should fit freely

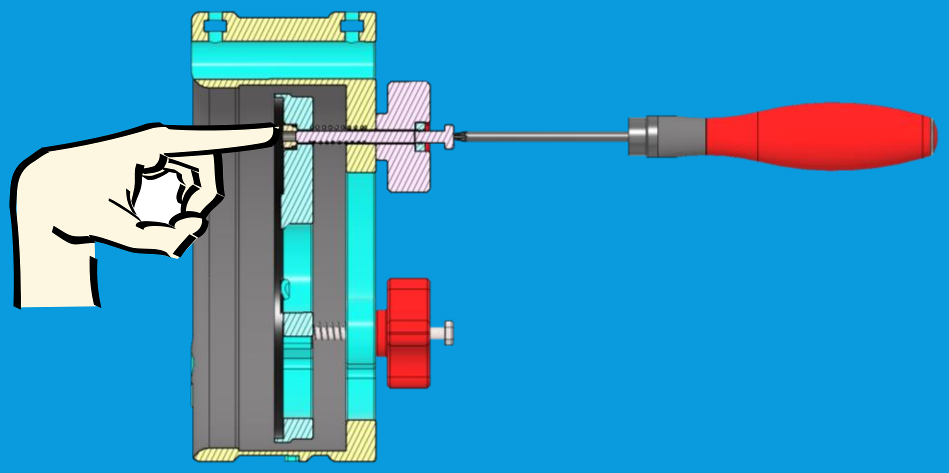


4.4: ASSEMBLE PRIMARY COLLIMATION CELL

- Slide a spring over the three loose screws (spring should fit freely)
- Carefully position mirror cell over the three screws
- Push the cell down over the springs, and tighten the screws into the nyloc nuts using a matching screwdriver or allen key
- Just a few turns should seat each screw into the nut

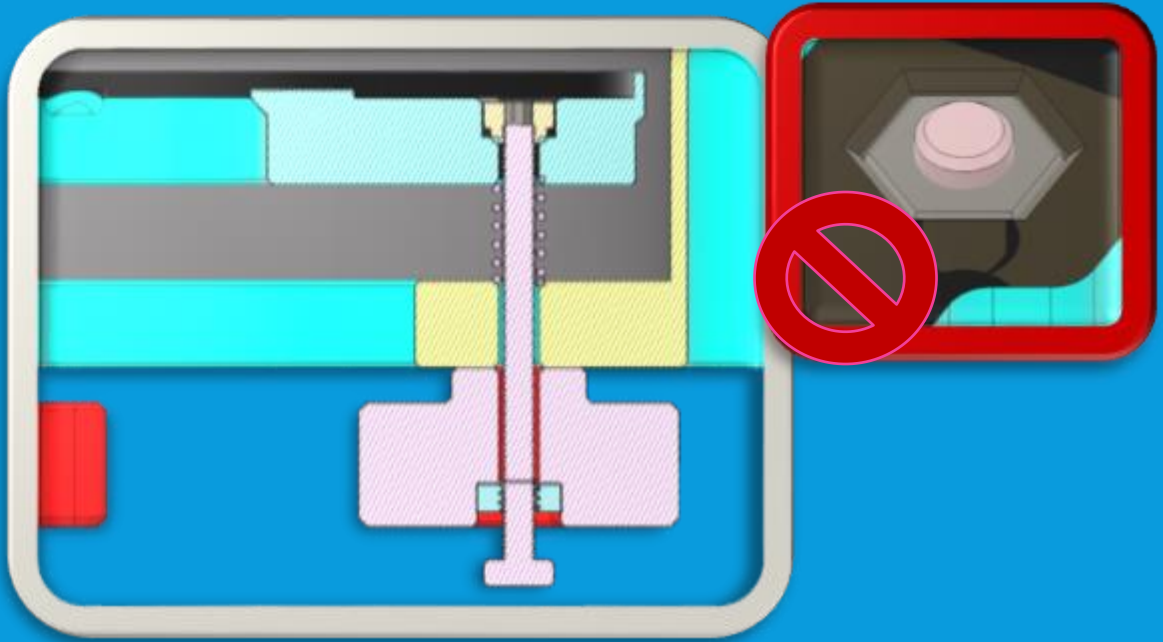




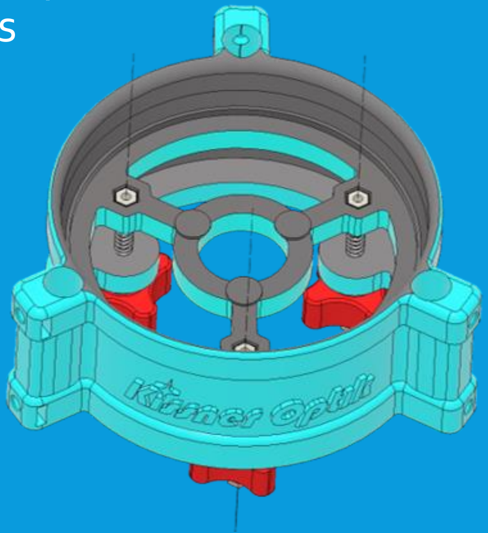


4.5: FINISH PRIMARY COLLIMATION CELL

- Once all three screws are seated, tighten each bolt further
- Ensure ends of bolt remain below the level of the end of the nut—it is acceptable if the bolt is several threads below the end of the nut



- Test the knobs turn freely, and clear the holes in the LTA for the rods
- Note adjusting the knobs compresses the screws, adjusting mirror position in/out and angle
- Precise setting of the knobs is not critical at this time, and will be handled in collimation
- Lower Tube Assembly is now complete, and can be set aside until final assembly begins

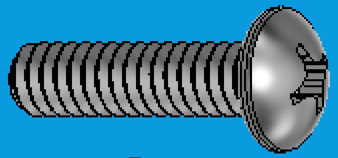


PART 5: MIDDLE TUBE ASSEMBLY

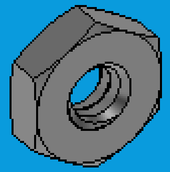
5.1: MIDDLE TUBE ASSEMBLY START

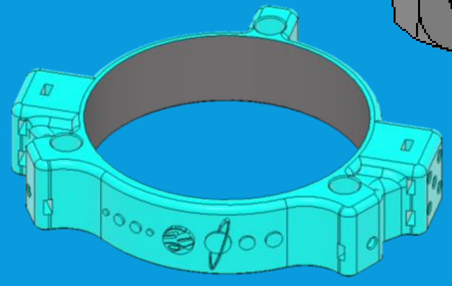
- Middle Tube Assembly, also known as the "mid-ring" or the "altitude bearing," provides a mounting spot for the two bearing wheels which allow it to turn on a mount
- Uses components listed below

(8x) #10-24 Screw ¾" Long



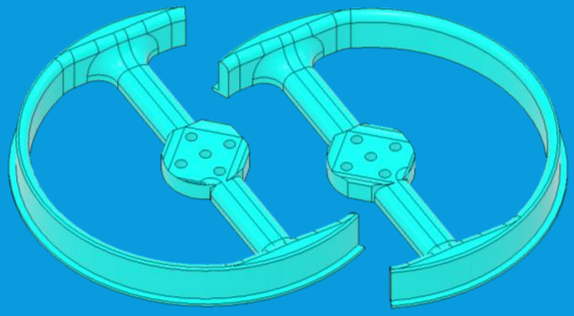
(8x) #10-24 Nut



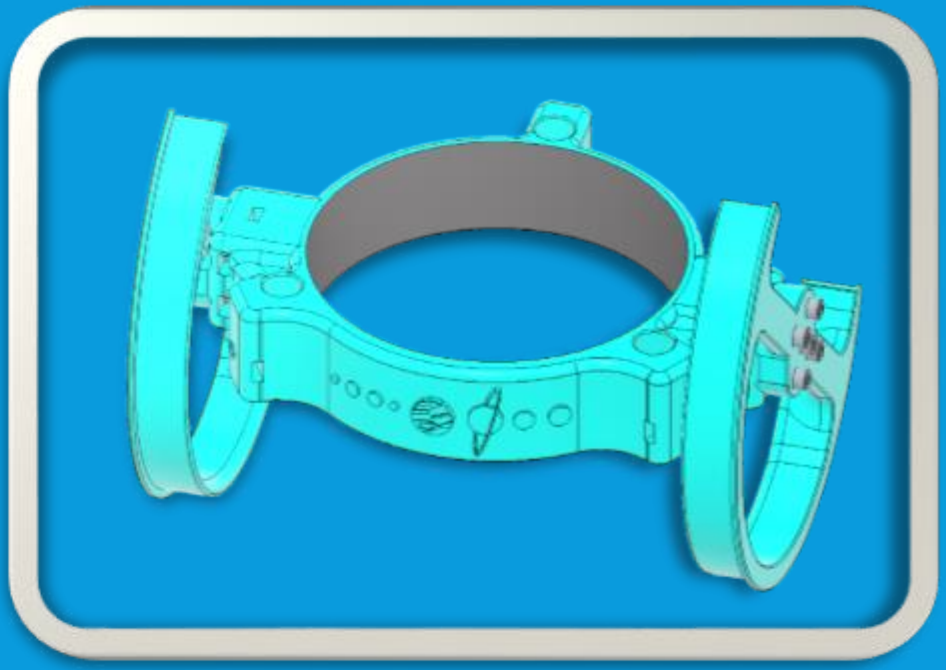


Middle Tube Assembly Ring

Left & Right Crescent Bearing

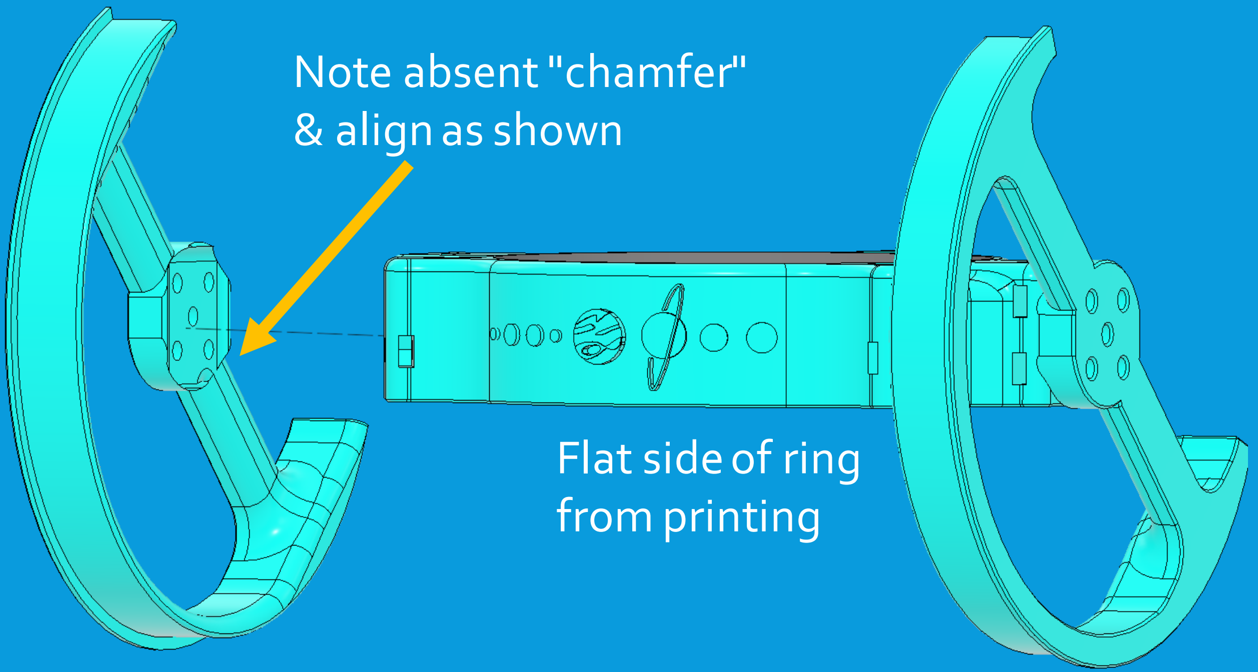


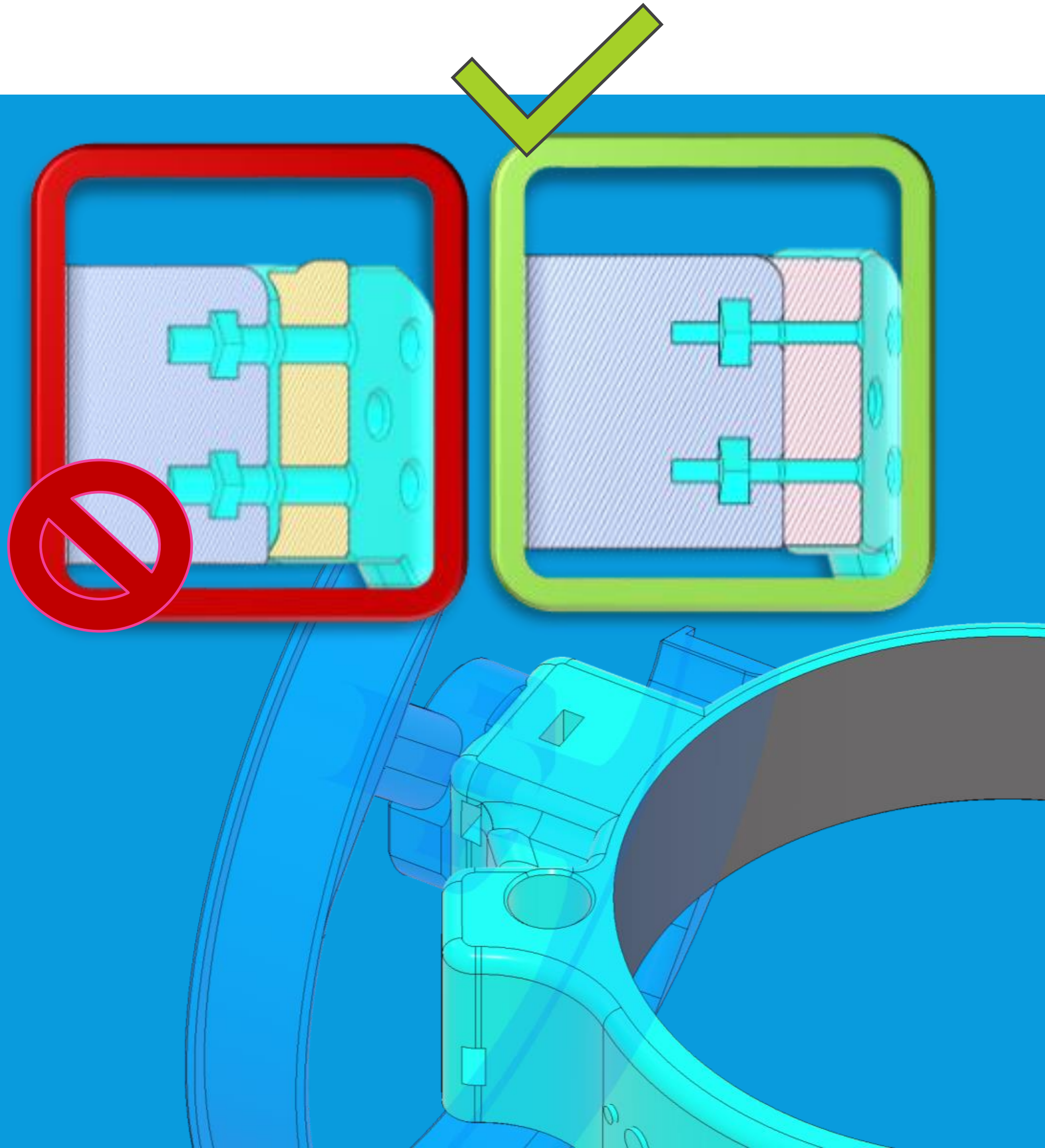
- If a tripod mount is being used as an interim solution, consult appendix for discussion of how to supplement or replace the MTA with a tripod mount
- Completed Middle Tube Assembly:



5.2: MIDDLE TUBE ASSEMBLY

- Test fit crescent bearings to the sides of the Middle Tube Assembly ring
- Note the curved surfaces, make sure the fit is perfect and the bearings are in the right orientation. Bearings should be flush

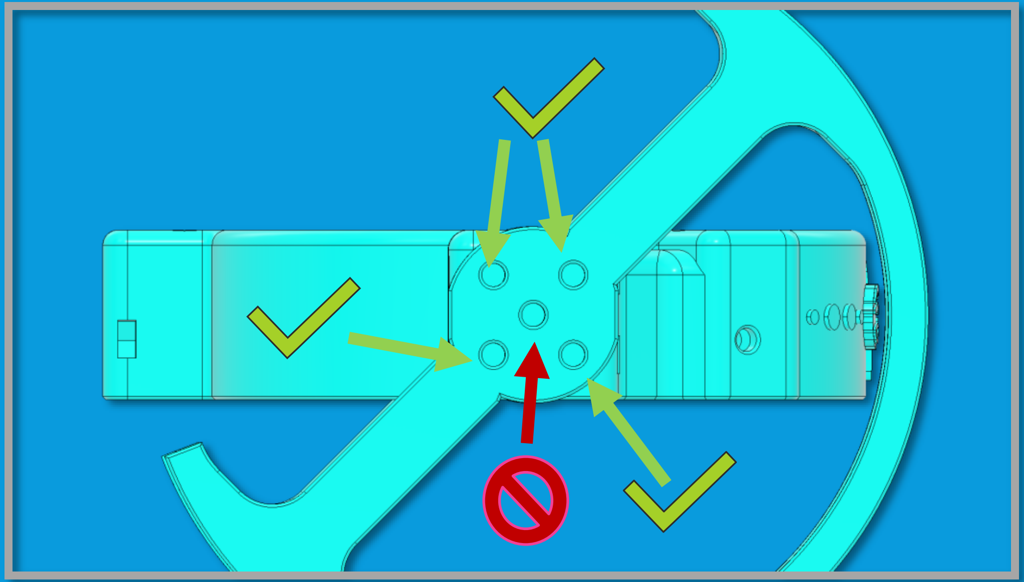


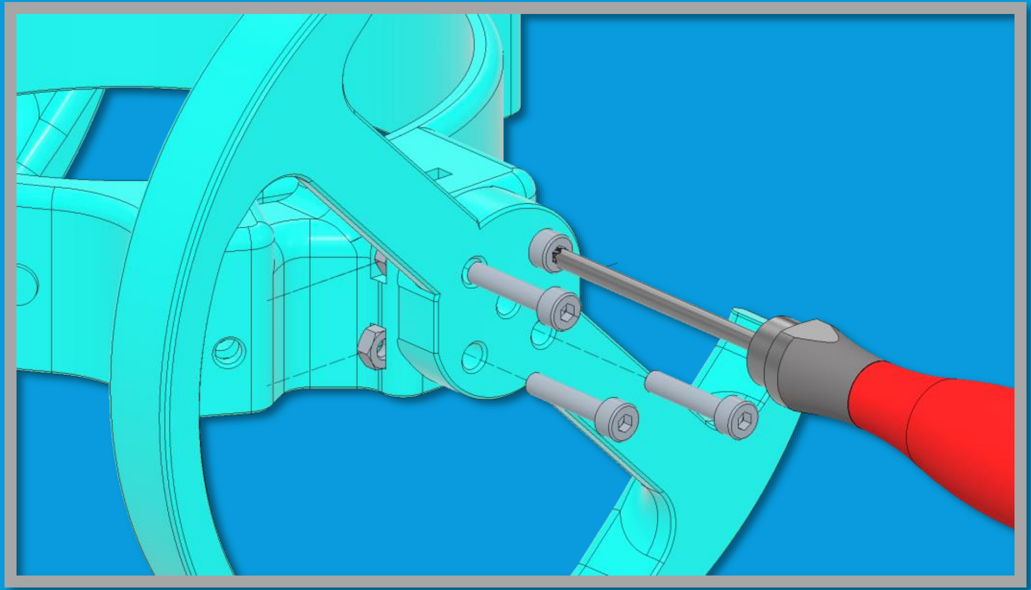


5.3: MIDDLE TUBE ASSEMBLY

- Insert nuts into each of the outer four nut pockets on the two bearing mounting pads (8 total)
- Center of pattern is for attaching accessories if desired (see appendix)

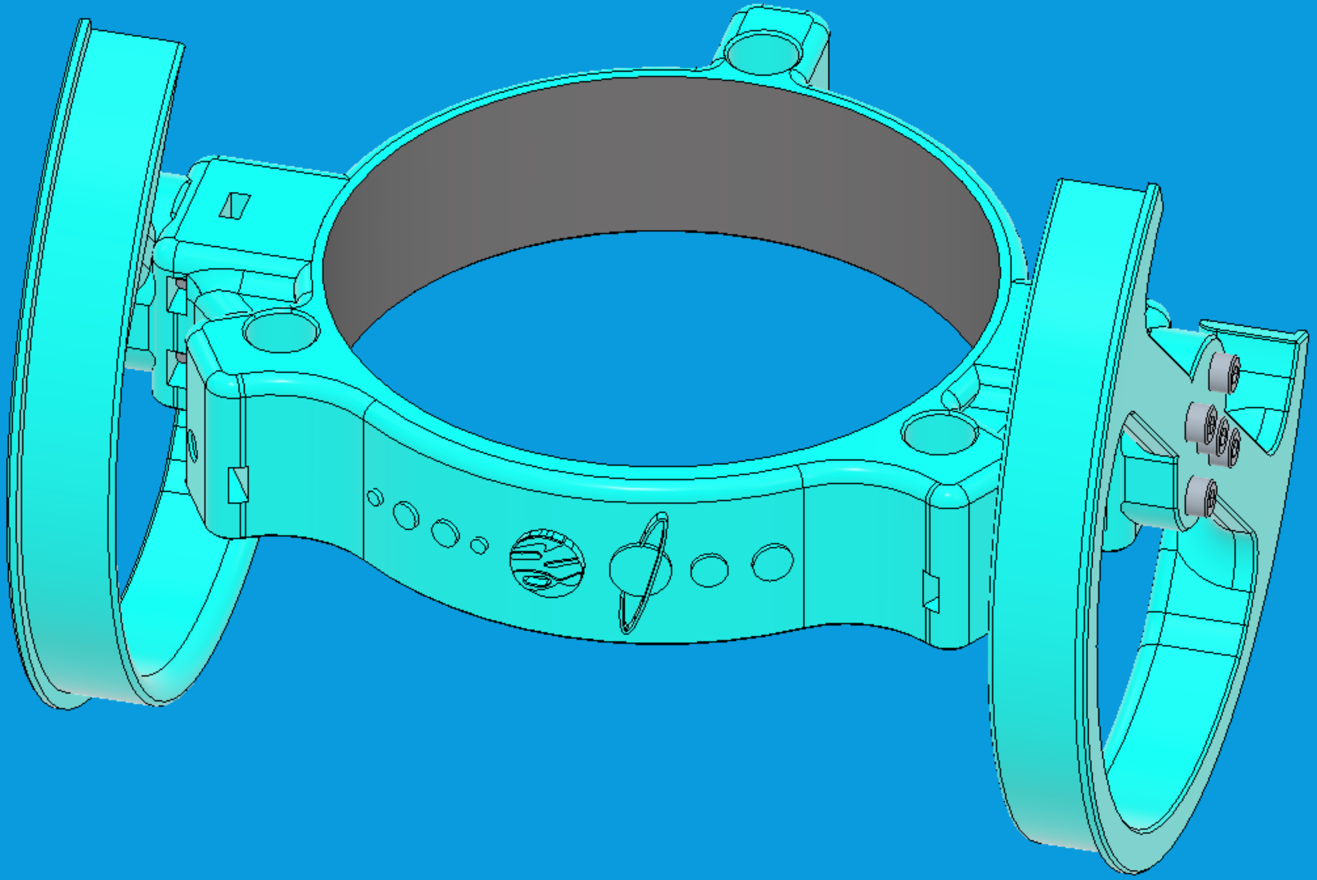
- Insert medium-length (¾"+) screw to matching holes through bearing (note orientation)
- Tighten all four on each bearing first loosely to engage nuts & hold bearing in place, then tighten fully until bearings are attached firmly





5.4: MIDDLE TUBE ASSEMBLY COMPLETE

- Middle Tube Assembly is now complete and can be set aside until final assembly



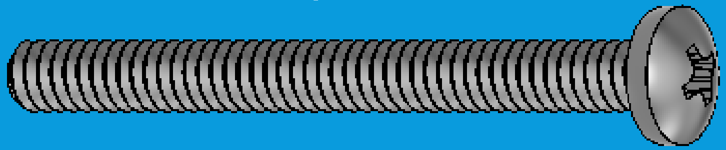
PART 6: SECONDARY MIRROR CELL

6.1: SECONDARY MIRROR CELL

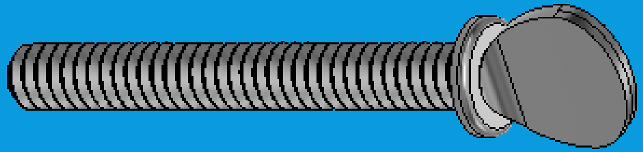
- The secondary mirror cell consists of:

1. Secondary mirror holder—supports the secondary mirror at an angle
2. Spider—holds the collimation assembly and mirror in the middle of the telescope
3. Collimation assembly—bolts used for controlling the position of the secondary holder

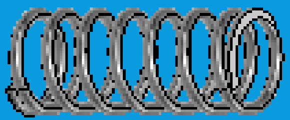
- Secondary Mirror Cell Parts List:



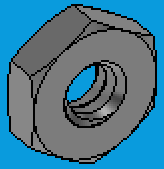
(1x) 2" Long #10-24 Screw



(3x) #10-24 Screw, At Least Length 1 ¼" (Ideally Thumbscrew)



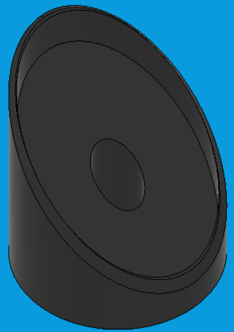
(1x) Spring



(1x) #10-24 Nut

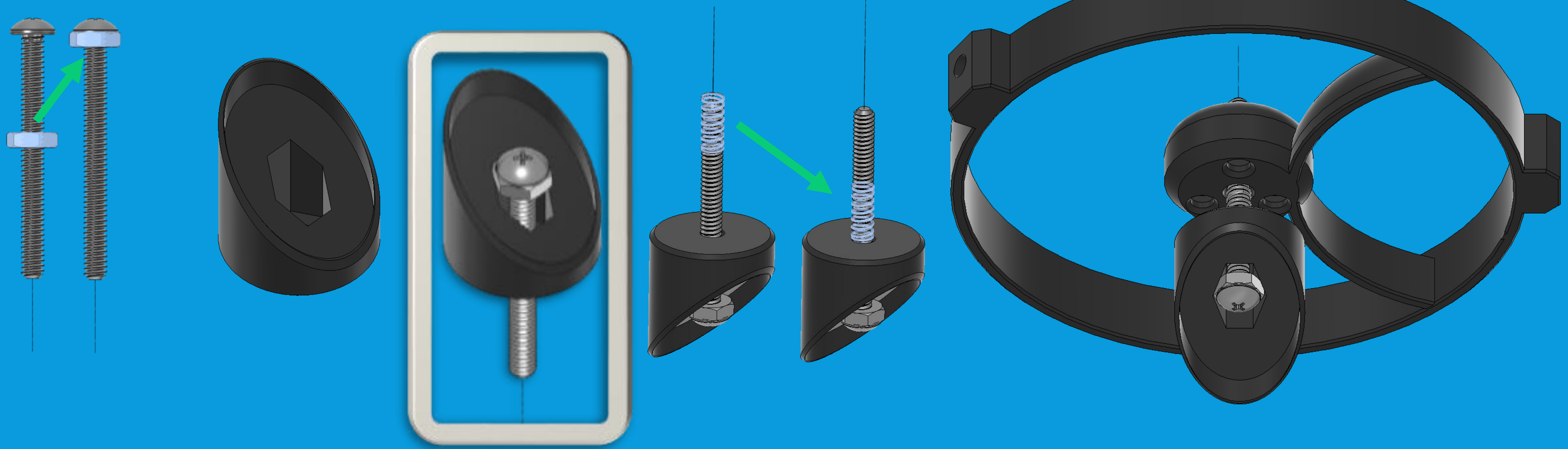
Selected Spider

Secondary Mirror Holder



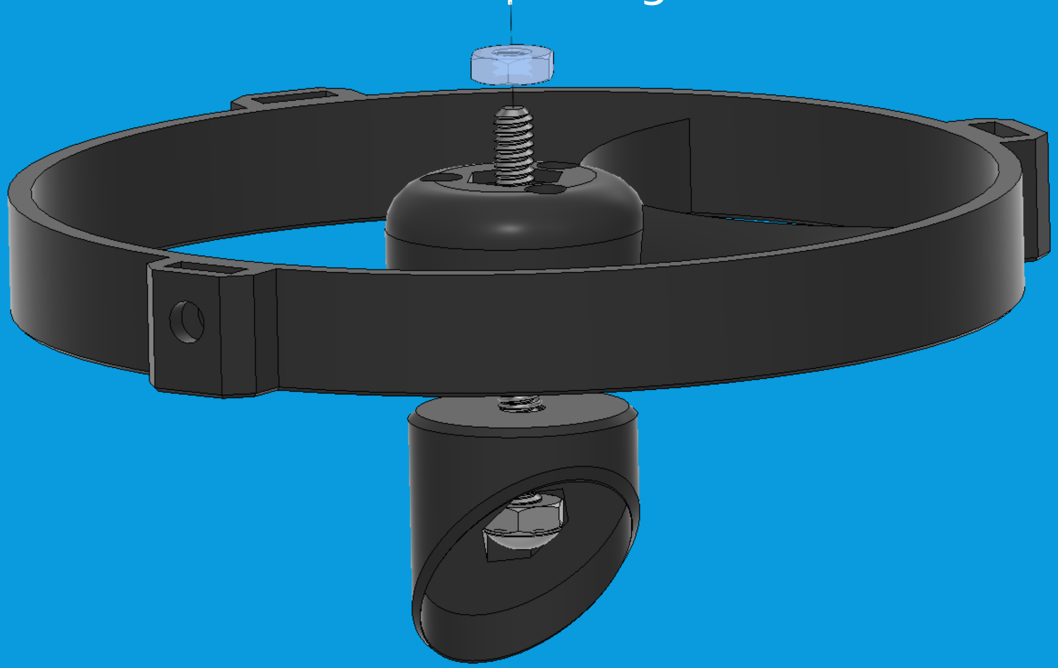
6.2: SECONDARY COLLIMATION CELL

- Take 2" screw and one standard #10-24 nut.
- Thread nut over screw until nut is jammed against head of screw.
- Insert screw and nut through secondary holder.
- Drop spring over the screw.
- Find the spider and place center hole of spider over end of screw, ensuring spring remains between spider and secondary holder



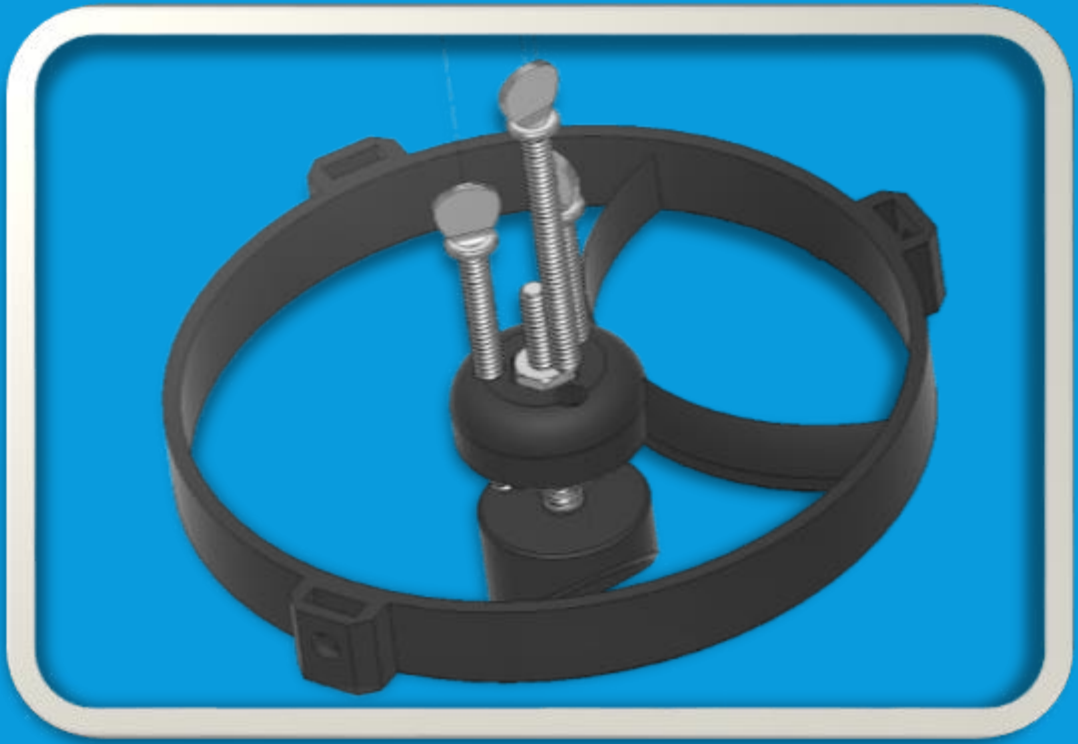
6.3: SECONDARY COLLIMATION CELL

- Twist a nut over the exposed screw
- Ensure it sits into the hexagonal cavity in the spider. Cavity is larger than nut, and nut should be free to wiggle without spinning
- Pushing on screw or secondary holder should allow slack to tighten nut and adjust position of secondary.
- Adjust nut enough to tension spring, fine adjustment of position will be carried out later.

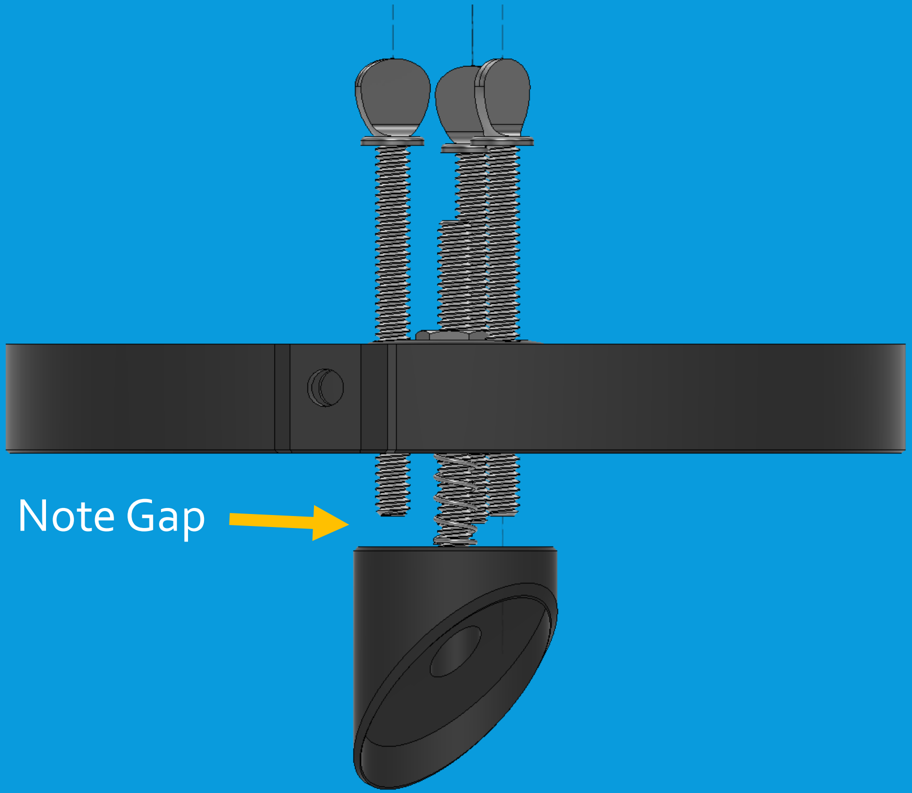


6.4: SECONDARY COLLIMATION CELL

- Add three secondary collimation bolts.
- Long thumbscrews or socket head screws are ideal, but any long enough screw works.

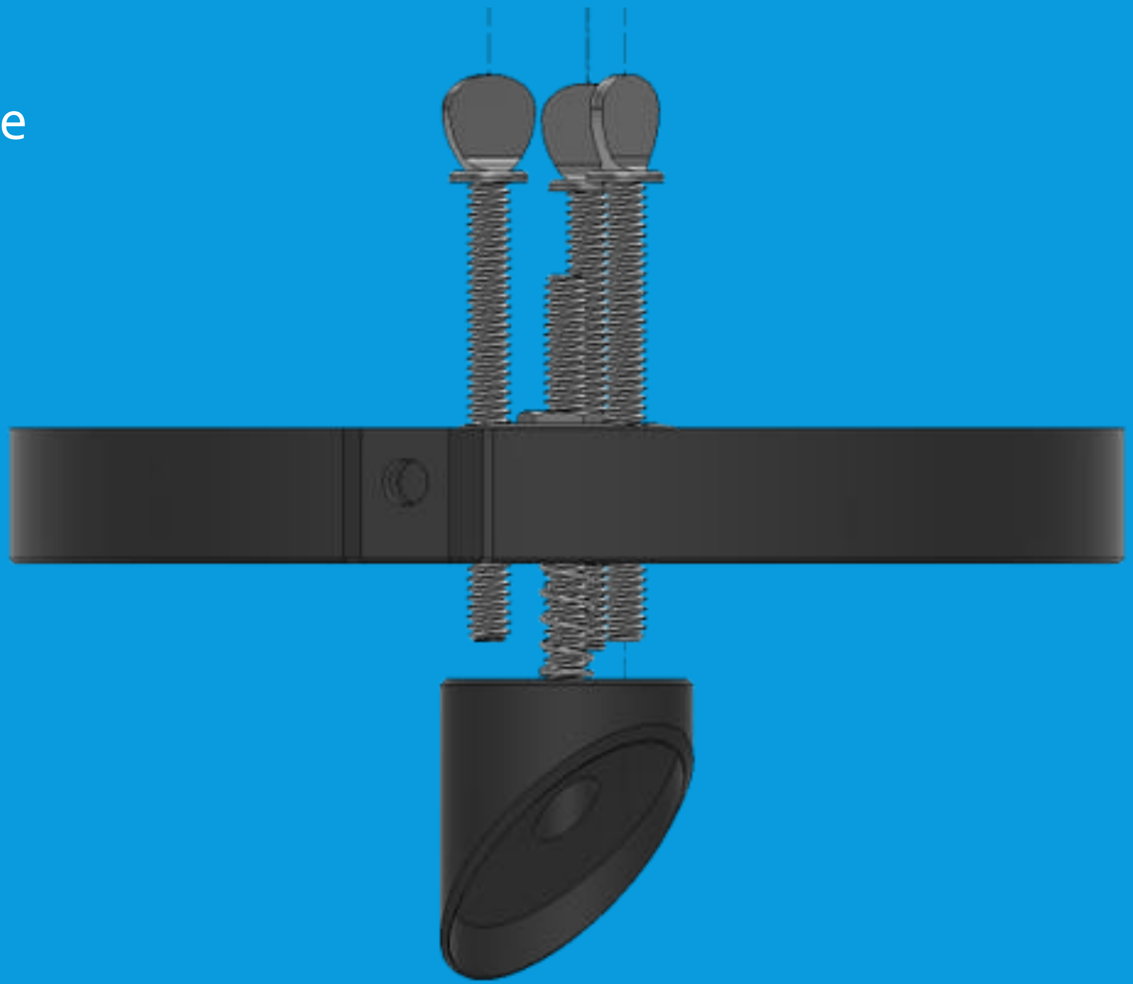


- Screw partway through the spider
- Leave a gap between screws and the secondary mirror holder for now



6.5: SECONDARY MIRROR CELL COMPLETE

- Assembly of secondary mirror cell is now complete, and it can be set aside until final assembly
- Gluing of mirrors and collimation process will be discussed after major assembly is complete



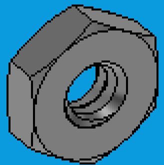
PART 7: UPPER TUBE ASSEMBLY

7.1: UPPER TUBE ASSEMBLY

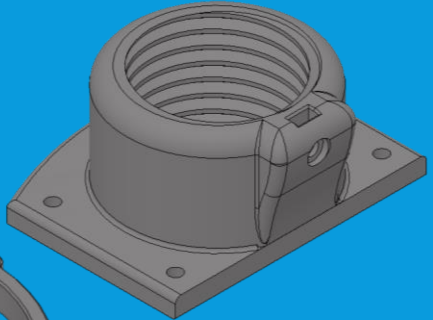
- Upper tube assembly attaches the spider to the rods, and contains the focuser, which is where eyepieces will be mounted and allows adjustment of the focus of telescope

- This will show the basic printed helical focuser. Some additional focuser options are discussed in appendix which require slightly different assembly

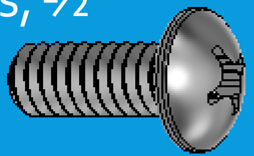
(6x) #10-24 Nuts



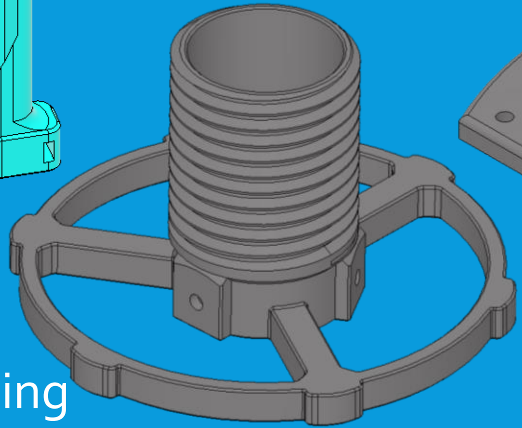
Focuser Base



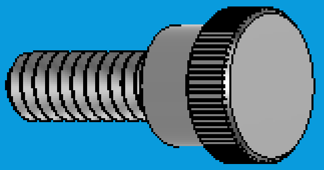
(4x) #10-24 Screws, ½"

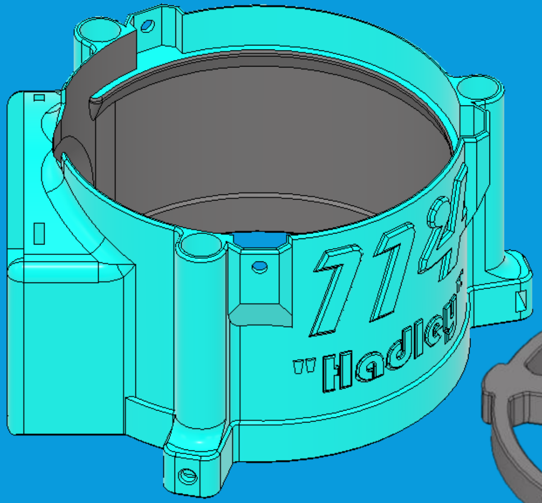


Focuser Tube



(2x) #10-24 Thumbscrews

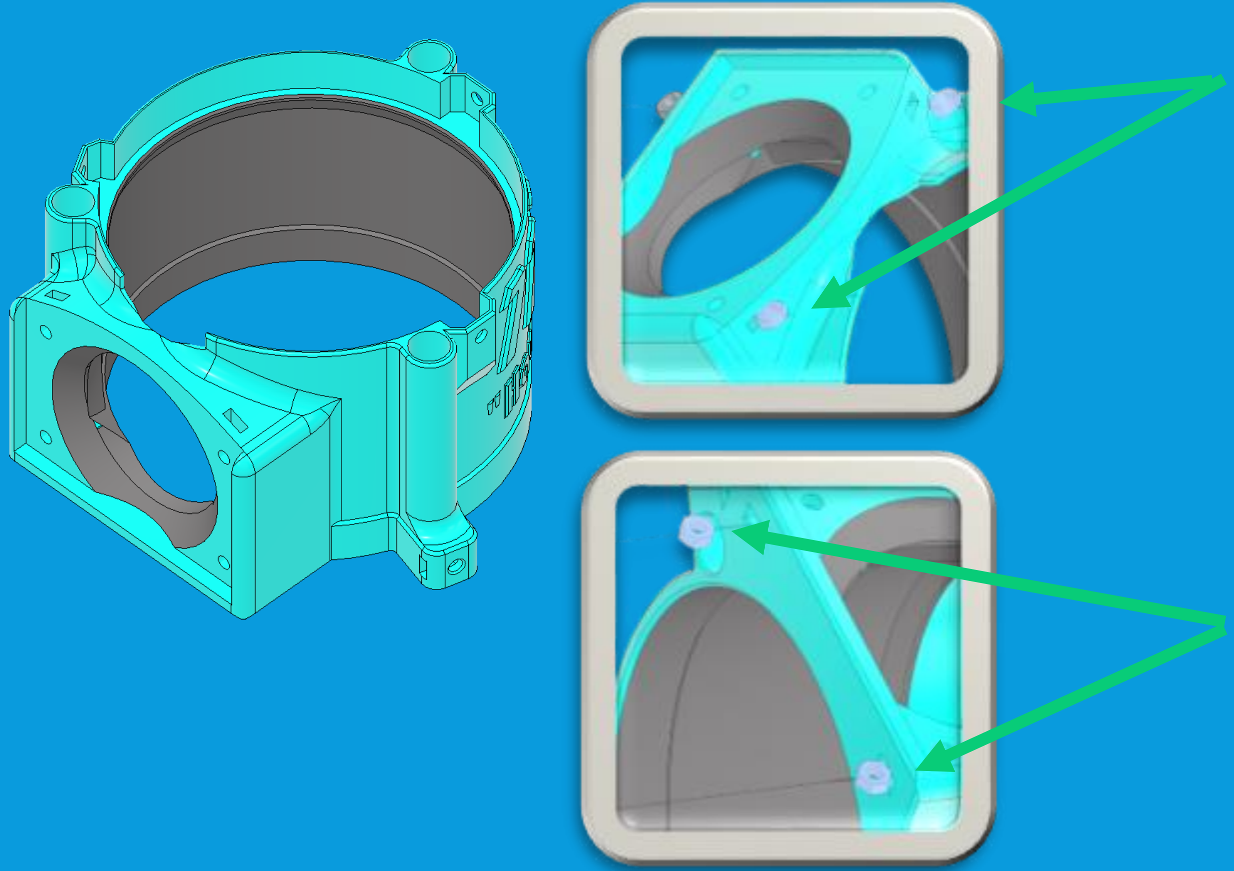




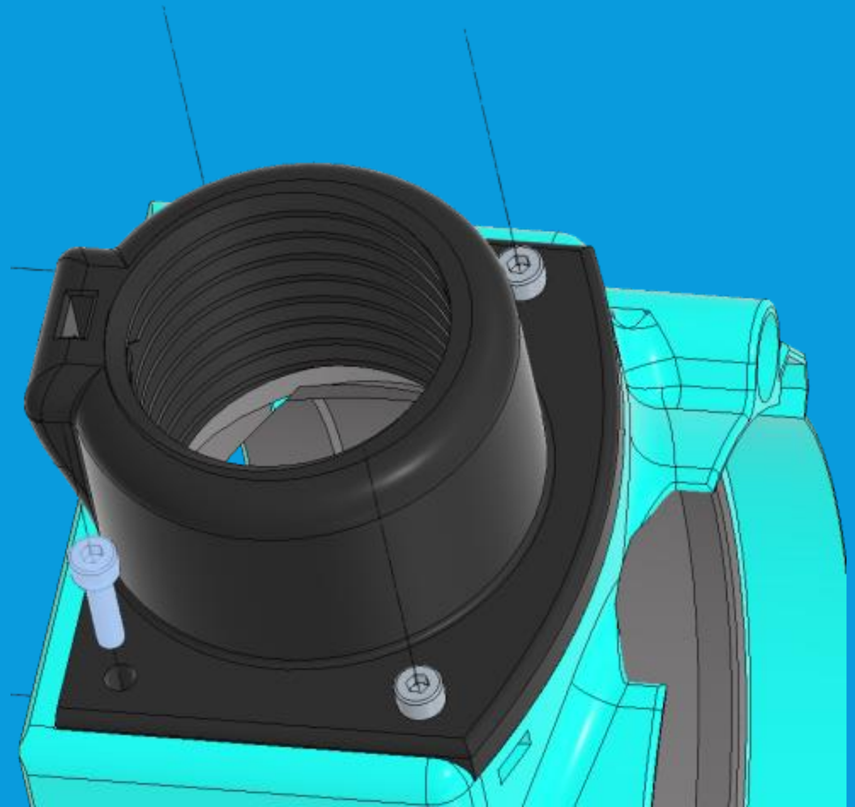
Upper Tube Assembly Housing

7.2: UPPER TUBE ASSEMBLY

- Locate Upper Tube Assembly
- Insert 4 nuts into the four pockets shown

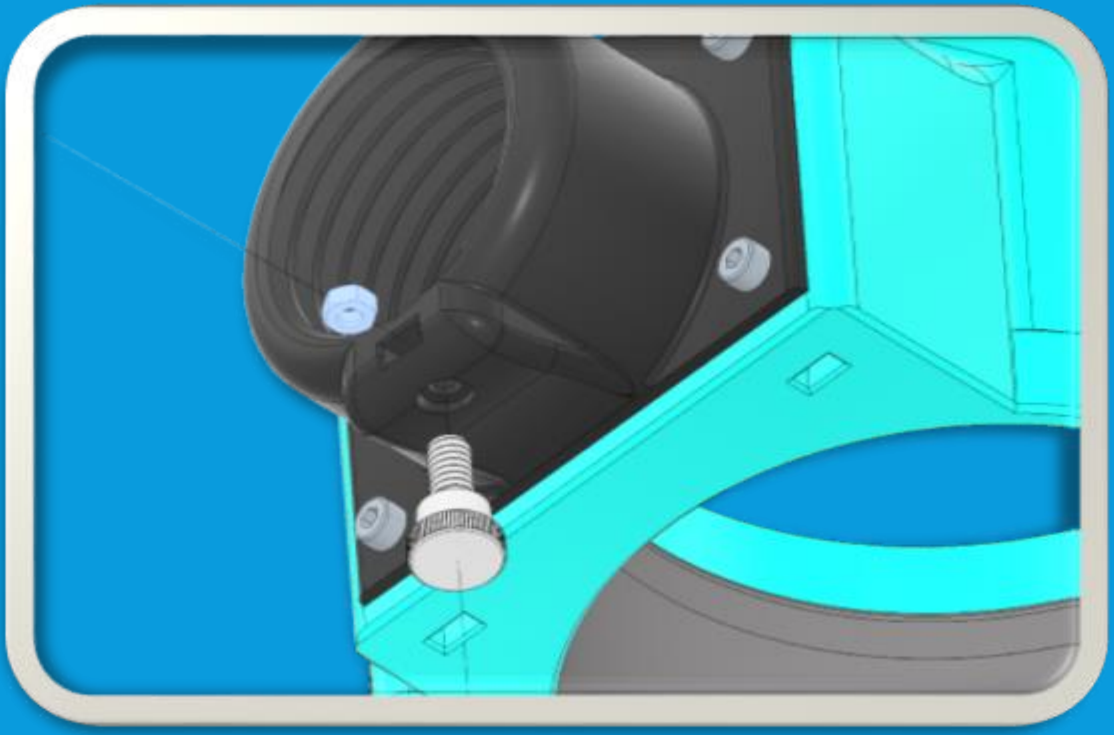


- Locate focuser base
- Assemble to Upper Tube Assembly and attach with four screws (medium length, ¾") through nuts

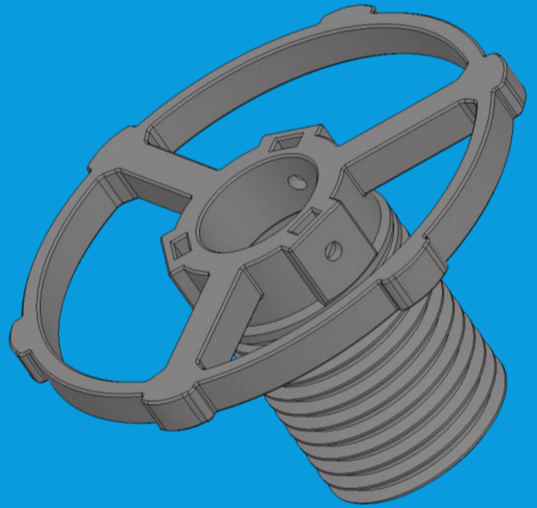


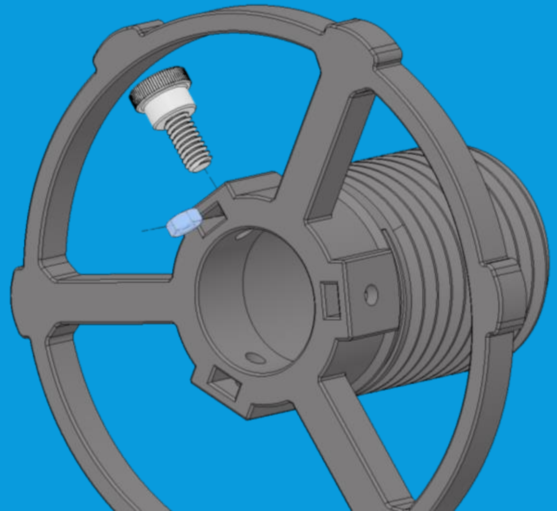
7.3: FOCUSER ASSEMBLY

- Insert a nut into pocket on focuser base, then insert ½" screw (ideally nylon thumb screw) into nut from outside base as shown



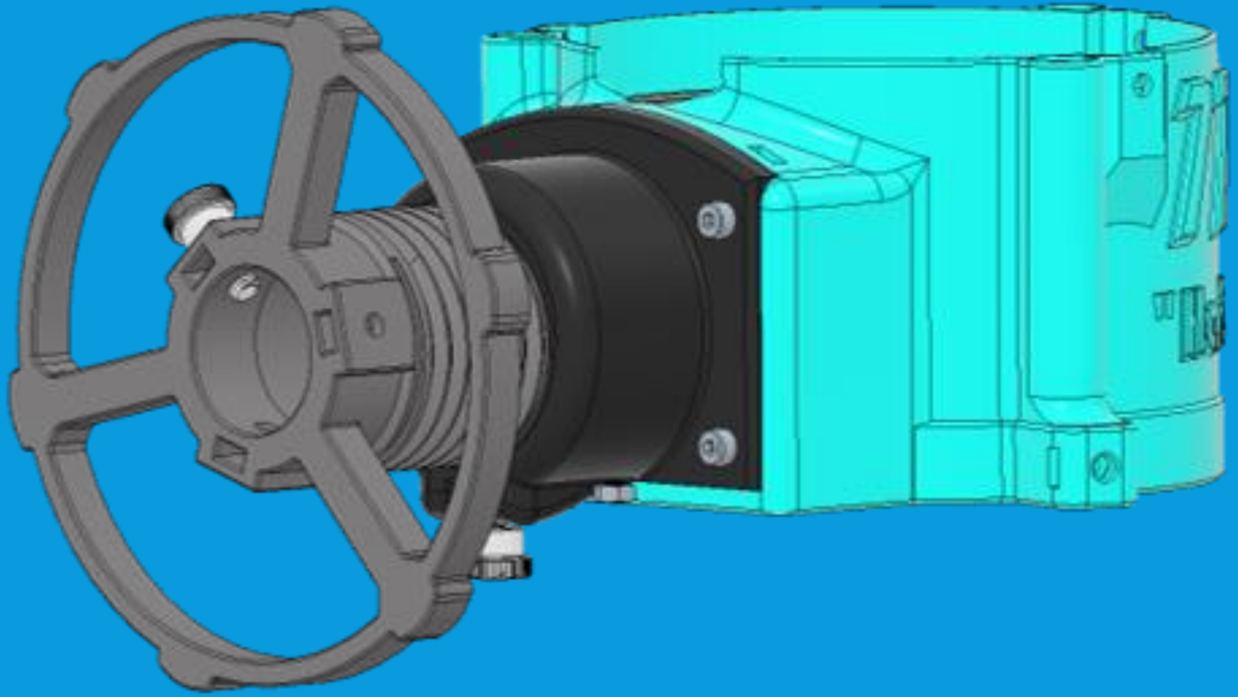
- Insert a nut into one of three pockets on focuser wheel, then insert ½" screw (ideally nylon thumb screw) into nut from outside base as shown
- Additional spots are for extra nuts & screws if desired and are discussed later. They can be inserted later with same procedure if desired. For now, leave off.





7.4: COMPLETING UTA ASSEMBLY

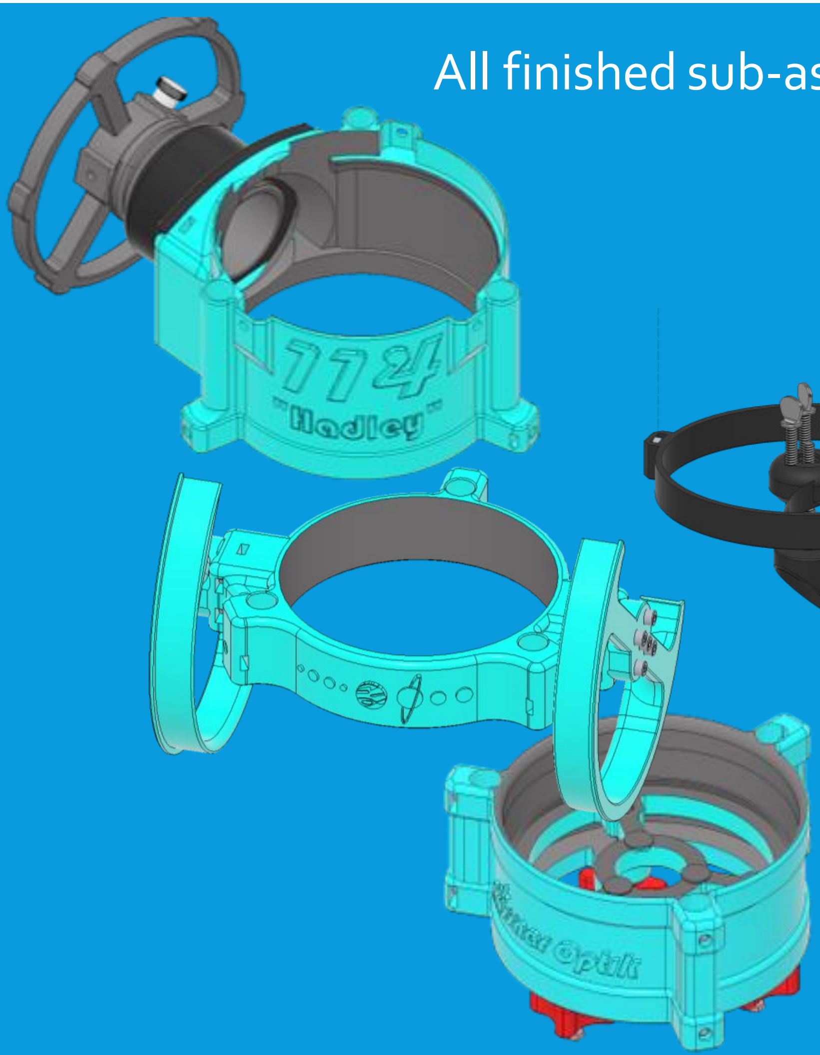
- Screw focuser handwheel into focuser base. Precise positioning does not matter at this point in assembly.
- Upper Tube Assembly is now complete, completing all major sub-assemblies. You are now ready to assemble the main Optical Tube Assembly from the sub-assemblies and remaining parts!

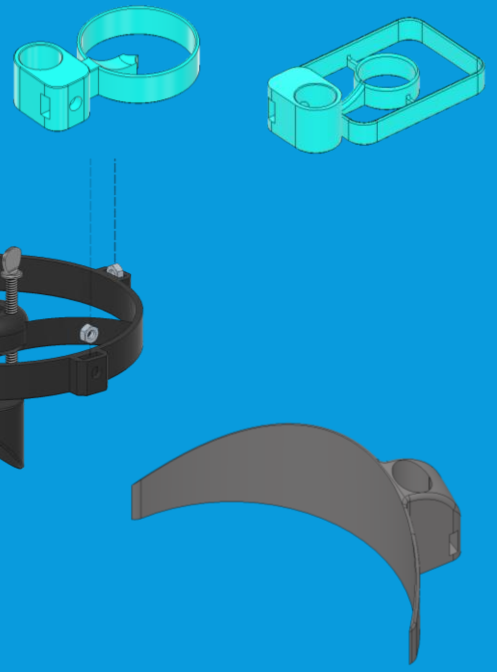


PART 8: OPTICAL TUBE ASSEMBLY

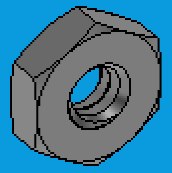
8.1: PARTS NEEDED

All finished sub-assemblies & all remaining prints

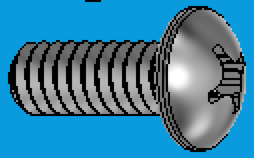




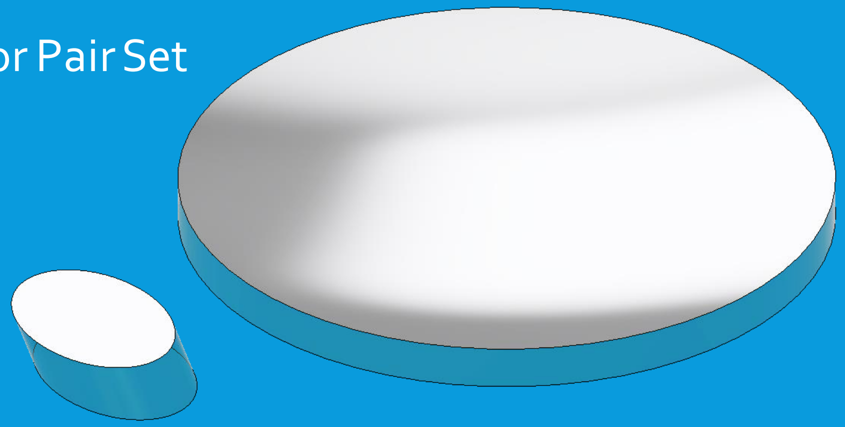
(18x)#10-24 Nuts



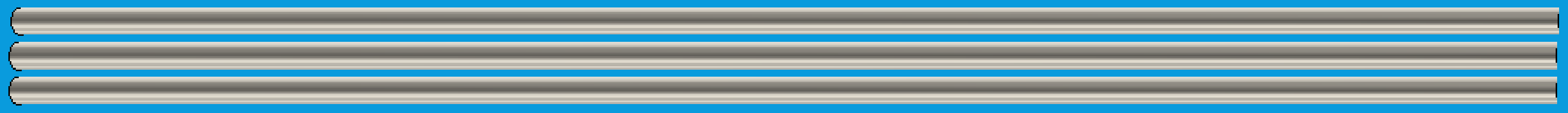
(18x) #10-24 screws, ½"



Mirror Pair Set

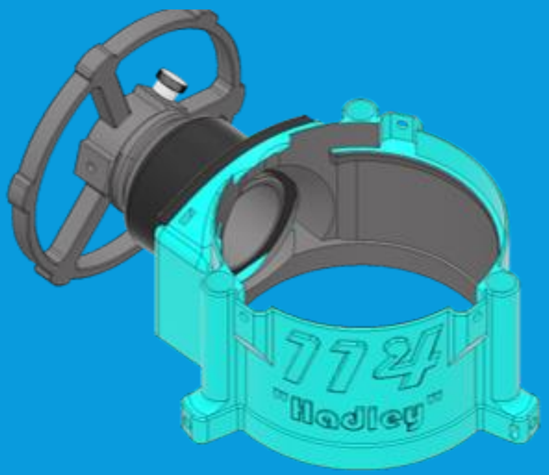


(3x) Aluminum or Steel Rod/Tube

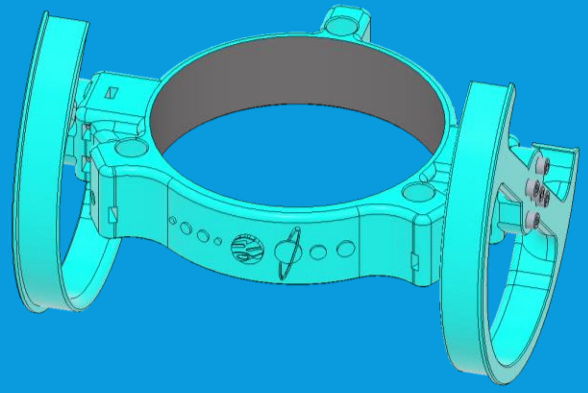


8.2: ATTACHING LTA TO RODS

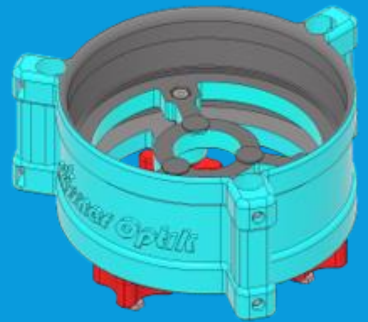
- Gather the UTA, MTA, and LTA and Three (3) ½" Rods or Tubes. (If using metric, 12mm rods.)
- Insert three rods into holes on LTA, allowing enough excess to be longer than collimation bolts. Measure excess to have same amount of excess on each rod.



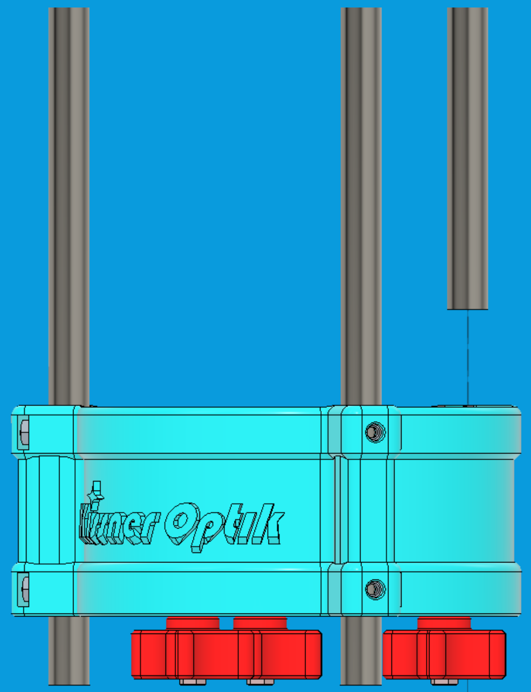
Upper Tube Assembly



Middle Tube Assembly

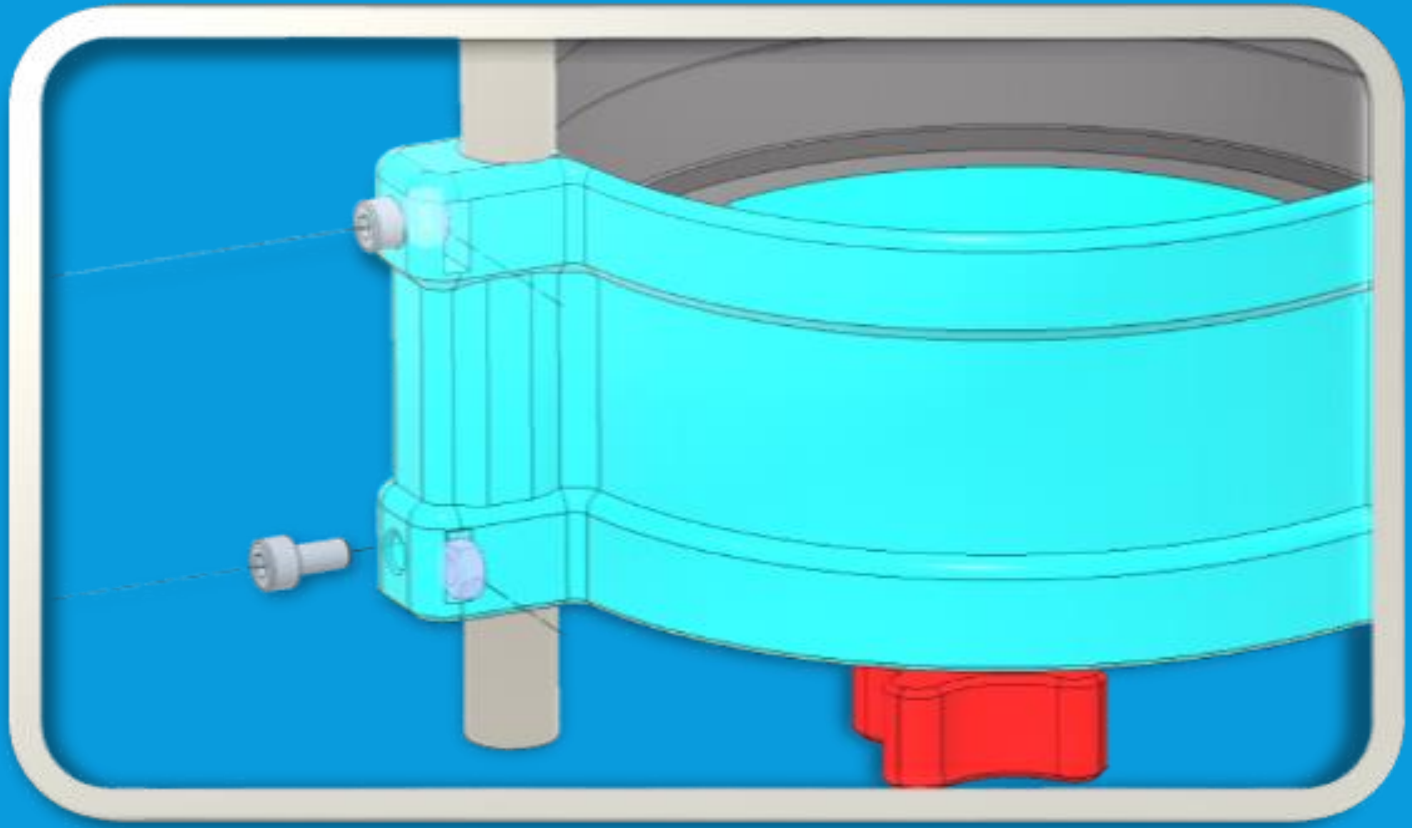


Lower Tube Assembly



8.3: LTA NUTS & BOLTS

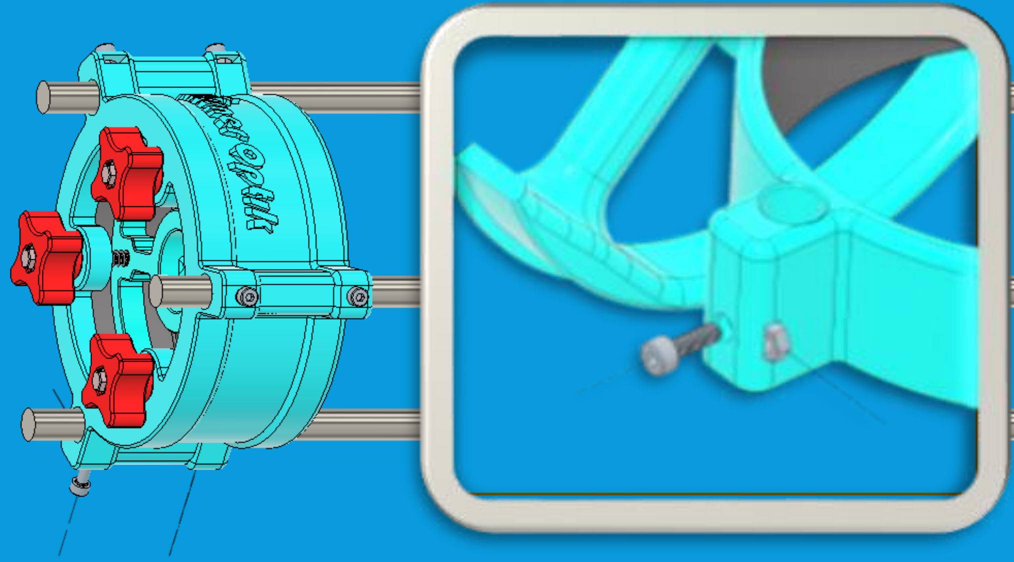
- Insert six nuts into pockets on rods as shown (two pockets per rod) Inert six short (½") screw into each nut/pocket
- Tighten this set screw against rod until tube will not slide up and down. Use care not to over-tighten. Excessive pressure can crack printed parts.
- Cracked parts may still be usable, especially if only one pocket per ring is cracked, but are to be avoided if possible.

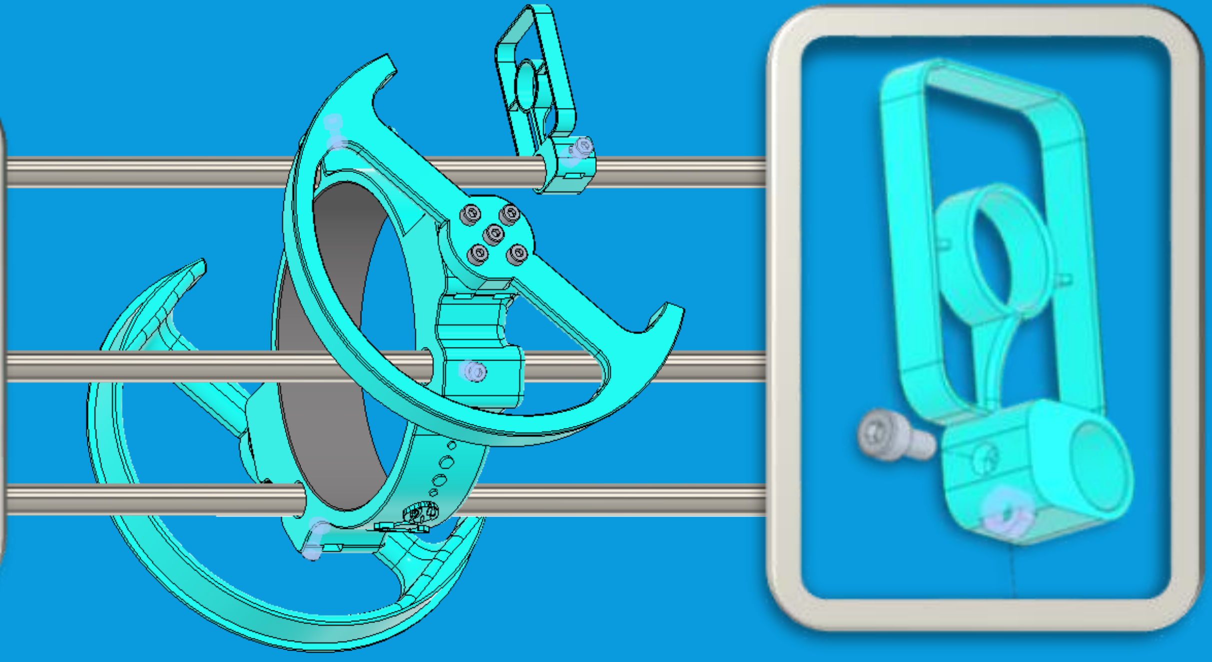


8.4: MIDDLE TUBE ASSEMBLY & LOWER SIGHT

- Slide MTA onto rods, positioning roughly at middle. (Precise position will be set later in balancing.)
- Attach to rods with three nuts and three short (½") screws in pockets as show, similar to with LTA.

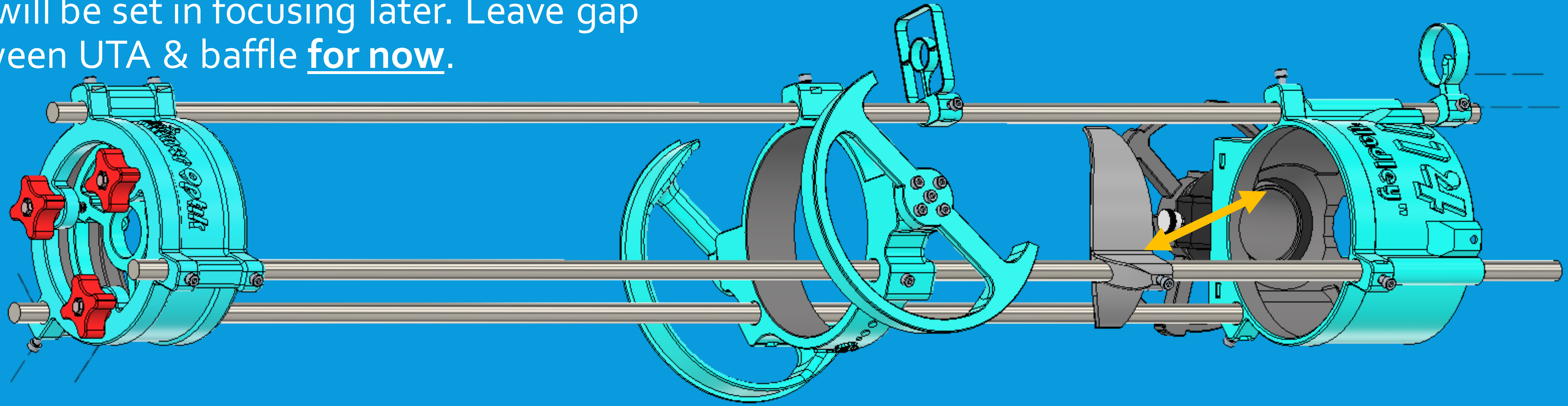
- Locate lower sight and slide onto one rod as shown.
- Attach to rod with one nut and one short (½") screw. Precise positioning will be set later.





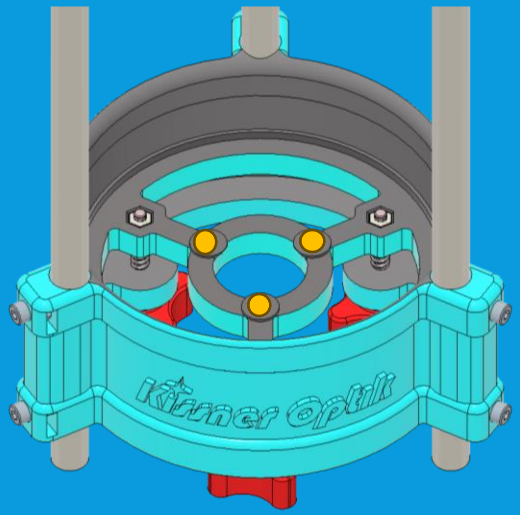
8.5: BAFFLE & UTA

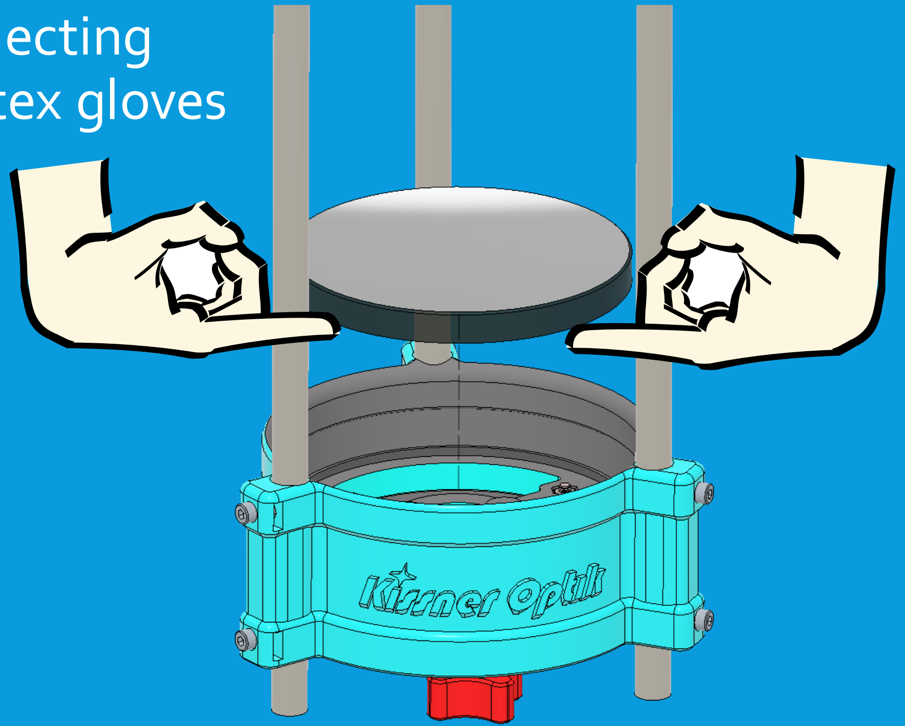
- Slide baffle onto indicated rod, note position opposite focuser hole in UTA. Attach with one nut in pocket and one short (½") screw as shown.
- Slide UTA onto rods, aligned as shown relative to baffle. Attach with three nuts and three screws as shown. Positioning will not need to be precise and will be set in focusing later. Leave gap between UTA & baffle **<u>for now</u>**.
- If alternate orientation of UTA and focuser is desired, move baffle and UTA to alternate arrangements as shown later in presentation.
- Slide upper printed sight onto same rod as lower printed sight, and attach with one nut and one short (½") screw.

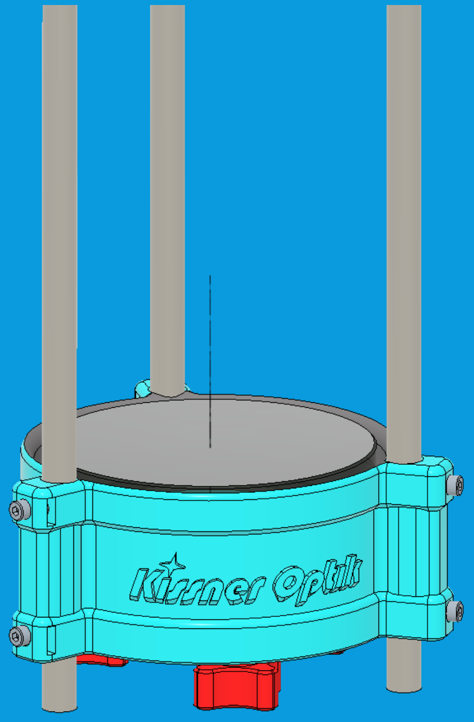


8.6: MIRROR GLUING—PRIMARY MIRROR

- Place three dots of glue (~¼") on pads as shown.
- Carefully lower mirror onto cell.
- Be very careful to never touch the reflecting surface of the mirror. Clean rubber/latex gloves may be useful.
- If possible, have a second person hold the telescope, reach up through the primary cell, and use your other hand to lower the mirror onto the first hand.

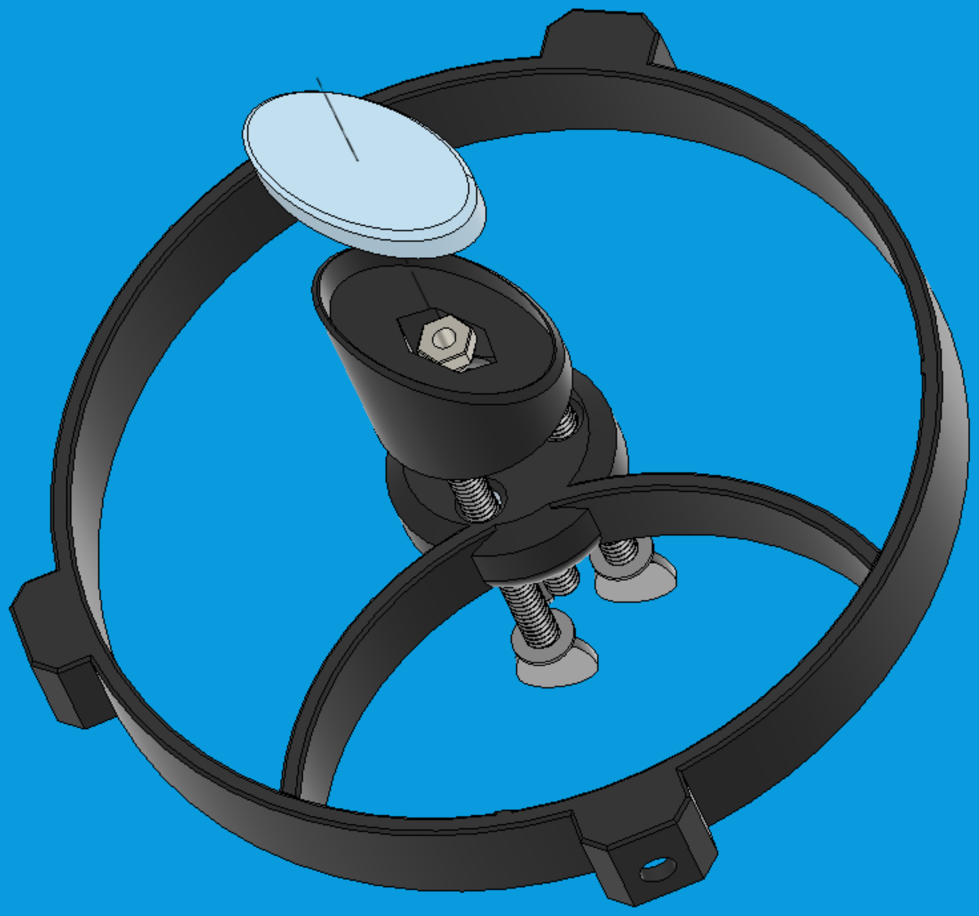






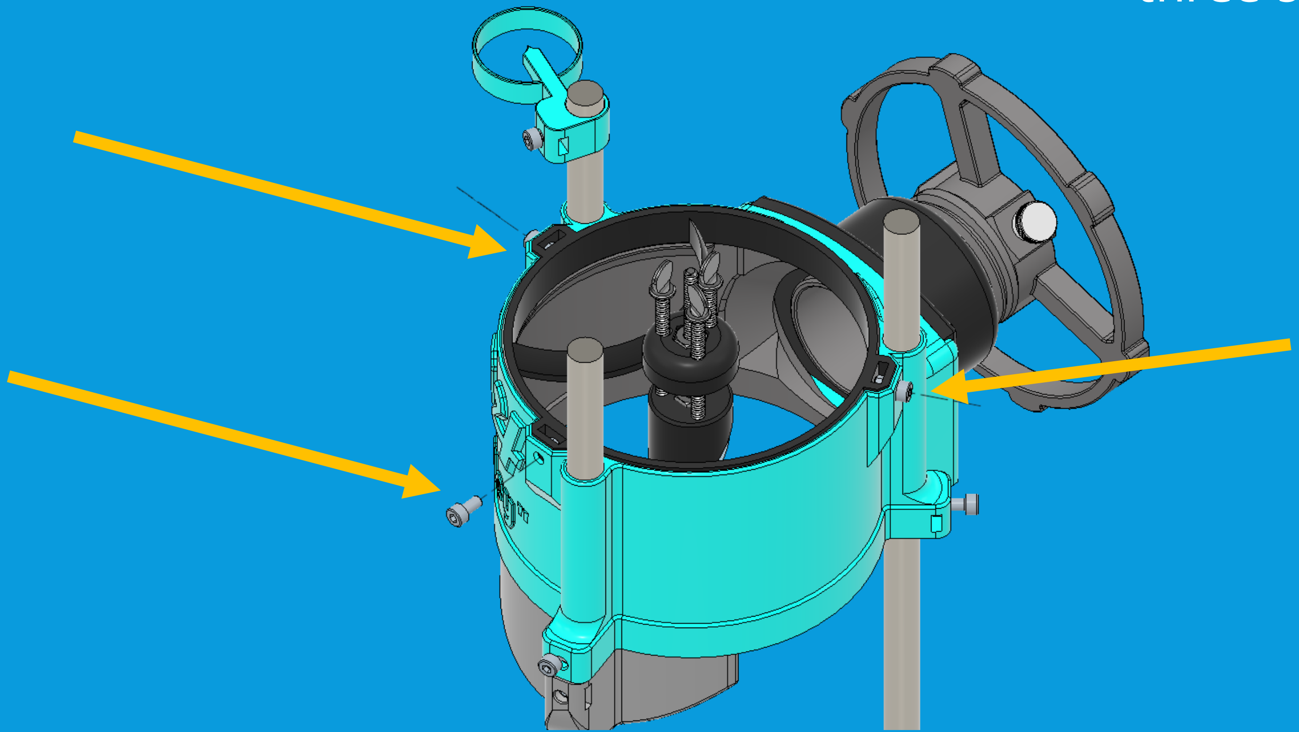
8.7: MIRROR GLUING—SECONDARY MIRROR

- Apply silicone adhesive to surface of secondary holder. Use a small blob (roughly 5mm in diameter) and spread across surface.
- Holding secondary mirror by edge, press into adhesive on holder for a few seconds.
- Avoid touching mirror surface as much as possible. Clean latex/rubber gloves may be useful.
- Leave aside for several hours to cure – prop it so the mirror is facing upwards.



8.8: MOUNTING SPIDER TO UTA

- After mirror adhesive has been allowed time to cure (1-2 hours) insert spider and secondary mirror assembly into Upper Tube Assembly

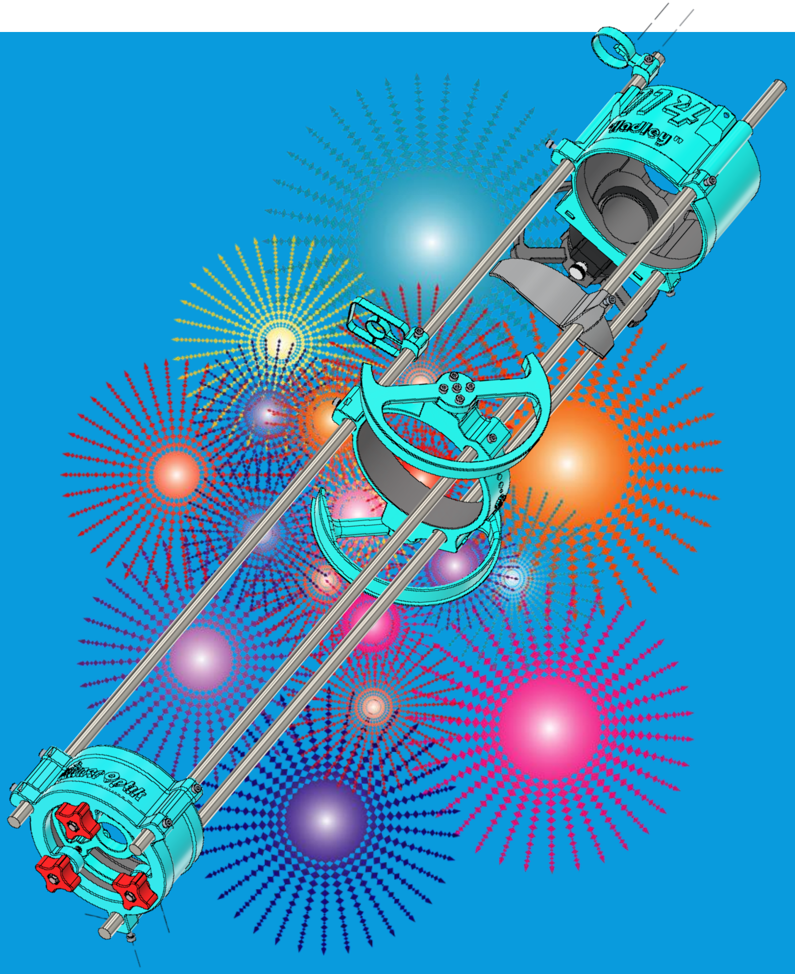


- Pocket positions will only allow one orientation.
- Attach with 3 nuts in pockets and three short (½") screws.

8.9: OPTICAL TUBE ASSEMBLY COMPLETE

Congratulations! With assembly complete, you now have a telescope!

Before it becomes a "useful" telescope, we need to align the mirrors and reach focus. This will be covered next.

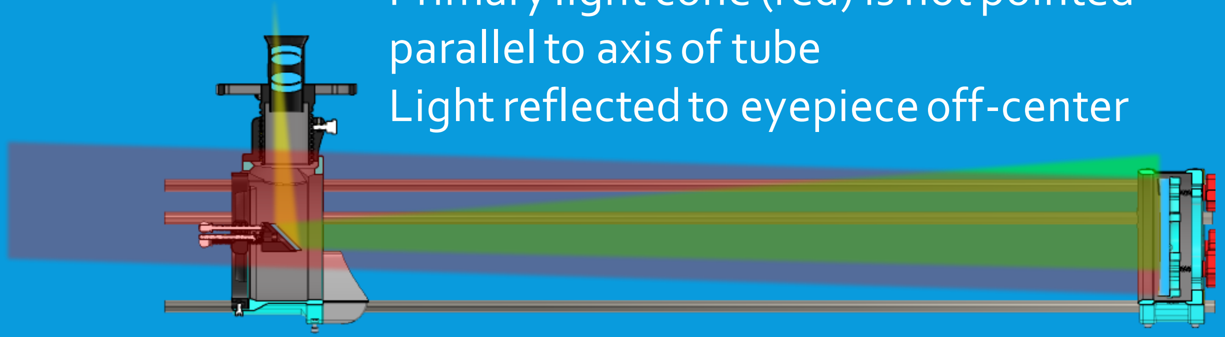


PART 9: INITIAL SETUP AND CALIBRATION

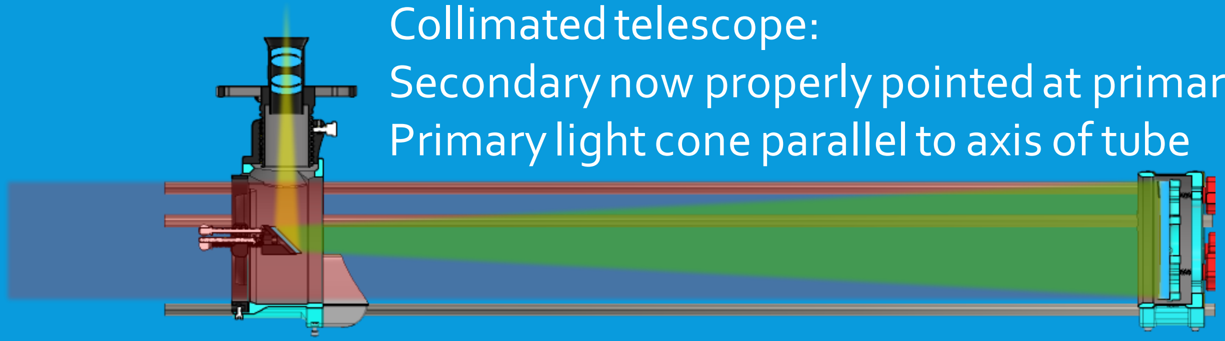
9.1: COLLIMATION PROCESS

- In a reflecting telescope, "collimation" is the term for aligning the mirrors to properly point light into the eyepiece
- Hadley is forgiving of small errors in collimation, so it can be adjusted by eye if needed
- If additional precision is desired, some useful tools for collimation (including one which can be printed) are explained on the next slides

Secondary light cone (green) does not properly point at primary
Primary light cone (red) is not pointed parallel to axis of tube
Light reflected to eyepiece off-center



Collimated telescope:
Secondary now properly pointed at primary
Primary light cone parallel to axis of tube

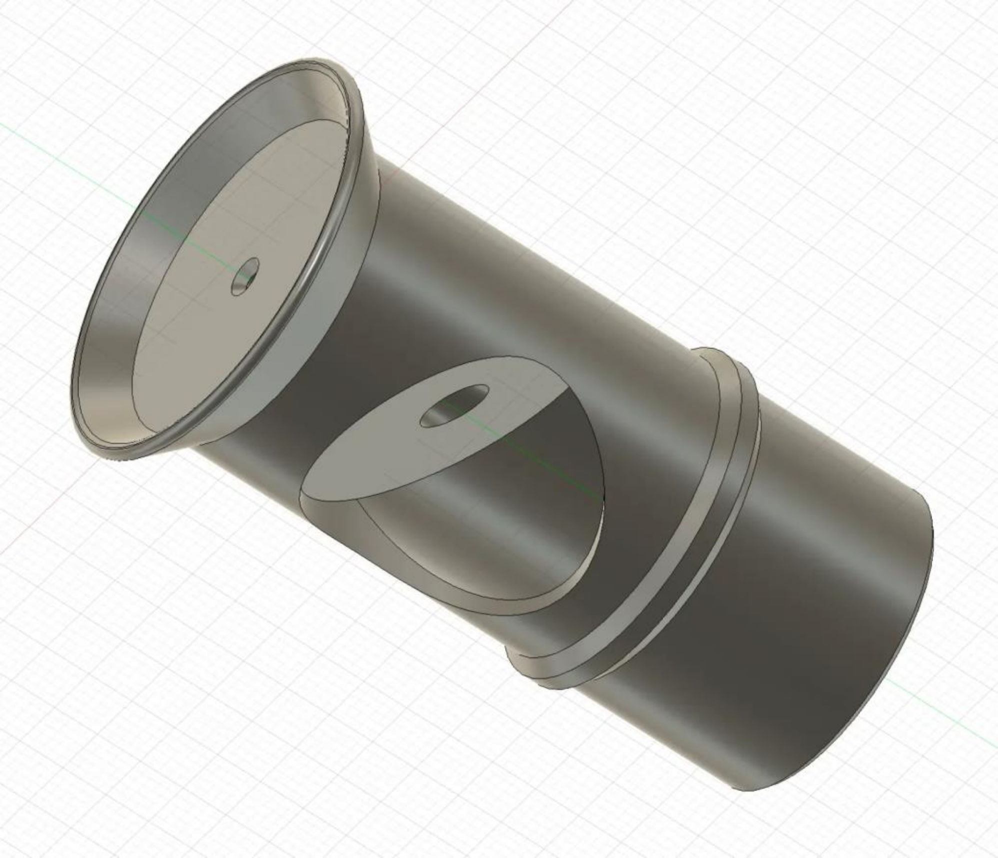


9.2.1: COLLIMATION TOOL—COLLIMATION CAP

- A collimation cap is an eyepiece with a small pinhole to look through and a reflective inner surface
- Eyepiece restricts the view when collimating, ensuring repeatable alignment of your eye and the mirrors of the telescope, while inner surface helps with finding center of secondary when looking through eyepiece
- With added features, becomes a Cheshire Eyepiece, seen on next slide

9.2.2: COLLIMATION TOOL—CHESHIRE EYEPIECE

- As with a collimation cap, Cheshire collimation eyepiece (right) has a small hole which ensures repeatable positioning of your eye relative to the telescope to judge alignment
- Angled portion can be lit from the side with a flashlight, creating a bright area around the center spot much like with a collimation cap
- This is visible when looking into eyepiece for collimation, and helps locate secondary and primary
- Useful to have and easy to print, & file available on Printables



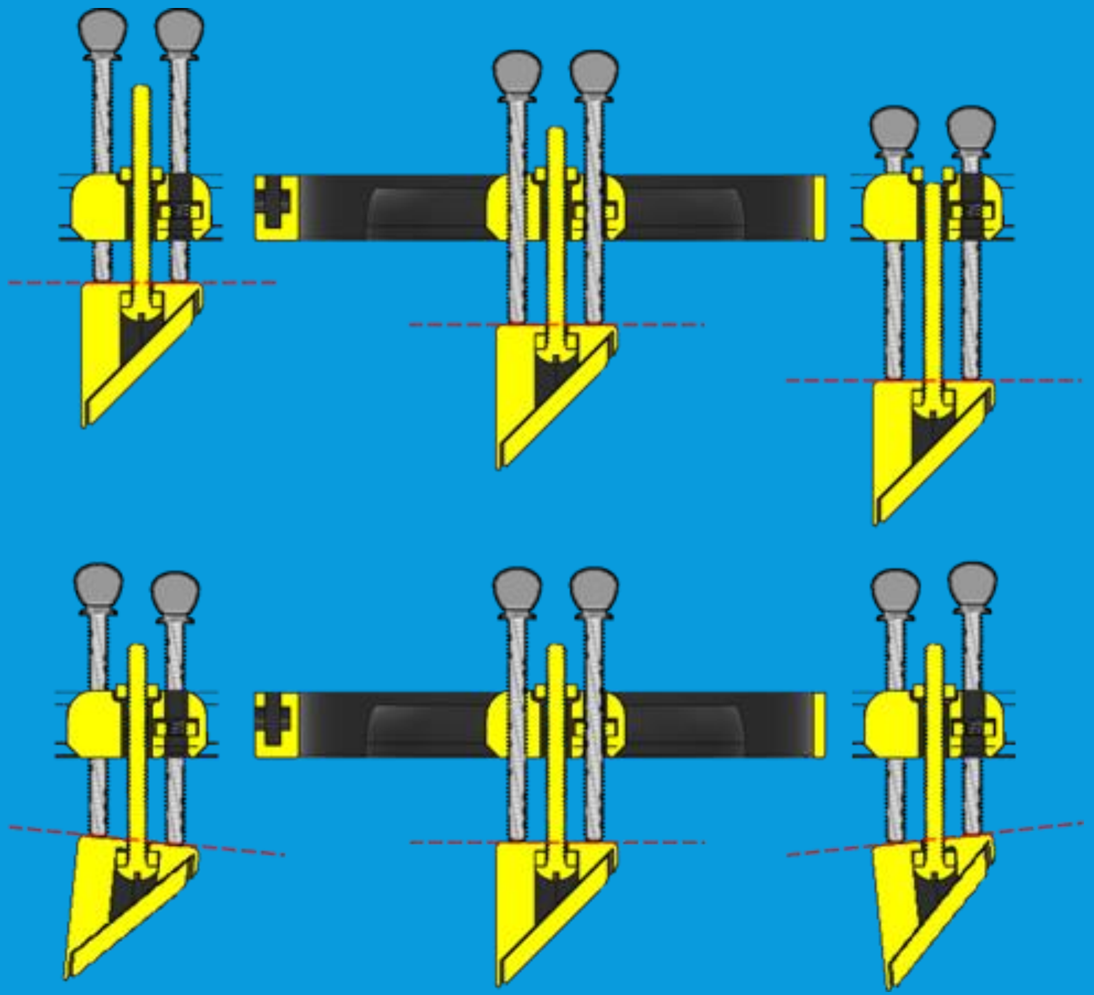
(https://www.printables.com/model/256728-cheshire-collimation-eyepiece-for-reflector-telesc)

9.2.3: COLLIMATION TOOL—LASER COLLIMATOR

- A laser collimator replaces looking down the eyepiece with a laser projecting a spot through the focuser, bouncing off the secondary, and onto the primary.
- Looking at the spot allows adjusting the bolts for collimation to center the spot on the primary mirror while looking at effects without also having to be looking through the eyepiece
- Laser collimation is more useful with telescopes requiring more sensitive collimation, and is overkill for Hadley.

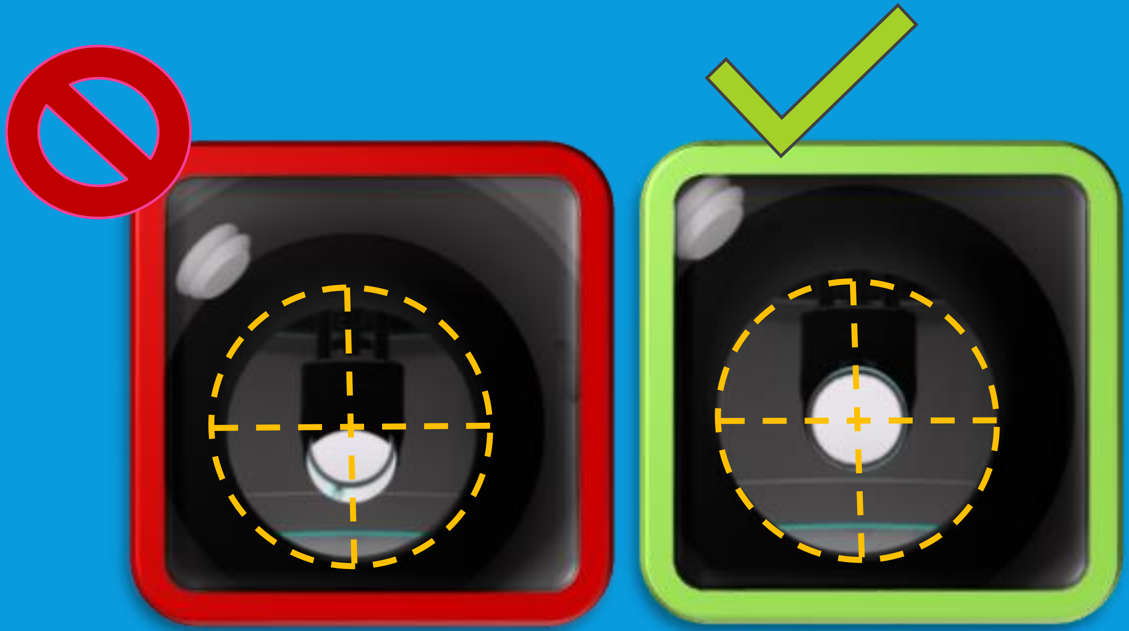
9.3: SECONDARY MIRROR ALIGNMENT

- Adjustment of the secondary mirror is carried out with the collimation bolts on the secondary mirror/spider assembly.
- Adjust the secondary mirror height by turning secondary holder to adjust the length of the center collimation bolt, compressing or releasing the spring
- Adjust the angle of the mirror by using any two of the three outer collimation bolts to press on the mirror and "tip" It slightly on the center bolt
- Note how a small difference between the outer bolt distances makes for a large change in angle (+/-6 degrees in images at right)

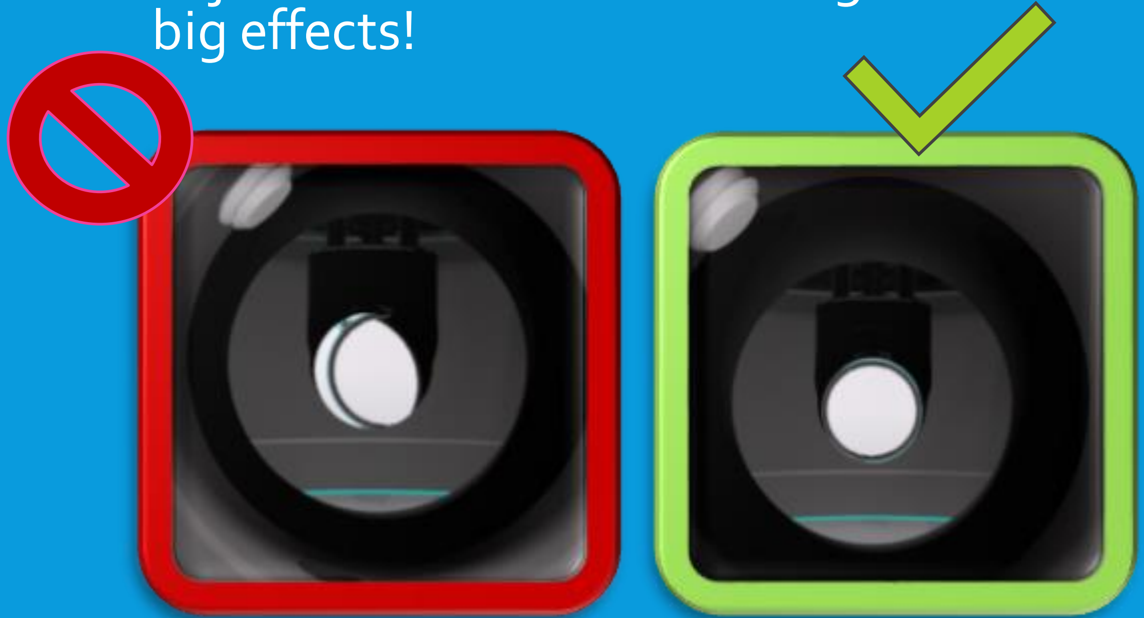


9.4: SECONDARY MIRROR HEIGHT & ROTATION

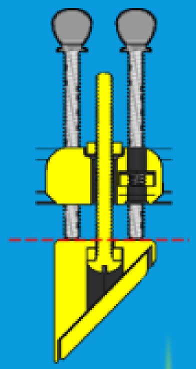
- Start by ensuring the secondary mirror is aligned properly vertically.
- Adjust nut on center bolt up or down until secondary mirror is centered in view through focuser



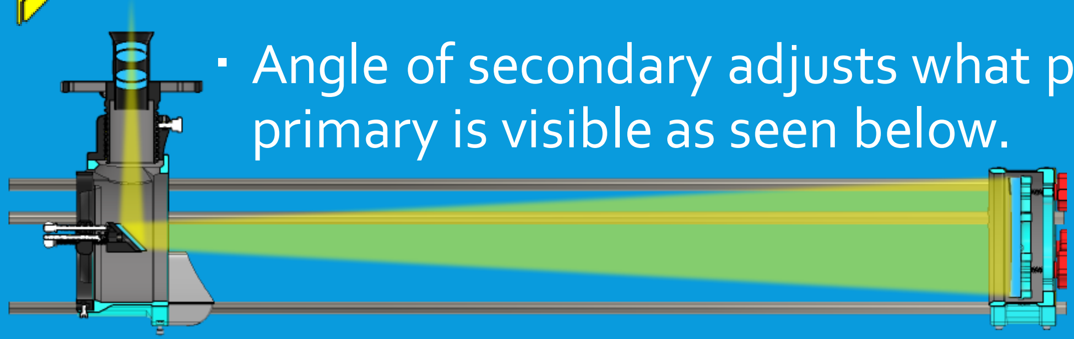
- Next, align the secondary mirror to aim into the eyepiece. Look for the circle of the primary mirror—it should be centered in the view.
- If the view of the primary is off-center, adjust the rotation. Small angles have big effects!



9.5: SETTING SECONDARY ANGLE



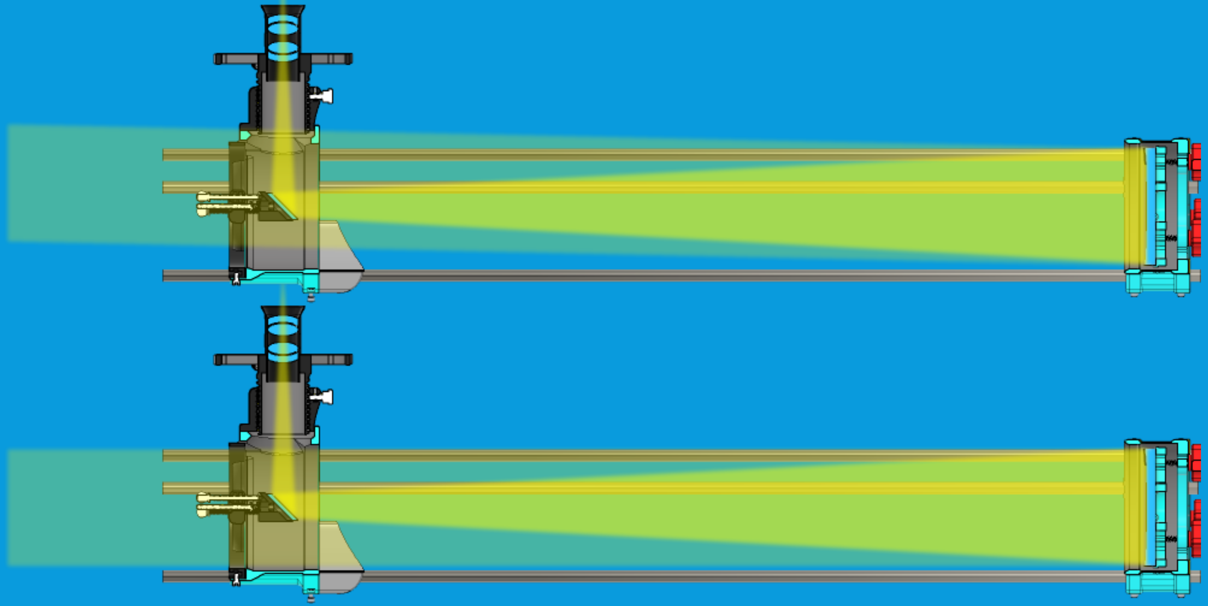
- Screw two outer collimation bolts on the spider until they touch the mirror holder. Further adjustment of these will now adjust the angle of the mirror as shown on slide 9.4.
- Angle of secondary adjusts what part of primary is visible as seen below.



- Looking through eyepiece, adjust collimation bolts until view of primary is centered in secondary as shown below
- Once centered, adjust third bolt to touch and stabilize secondary holder

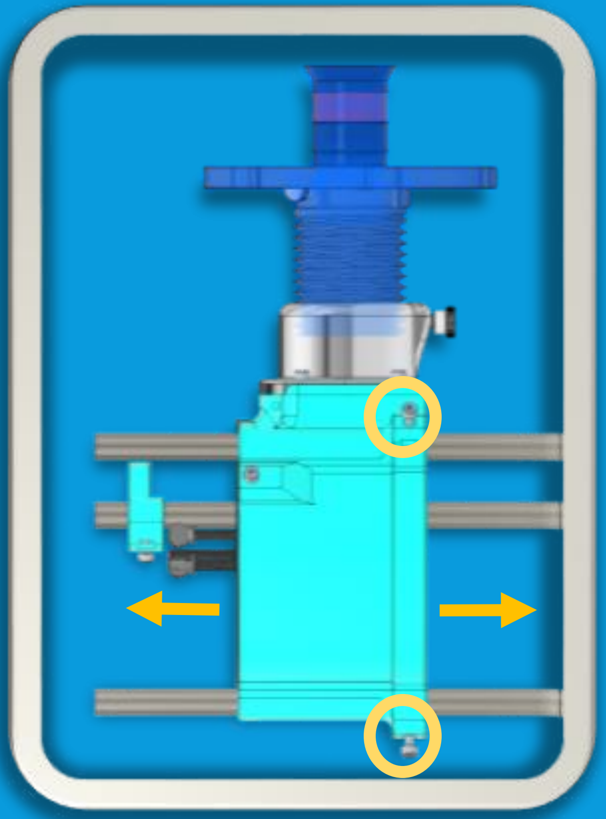
9.6: ALIGNING THE PRIMARY

- Final step of collimation is to adjust the primary cell so the primary is looking out along the axis of the telescope as shown below

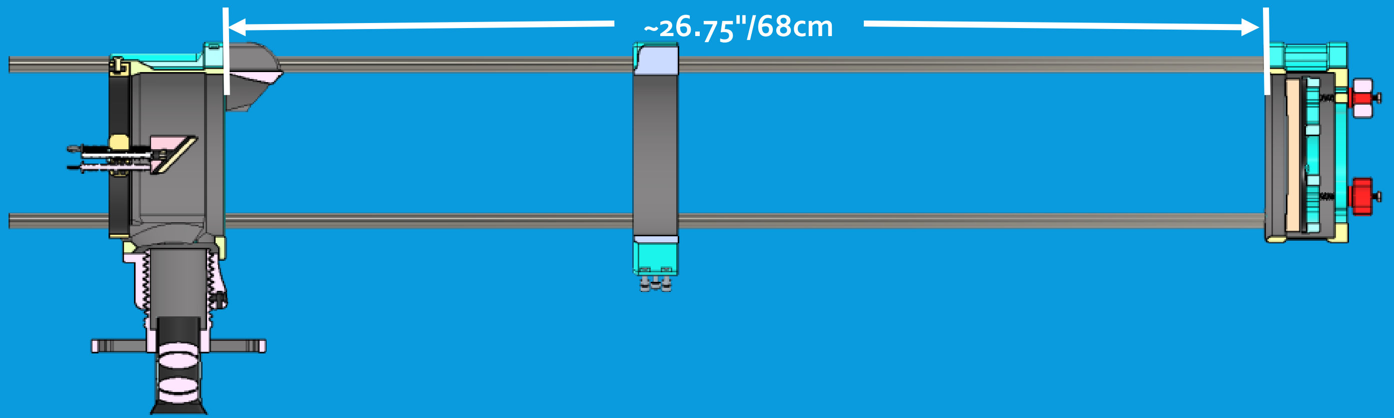


- Adjust one knob at a time on the primary mirror cell until image of the spider is centered in the image visible on the secondary as shown below
- Perfection is not required

9.7: SETUP FOR INITIAL FOCUS

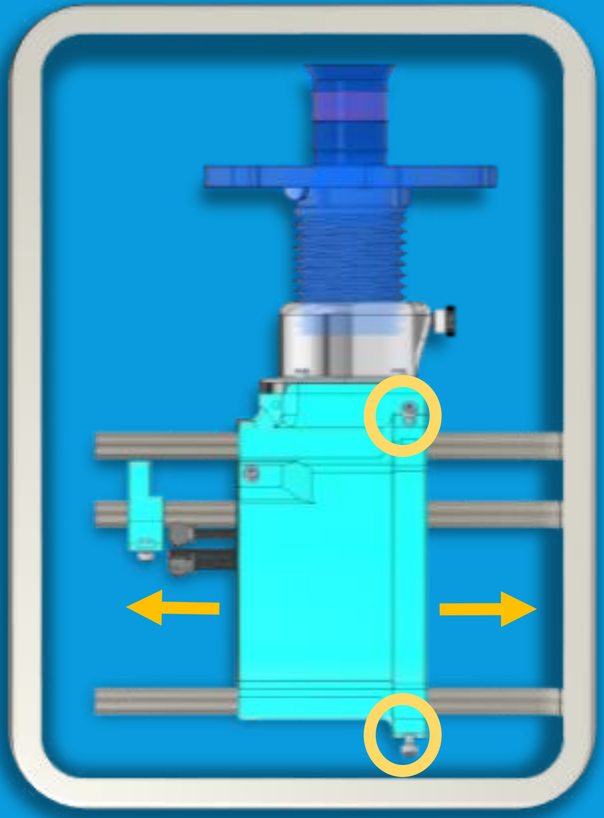


- Insert an eyepiece, secure it in the focuser.
- Position focuser near the upper end of its travel range (screwed nearly entirely out) & loosen the screws securing the UTA so it slides freely
- Adjust UTA to LTA distance to baseline of roughly 26.75"/68 cm

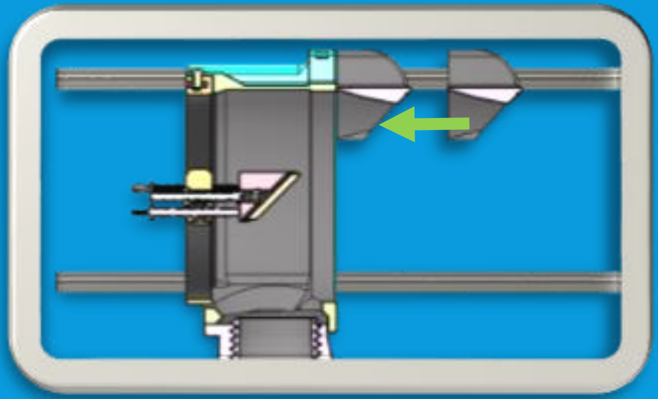


WARNING: This telescope is now a complete light concentrating system. Do NOT point it at the sun! The risk of fire, vision damage, or blindness is severe!

9.8: INITIAL FOCUS PROCESS



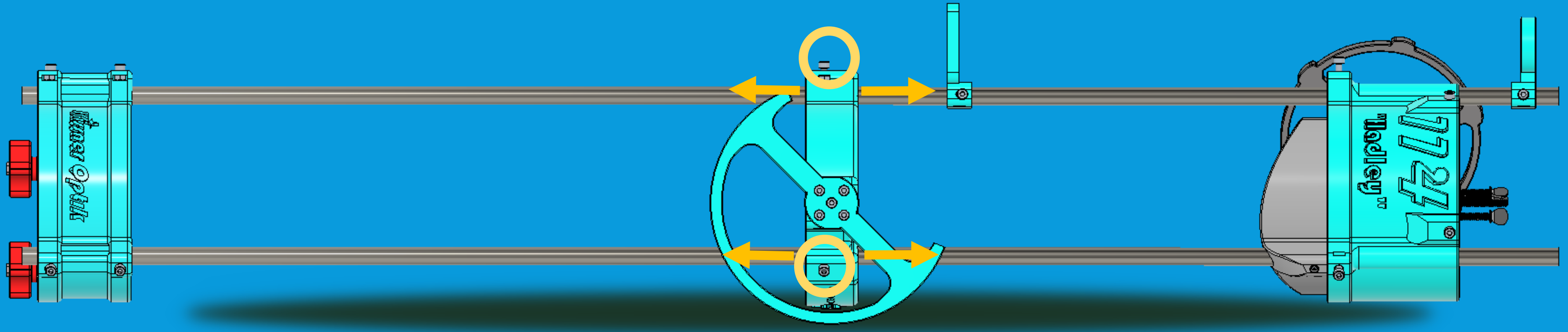
- Rest the telescope in your lap or on a table, pointing it at as distant an object as you can comfortably find, ideally at least half a mile
- Look into the eyepiece. Slide the upper cage until an image comes to focus. Fine tune the sharpness, and then tighten the screws by hand
- Adjust baffle to snugly fit against base of UTA in final position



WARNING: This telescope is now a complete light concentrating system. Do NOT point it at the sun! The risk of fire, vision damage, or blindness is severe!

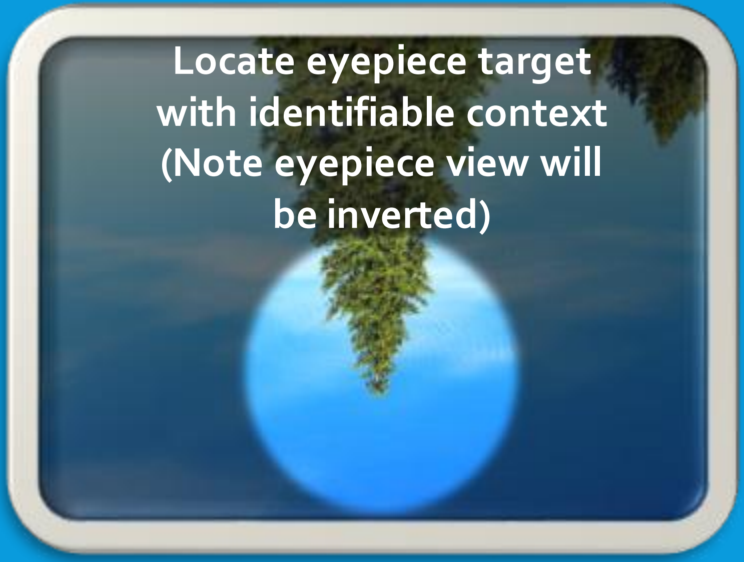
9.9: BALANCING THE TELESCOPE

- In aiming up/down ("altitude)", the telescope functions like a see-saw, with its fulcrum at the bearings
- To balance the Hadley, the middle ring (or tripod mount) should be at the balance point
- Loosen screws on the middle ring so it can slide freely. **Insert eyepiece.**
- Rest telescope on flat surface
- Adjust center ring until it balances, resting on only the bearings.
- Tighten screws. Your scope is now balanced!



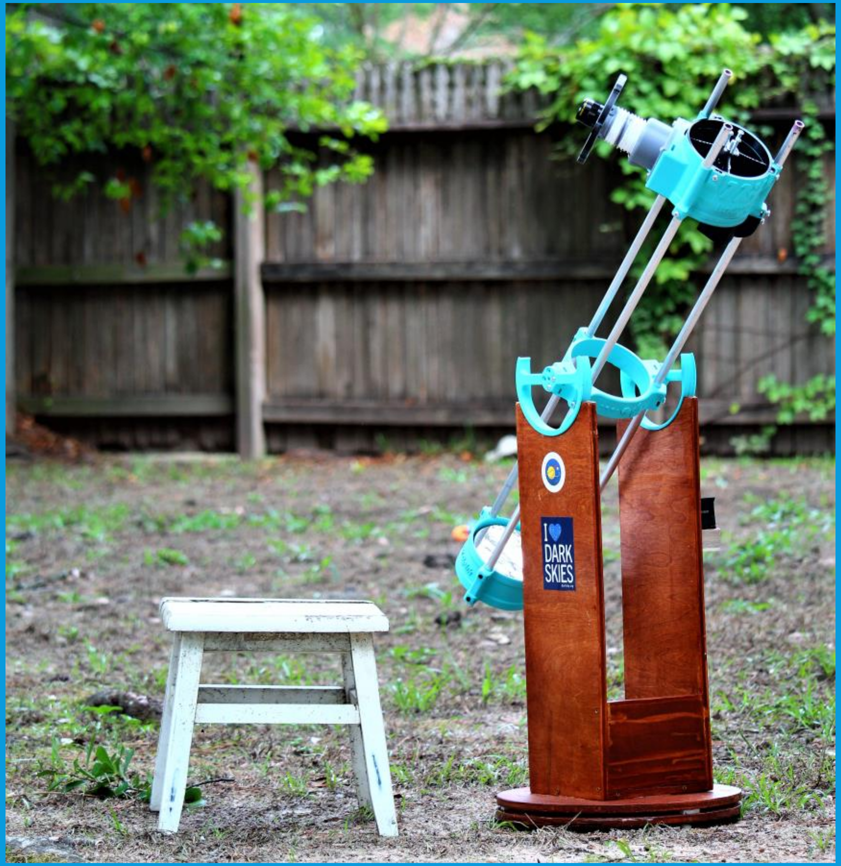
9.10: ADJUSTING SIGHTS

- Insert lowest-power eyepiece into the focuser and point Hadley at an identifiable target (such as a tree, building, or the moon)
- Loosen the screws on the lower sight
- Adjust lower sight up and down on rod until lower sight and upper sight circles are roughly the same size
- Rotate lower sight around rod until the two rings capture the same image as seen through eyepiece. Tighten.



9.11: CONGRATULATIONS (AGAIN)

- You now have a functional telescope, ready to attach to the mount of your choice (printed or wood)
- Mounting options are discussed in Appendix B



A. EYEPIECES IN DETAIL

A.1: WHY FOCUS ON EYEPIECES?

- Eyepieces are necessary to magnify/project the image telescopes form into something usable by our eyes.
- Poor quality eyepieces can worsen the image, and premium eyepieces can be quite expensive
- Hadley is forgiving on inexpensive eyepieces, and this is an area of diminishing returns. Note that premium eyepieces can cost more than the whole telescope.

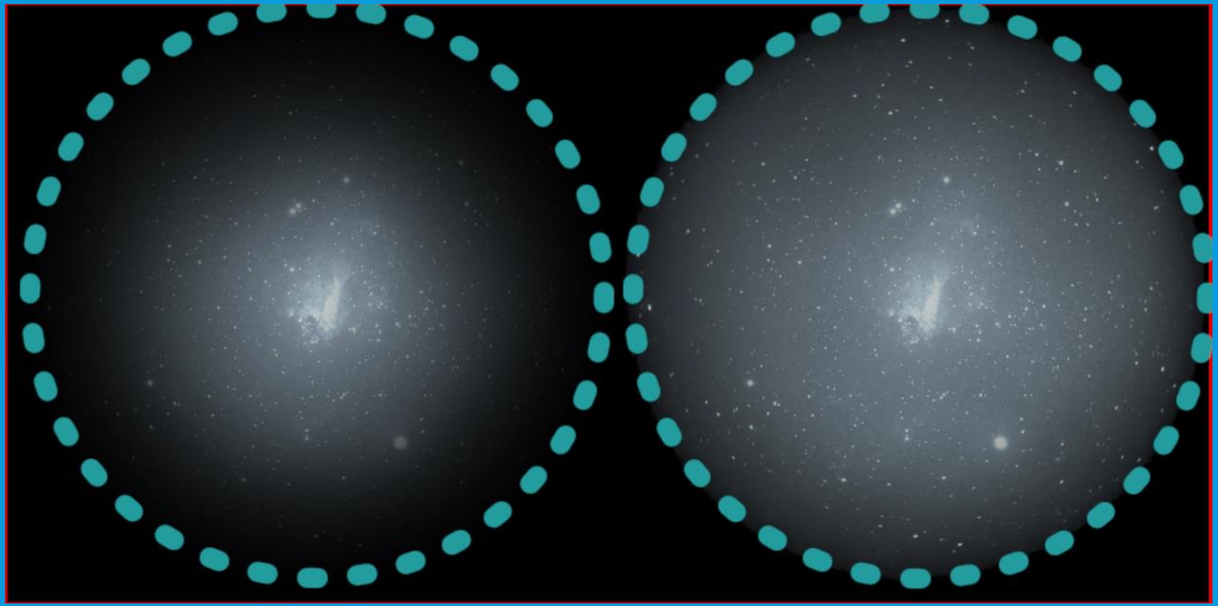
Author's note: It is my opinion that an eyepiece is about a 40% contribution to the view. It takes a decent telescope to get a good view of anything, but where do you go from there? **Closer/further/wider/sharper, the rest is up to the eyepieces.**

A.2: EYEPIECE DIAMETER

- 1.25" is a common standard, and good eyepieces are "future proof" even if you get a nicer telescope
- **2" eyepieces exist; Hadley cannot use these.** This is for several practical reasons, including the illumination profile.
- Quality is unrelated to barrel size; most 2-inch eyepieces have 1.25-inch cousins that are just as good.
- 2-inch eyepieces are a consequence of internal optics size; very wide fields require a larger unobstructed path.

2-inch eyepieces require a large secondary to illuminate the off-axis parts of the field. (Effect of undersized secondary below)

Larger secondaries (slightly) worsen contrast, however.



A.3: FIELD OF VIEW

- "Field of view" can mean two different things – "true field" or "apparent field."
- True field (TFoV) is a measure in arc-size of the sky. A 25mm plossl has a 1.4 degree true field in Hadley.
- Apparent field (AFoV) is how wide a view the eyepiece presents for your eye. A 25mm plossl has roughly a 50-degree apparent field.
- Apparent field is *nice* to have a lot of. It is one of the most expensive qualities of any eyepiece.

- Magnification relates the apparent field of view of the eyepiece to the true field you will see through it in your telescope: a 25mm plossl has a 36x magnification in Hadley 114, and thus the 50 degree apparent view gives 1.4 degree TFoV

A.4: BUDGET LOW-POWER EYEPIECES

25mm Plossl

- Comes in Hadley kit, roughly $20 USD
- Sharp, good all-round eyepiece
- FoV tends to be ~50 degrees (often mis-stated)
- Recommended mid-cost option

32mm Plossl

- Slightly pricier than the 25mm, costs $25-30 USD
- More "zoomed out" for easy finding

23mm Aspheric

- This eyepiece is the best you will find for $12 USD. It *works* but softens the image.
- Recommended minimum-cost option

A.5: BUDGET HIGH-POWER EYEPIECES

"Redline"/"Goldline" series

- These two have the same internal optics, but are listed differently
- They come in 6, 9, 15 and 20mm; the 6 and 9 are much better than the 15 and 20
- **Redline:** "68-Degree Ultra Wide Angle)"
- **Goldline:** "66-Degree Ultra Wide Angle HD"

6mm vs 9mm Focal Length

- 9mm makes 100x power and 0.68 degrees true field of view
- 6mm makes 150x power and 0.44 degrees true field of view
- 6mm gives more power for larger views of planets or lunar features but smaller field of view makes finding targets more challenging and motion of objects across the sky cross the view faster - requiring more frequent adjustment.

A.6: "FANCY" GLASS FOR SPECIAL EYES

82 degree 16mm UWA

- Notably more expensive, $99 USD, similar in cost to entire rest of 'scope
- Exceptionally wide apparent field of view makes for remarkable viewing with a very sharp image
- Similar true field of view as 25mm Plossl, apparent field of view much larger (see graphic on page 73)

82 degree 7mm UWA

- Similar cost, ~$99 USD
- Similar power to 6mm or 9mmm eyepieces, but much larger field of view
- More of lunar surface visible at once, or easier time finding planets and longer without repositioning due to motion of sky objects

These are the "Rolls-Royce" of Hadley eyepieces, and very future-proof for use with larger telescopes. Given expense, consider deferring as an upgrade to initial kit, but very worth the money if you stick with the hobby

A.7: HIGH POWER/ EYEPIECES TO AVOID

Avoid Short Focal Length Aspheric or Plossl Eyepieces

- Aspherics have bad focus and bad distortion at short focal lengths due incorporating a lens made of acrylic.
- "Scaling designs" such as the plossl, kellner and others may still produce sharp images -
- But short (<10mm) focal length versions have tiny eye lenses and are uncomfortable to use.

Avoid Eyepiece "Kits"

- These include the worst versions of any series, including those mentioned at left
- They tend to include a bunch of unnecessary accessories like cleaning cloths, color filters, or cheap barlows

A.9: BARLOW AND ZOOM EYEPIECES

Barlow Eyepiece

- Barlow lenses go between the eyepiece and focuser
- They effectively multiply the magnification of the telescope, steepening the light cone
- Common sizes are 2x, 3x, 5x, etc
- More than 2x is probably overkill for Hadley—a 25mm will behave like a 12mm, a 9mm like a 4.5mm, close to the limit of Hadley's usable magnification (~225x)
- Be mindful that high magnification is challenging and requires a *good* mount

Zoom Eyepiece

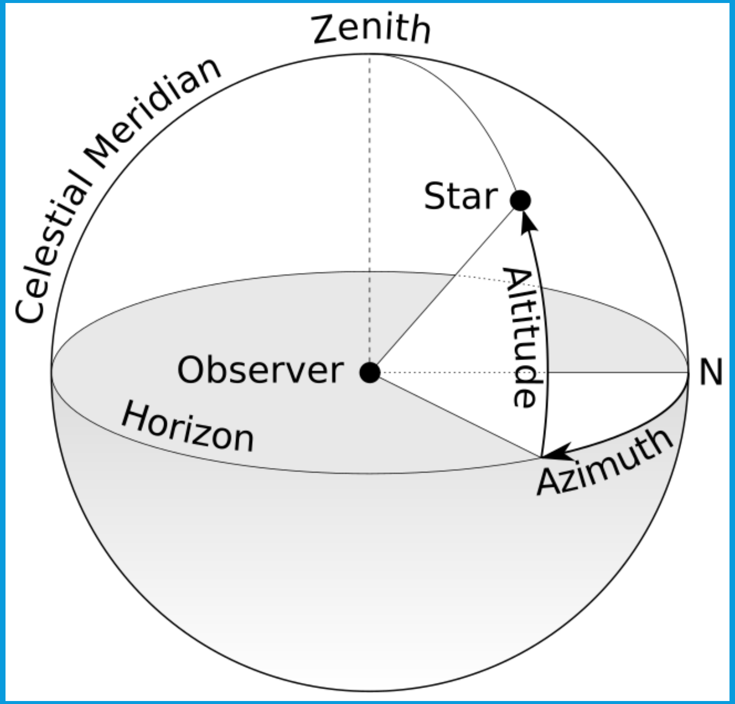
- There are eyepieces that offer a variable focal length, resulting in variable magnification
- These tend to trade field size and sharpness for convenience, offering worse performance than any discrete eyepiece. They are also pricier.
- Still, many people enjoy them greatly, including the feeling of "zooming in"
- They are rarely "parfocal" - this means focus must be adjusted with each change in zoom for a sharp image.

B. TELESCOPE MOUNTS

B.1: WHAT MOUNTS ARE FOR

- A telescope needs a mount to hold off the ground and allow it to be pointed at the sky in both altitude (up or down) and azimuth (rotation around the vertical)
- Mount options typically seen include tripods and "Dobsonian" mounts.
- Hadley is set up for a Dobsonian mount solution, as these offer the best combination of stability and ease of construction for an amateur project.

https://commons.wikimedia.org/wiki/File:Azimuth-Altitude_schematic.svg



B.2: WHY NOT A TRIPOD?

- Tripods are often very unstable and have difficulty pointing accurately with the weight and length of a Hadley even if they are rated for the weight (~5 lbs) of the telescope
- It is often nonviable to make adjustments on the order of 1/2 a degree with a tripod, but hand-tracking (which is easy on a proper mount) requires this.
- If you have an expensive tripod already, or no other option, you may be able to use a remix mounting option for a tripod, but a Dob will tend to be better and some modification from the instructions and default parts will be required

B.3: DEFINING THE DOBSONIAN

What is a "Dobsonian?"

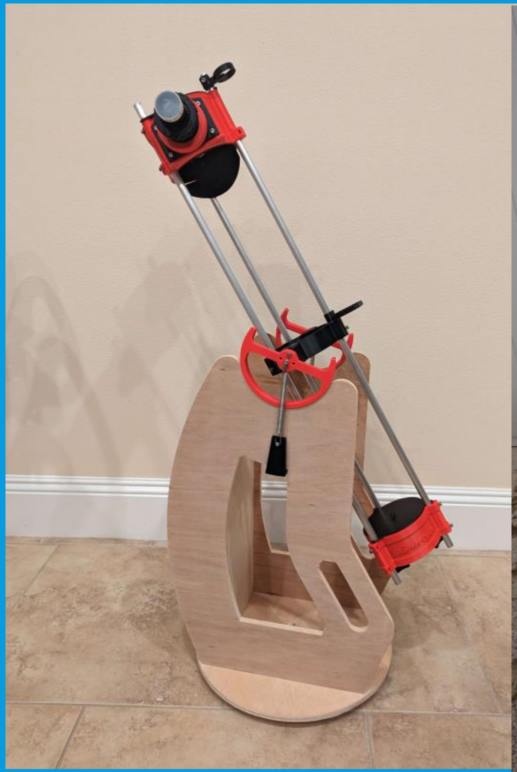
- This is a simple telescope mount that works in two axes; left-right (azimuth) and up-down (altitude), not unlike an old naval cannon.
- "Dob" describes both the mount and telescope-mount pair; almost always a Newtonian reflector.
- It was named for John Dobson, who pioneered large DIY telescopes any amateur could make.
- The hallmark feature is a wide "bearing" (a circle or semicircle) with small contact points, resulting in a stable mounting with substantial friction.

John Dobson with one of his telescopes.

- For Hadley, there are many paths to a viable mount, which is why there's no single "official" mount at this time—discussion on next slides

B.4: WOODEN DOBSONIAN MOUNTS

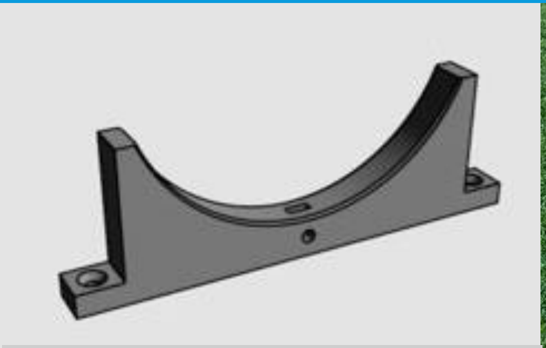
- Wooden mounts offer an excellent combination of stiffness and stability, and thus are recommended if the Hadley builder has access to the right tools and skills
- They can be as simple or complex as the user's skill and tools allow, ranging from straight cuts with a hand or tablesaw augmented by plastic bearings to very elaborate aesthetic creations.
- Wood mounts using material over ½" thick may need spacers or the "wide" middle ring mount option.

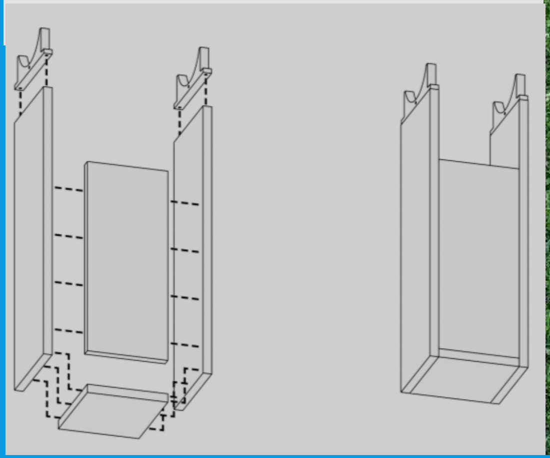


B.5: SIMPLE WOOD "PELF MOUNT" PLANS

- Community member Pelf has designed these plastic bearing supports, enabling simple straight-cut mounts
- Includes as a base plans for how to cut a 4 foot by 2 foot sheet of plywood with just straight cuts into the pieces for a simple mount using the bearing supports
- Note "sandwiched" lower azimuth mount for turning the telescope on the ground.
- More complete plans available on the Discord and at this page:

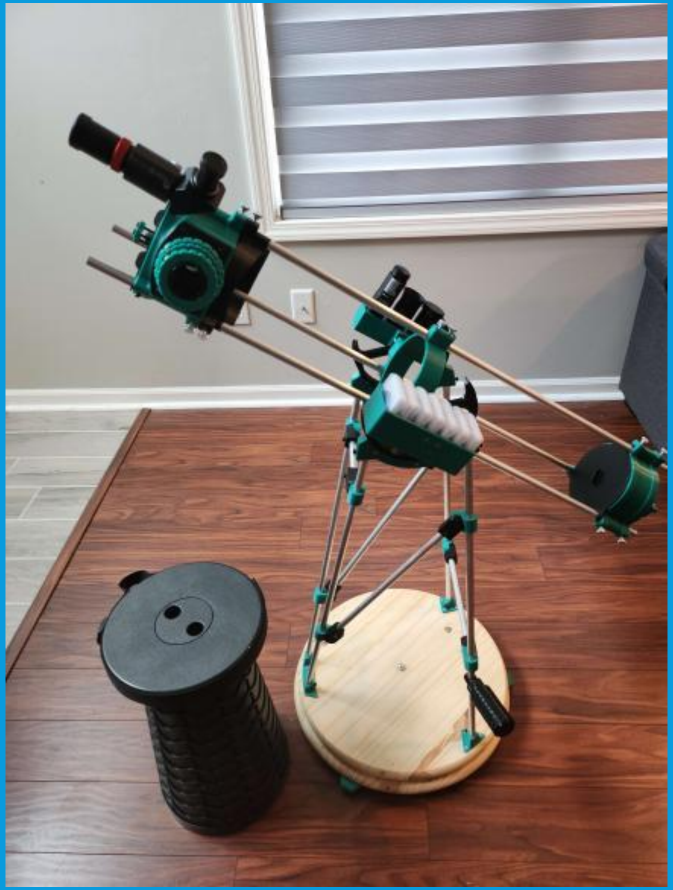
  https://www.printables.com/model/525530-hadley-bearing-support.





B.6: DIAGONALIZED PRINTED MOUNTS

- For mounts built with rod/tube and printed pieces, diagonal bars are recommended to make the mount stiff and stable.
- Examples of community projects are shown at right and discussed on subsequent slides, offering options for metric and imperial projects, and including heat set or inserted-nut solutions for assembly.
- With all options, a wood disc azimuth bearing is still recommended, though it's not required for minimum usability and can be installed later.



B.7: COMMUNITY PRINTED MOUNTS

- Hillexed's mount (far left): uses cheap, sturdy EMT conduit pipe, imperial unit friendly with nuts & bolts
- Ivan's mount (mid-left), metric fittings with nuts and bolts, with two-part upper suitable for smaller printers
- Marci's mount (right), metric fittings with bolts and brass heat inserts, wood disc base (suitable for sandwich)

B.8: MAKE ME AN AZIMUTH SANDWICH

- The ideal left-right (azimuth) steering mechanism is simply a large friction bearing, a **sandwich of two wooden discs and a material to limit friction.**
- The top disc attaches to the rest of the mount/telescope.
- The bottom disc stands on three feet, for tripod stability.
- Between the two is a friction interface, with three contact points for maximum stability.
- **This is not a Hadley-specific design**

This type of bearing is ubiquitous, appearing on many commercial, amateur and high-end telescopes.

B.8.1: AZIMUTH SANDWICH MATERIALS

- For Hadley, the bearing is ideally a pair of disks 12-18" in size.
- For the center bearing, a threaded T-nut installed in the lower disk or a nut below the lower disk serves to anchor the bolt which serves as the axis passed through both discs
- Compression control and holding the mount together when carrying the mount around can be provided with the bolt directly, or a knob or wing-nut can be used on the bolt to control the compression with easier adjustment.
- Little tightness is needed, gravity should do the work. Compression more to hold mount together when moving it.
- Three pads of slick material let the upper rotate on the lower disc

Teflon is the ideal slip pad material, as against various surfaces it exhibits ideal friction characteristics – namely, very little difference between static and kinetic friction. For us, that means no "start-up jerk" which translates to very smooth steering action.

PTFE or other furniture sliders may substitute for cheaper/more accessible materials

B.8.2: AZIMUTH SANDWICH ASSEMBLY 1

(1) Find the center of your two disks (there are a number of ways including jigs)

(2) Carefully drill this out on both disks.

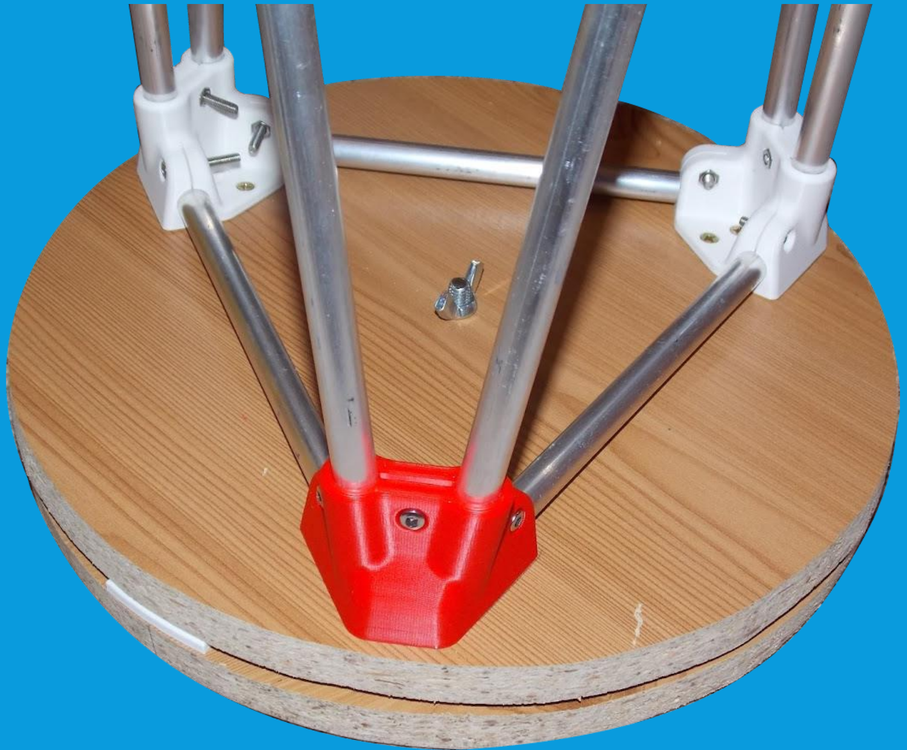
(3) Attach raised feet to underside of bottom disc in triangle pattern.

(4) Cut three small (0.5") Teflon pads and drill center holes for mounting, or obtain three PTFE furniture sliders. Mount these on the other side of the bottom disk, right above the feet.

B.8.3: AZIMUTH SANDWICH ASSEMBLY 2

(4) Prepare upper disk with center hole and mount to vertical supports (printed or wood). As an additional option, cut laminate or a large record can be glued to the under side of the top disc for improved friction behavior.

(5) Assemble stack: lower disc, upper disc, bolt through upper disc, optionally knob or wingnut on bolt before inserting. Tighten knob, nut, or bolt only until there is no room to wobble.



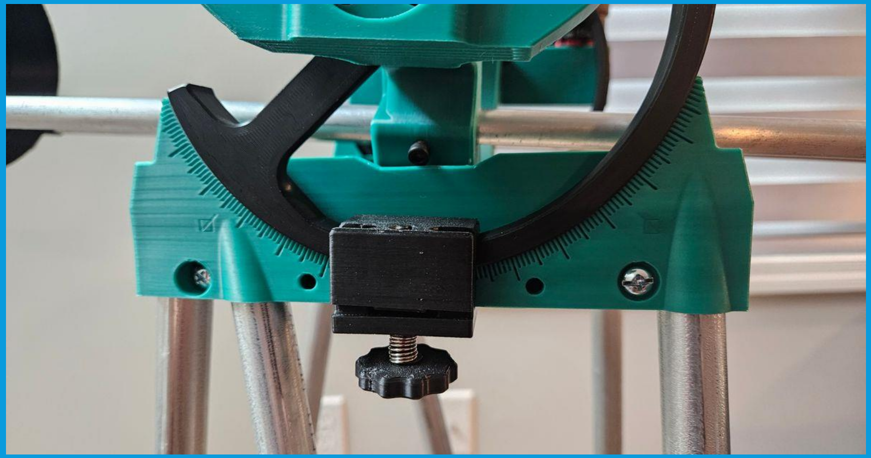
Images of bearing assembly provided by Hayden on the Hadley Discord

B.9: COMMERCIAL MOUNTS

- If you have an old equatorially mounted telescope, such as one of the Astromaster or Powerseeker EQ's, you may (with some mods) use this for Hadley. It will work better than with the original telescope, as the Hadley is very lightweight
- If you have a large ($400-900) budget for attempting astrophotography, know that "deep sky" is a more challenging rabbit hole than visual and planetary photography.
- However, motorized mounts (Skywatcher AZ-GTI, various other skywatcher/ioptron etc.) are very useful for visual astronomy as well.

B.10: TENSION, COUNTERWEIGHT, CLAMPS

- **Some mods to improve altitude action are discussed here.** Always start from a place of balance.
- Ideally, your Hadley is balanced at the altitude bearing around your average eyepiece and demonstrates enough balance insensitivity to not slide.
- However, printed friction bearings can be "slippy" and allow freedom to slide , especially if bumped by users.
- Tension springs and tensioners add more downforce to hold the telescope as balance changes and moves, and can act as a brake to stop inadvertent movement from clumsy users



- Counterweights can be adjusted to keep fixed center of balance
- Tension springs, counterweights, and clamps can also help compensate for change in balance as you swap eyepieces or add cameras

C. MIRROR TESTING AND PERFORMANCE

C.1: "WHY IS JUPITER BLURRY?"

Several factors contribute to a sub-par image. While Hadley is "easy to use", environmental conditions will limit the view especially at high power; moreover, improper alignment of the mirrors, or improper "collimation", can have significant impacts on performance.

In some cases, manufacturing defects have been identified in mirrors sourced from various online marketplaces. These defects cause the Hadley to underperform, sometimes quite severely, even when observing under good atmospheric and collimation conditions. This guide will show you how to identify these defects, and what you can do to correct them.

Before assuming bad optics:

- Collimation makes or breaks high power performance
- "Are the stars twinkling or pinpoints?" Like looking through water, stable air is needed for high power to work well – planets are blurrier if the "seeing" is bad.
- It takes practice to eke out color detail from the planets.
- Humidity (fog on optics), mirror optics, local air currents and more can impede the views.

C.2: WHAT MAKES A MIRROR BAD

This appendix briefly discusses how to narrow down optical problems, determining if the optics are the culprit.

- The process of grinding and polishing a "perfect" concave sphere is self-correcting, meaning a concave sphere can be precision ground quickly, cheaply and accurately on the order of a light wavelength
- In the case of Hadley's dimensions, spherical mirrors are a suitable stand-in for parabolic.

The Problem:

- Even nanoscopic deviations from the ideal shape will reveal themselves in the image
- In spite of "self-correcting", some factory mirrors have defects that are visible in a Hadley's final image
- Some optical defects can be "masked," but other defects cannot be addressed without replacing the mirror

C.3: STAR TESTING YOUR HADLEY

- Many optical issues can be diagnosed with a "star test" - observing a star slightly out of focus.
- To carry this out, point your telescope at a very bright star using the highest magnification you can muster. Lower magnification makes the star tests harder to discern.
- Ensure view is correctly focused, producing a pinpoint star*, then take the telescope slightly out of focus (in either direction).
- Slightly defocused *with the star centered,* the telescope should produce a concentric bullseye pattern.
- Problems with this pattern present a powerful diagnostic tool for your optics.

*pinpoint, with appropriate patterns for your choice of spider

Note that defective mirrors may be flawed in multiple ways, EG astigmatism and rolled edge.

C.4: STAR TESTING YOUR HADLEY

A bright star, taken slightly out of focus, is a powerful diagnostic tool.

- It should produce a series of concentric rings.
- If the image is turbulent and unsteady, the problem is unstable atmosphere or a hot mirror; try a calmer night.
- If the rings are not concentric, this generally indicates miscollimation of the primary, which wrecks the view.
- If one ring is extremely bright, it can be a mirror which is defective, or just still cooling. Leave outside for at least 15 minutes, then try again.

Above: examples of an acceptable "star test" in turbulent air

- If a single bright ring persists or bullseye pattern is elliptical or otherwise deformed, it likely indicates idenitfiable optical problems

C.5: TURNED-DOWN EDGE, "ZONES"

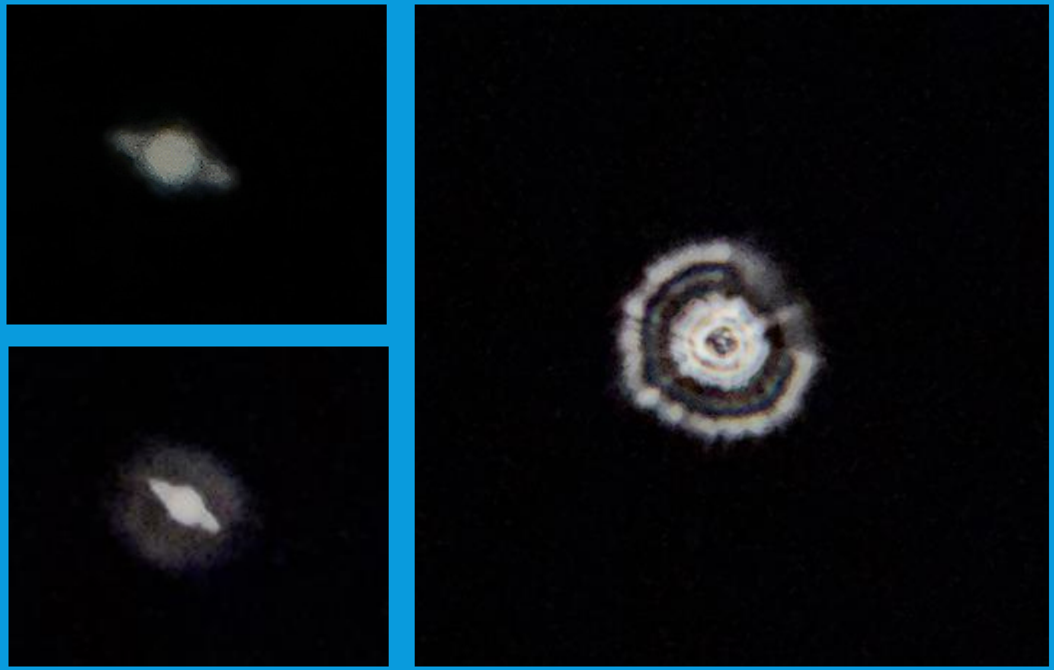
Turned-down edge, or TDE, is when the outer few millimeters of the optical surface are lower, scattering light *near* focus. This manifests as a halo or fuzz around stars and planets.

- "Zones" are when small "rings" of your mirrors surface focus light to the wrong distance. A zone-y mirror is best replaced.
- TDE is the most common defect in commercial optics, often seen in factory mirrors.
- It can be "masked" with a ring that covers only the bad part of the mirror
- Seen at right: a star test of a mirror with extreme zonal defects; Saturn pictured with and without a masking ring.

C.5.5: TURNED-DOWN EDGE, "ZONES"

Zonal defects are where portions of the mirror focus light differently. Edge zones can be masked, center zones are often inconsequential. Major zones prevent a clear image completely.

- They are diagnosed with a very strange star test / evidence of multiple focus points
- The Saturn shown at right is masked (top) and unmasked (bottom); notice how even when Saturn is "focused" there is a large area of defocused light.
- This is an example of a very bad mirror still putting up a decent view when the defective area is masked.



C.6: ASTIGMATISM

Astigmatism is a defect where something distorts the wavefront to be more cylindrical. Caused by a primary that is 1% too "pringle-shaped" or a secondary that is not perfectly flat.

To isolate it to the primary, note the angles of the ellipses; do they rotate with the primary? (Pull out cell, rotate, replace cell)

- Seen at right: a mirror with extreme astigmatism; the star is seen as a warped diamond shape, and the star test is extremely oblate.
- The telltale sign (whether primary or secondary) is that the ellipse turns 90 degrees when crossing focus inward or outward
- Minor astigmatism is livable; major astigmatism is not.

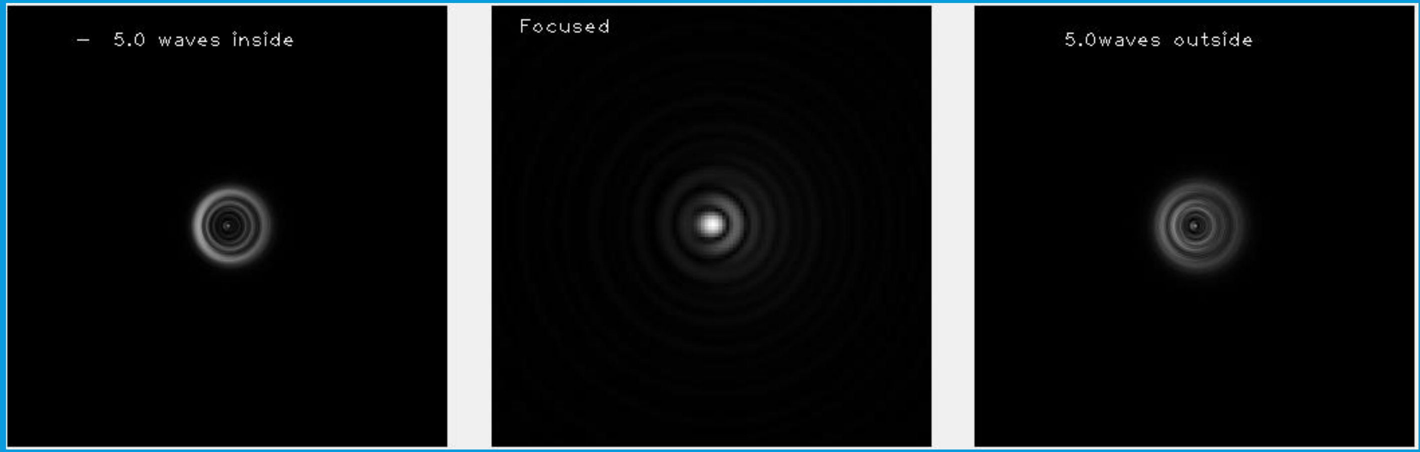
Above: the "inside" and "outside" star tests on a *very* astigmatic Hadley mirror;
Above right: a zoomed in view of the non-pinpoint star – the cross is distinct from the diffraction spike.

3.7.1: LIBRARY OF STAR TESTS

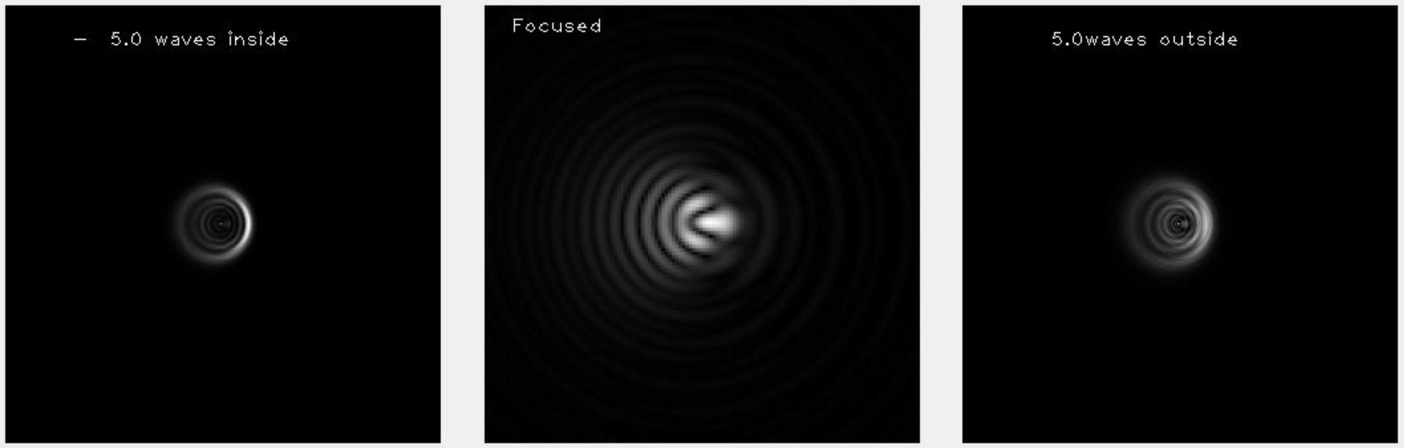
These star tests have been simulated by Matt Baker, generated in DFTFringe

Top: Slightly Miscollimated

Bottom: Severely Miscollimated
(Note how much this distorts what should be a single "point" of detail)

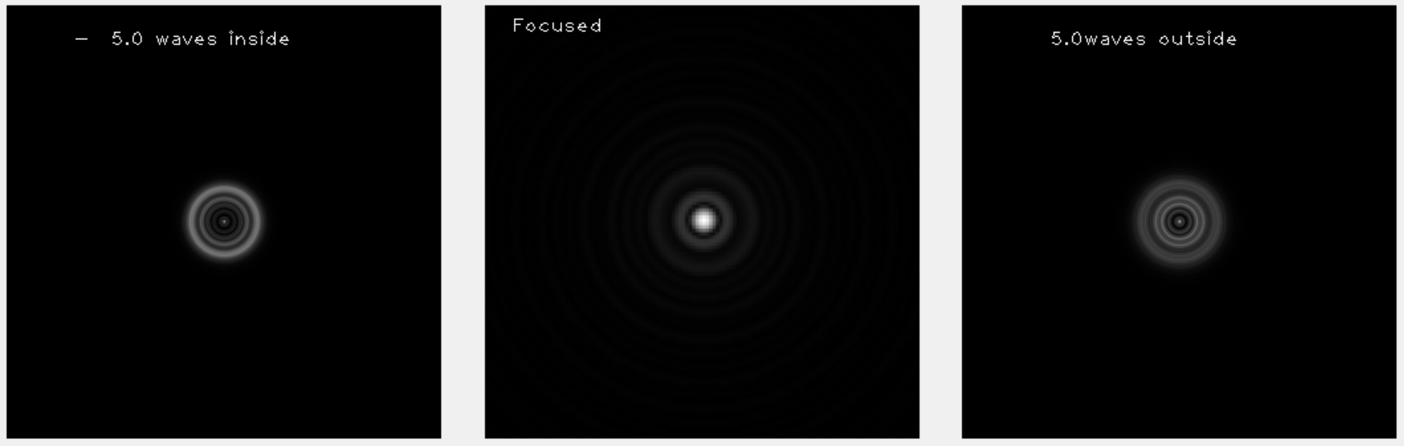


Reading these: The center is a vastly zoomed in view of what a "star" (without spikes from spider) looks like under these conditions/aberrations; the left and right slides are non-zoomed images of what the defocused star looks like.



3.7.2: LIBRARY OF STAR TESTS

Top: perfect mirror.
Note that the star test varies with aperture, secondary size/telescope type and more. These are all simulated spiderless Hadley tests. Other scopes will show more/fewer rings and produce slightly different star tests. But the nature of the defects is universally applicable.

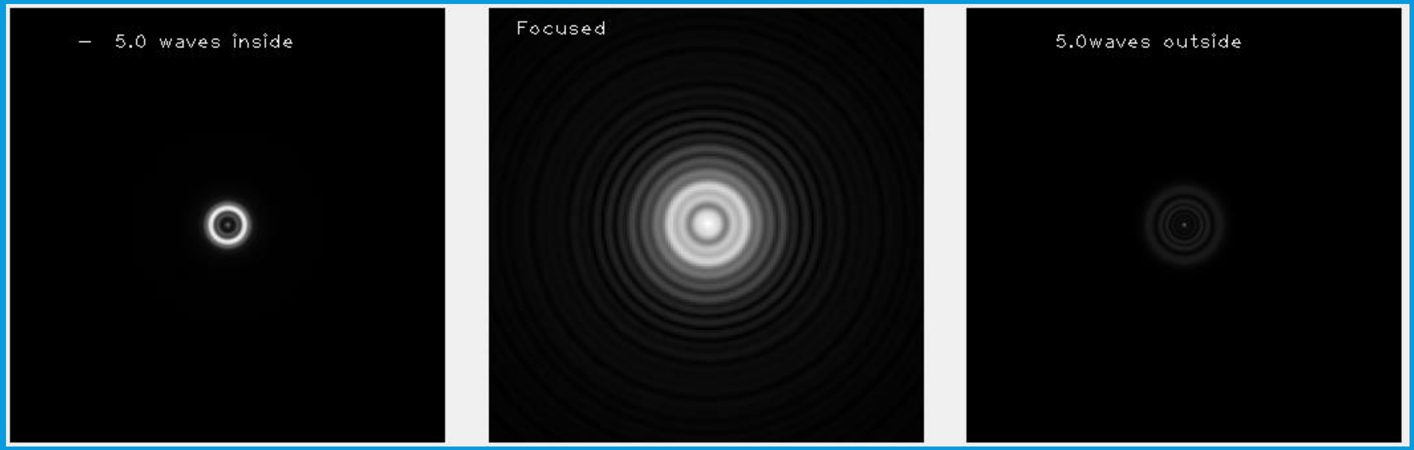


Bottom: Minor turned down edge.
Notice how (slightly) more light is scattered into diffraction rings *around* the star or point of interest.

Mostly harmless; mask the outermost portion of mirror.

3.7.3: LIBRARY OF STAR TESTS

Top: Very bad edge.
Note how the star is massively fattened, and much of the starlight is thrown into a "haze" round the image.
Mask down to somewhat salvage, but replacing the primary may improve views noticeably.



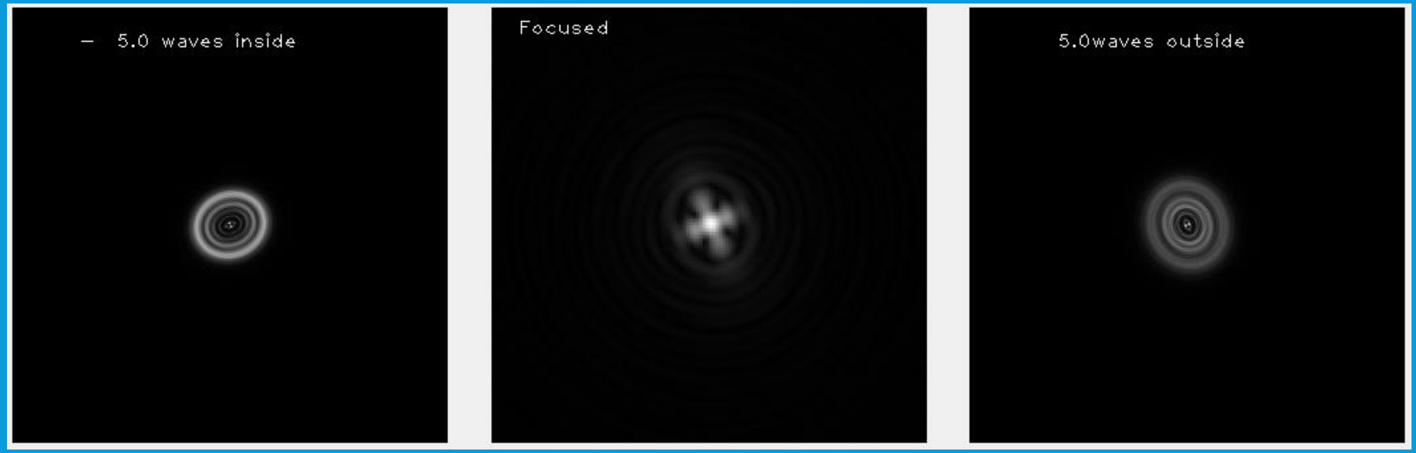
Bottom: "Zone" defect.
The appearance is of a superimposed, defocused star/view on top of the focused star; this manifests as halos and overall a severely worsened image. **Replace Primary.**

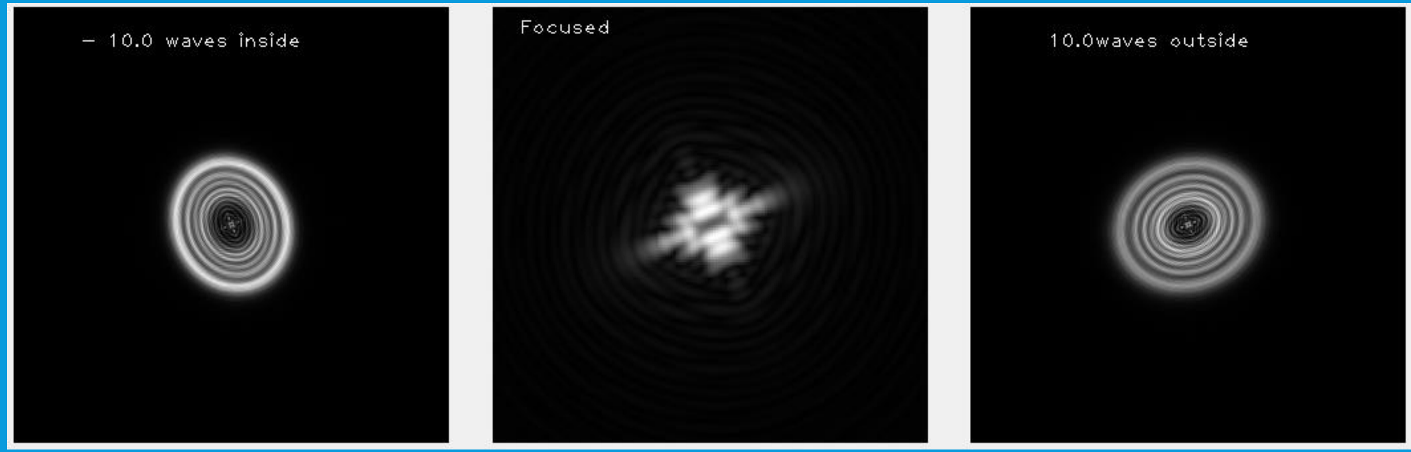
3.7.4: LIBRARY OF STAR TESTS

Top: Minor astigmatism
Bottom: Major astigmatism.

Astigmatism must first be correctly traced to the primary – note the angle of the ellipses. Rotate the primary in place, and see if the angle also rotates – if so, it is in the figure of the primary.

Very minor astigmatism is livable, major astigmatism will degrade views at high power.





APPENDICES & SECTIONS TO COME

- Important need-to-knows (sun safety, no touching or cleaning mirrors)
- Hadley protips: mark rods if you have to slide stuff around a lot, and other such tips.
- Parfocal rings for most common eyepieces?
- How to add and use finderscopes
- Fitting a camera
- More about spider options
- UTA Orientation Selection
- Other Hadley family telescopes
- Responsible management of "aperture fever"
- For questions in the meantime, ask at the Hadley Discord available on the Printables page